AP Statistics Prep Book 2027

The Complete Study Guide & Textbook for High School Students Featuring Practice Workbooks, TI-84 Playbooks, and 2027 Full-Length Exams

AP Statistics Master Review Series — Book 2

PR

2027 Exam Ready Edition

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Contents

How to Use This Textbook for Score-5 Mastery

Welcome to the **AP Statistics Prep Book 2027**. This complete textbook and study system is engineered to guide high school students from foundational statistical principles to complete Score-5 mastery on the official AP Exam.

The 4-Pillar Pedagogical Architecture

1. **Core Concept Lessons:** Every unit breaks down the official College Board Course and Exam Description (CED) topics into intuitive, digestible modules with zero confusing academic jargon. Definitions, formulas, and conceptual rationales are clearly demarcated.

2. **Guided Practice Workbooks:** Each chapter includes hands-on drill sets ranging from foundational calculations to challenging multi-step questions, complete with step-by-step grader-approved solutions.

3. **TI-84 Plus CE Playbooks:** Exact keystrokes and display menus showing exactly how to execute probability calcula-

tions, regression models, summary statistics, and hypothesis tests.

4. **Score-4 Rubric Templates:** Grader-approved fill-in-the-blank sentence frames designed to guarantee maximum credit on the 6 Free-Response Questions (FRQs) across every inference and analysis scenario.

FREE DIGITAL WEB COMPANION & 23 EXAMS SIMULATOR

Official Online Testing Suite (25 Exams Total: 2 in Book + 23 Online):

- **23 Full-Length Online Practice Exams:** 42 MCQs + 4 FRQs each with live 90-min timer
- **2027 Bluebook Simulator:** Embedded Desmos Calculator & QWERTY notation assistant
- **Instant Automated Grading:** Detailed rationales and score reports for all 966 online MCQs
- **1-Click Printable PDF Mode:** Print test booklets and answer keys on any device

Official Web Simulator: `https://eduprosuite-org.github.io/narayana-kdp-vault/practice-exams.html`

Official AP Statistics Exam Format (2027)

Section	Question Type	Questions	Time	Exam Weight
Section I	Multiple-Choice Questions (MCQ)	40 Questions	90 Minutes	50%
Section IIA	Short/Multi-Part Free-Response	5 Questions	65 Minutes	37.5%
Section IIB	Investigative Task (Extended FRQ)	1 Question	25 Minutes	12.5%

The 10-Week Master Score-5 Study Roadmap

- **Weeks 1–2 (Exploring Data):** Master Units 1 & 2. Focus on C-U-S-S for one-variable data and D-O-F-S with LSRL residual analysis for two-variable data.
- **Week 3 (Data Collection):** Master Unit 3. Drill SRS, stratified vs. cluster sampling, bias types, and randomized block experiments.
- **Weeks 4–5 (Probability & Distributions):** Master Units 4 & 5. Drill random variables, combinations, Binomial ($BINS$), Geometric, and the Central Limit Theorem.
- **Weeks 6–7 (Inference for Proportions & Means):** Master Units 6 & 7. Memorize PANIC (Confidence Intervals) and PHANTOM (Significance Tests).
- **Week 8 (Chi-Square & Slopes):** Master Units 8 & 9. Goodness-of-Fit, Homogeneity, Independence, and regression t-tests.
- **Weeks 9–10 (Full-Length Exam Simulation):** Timed completion of Practice Exam 1 and Practice Exam 2 under strict exam conditions.

Introduction: The Master Blueprint for AP Statistics Success

1. The 2027 Official AP Statistics Exam Architecture

The Advanced Placement Statistics Examination evaluates students on their conceptual understanding, data analysis acumen, experimental design competence, and statistical inference skills. The 3-hour examination is administered digitally on the College Board Bluebook application and is divided into two equally weighted components:

Exam Component	Format	Time	Raw Score	Scaled Weight
Section I	42 Multiple-Choice Questions	90 Minutes	42 Points	50% of Total Score
Section II	4 Free-Response Questions	90 Minutes	16 Points	50% of Total Score
Total Exam	**46 Questions**	**180 Minutes**	**58 Raw Pts**	**100% Composite**

Section I: Multiple-Choice Strategy (Questions 1–42)

With 42 questions in 90 minutes, you have an average of **2 minutes and 8 seconds per question**. Multiple-choice questions are weighted equally; there is no penalty for incorrect answers.

- **Question Structure:** Contains both individual discrete multiple-choice questions and **Item Sets** (sets of 2–3 questions sharing a common data table, scatterplot, or study description).

- **First Pass (Questions 1–42):** Answer all straightforward calculation, vocabulary, and graphical interpretation questions immediately.

- **Second Pass:** Return to flagged questions. Eliminate illogical distractors (such as negative variances, correlation coefficients outside $[-1, 1]$, or p-values greater than 1).

- **Zero Blanks Policy:** Fill every single option before the 90-minute countdown ends.

Section II: Free-Response Strategy (Questions 1–4)

Section II consists of **4 multi-part questions** in 90 minutes (approximately **22.5 minutes per question**). Graders evaluate each question on a 0–4 point holistic scale:

- **Question 1 (Multi-Focus on Practices 1 & 2):** Evaluates your ability to **Formulate Questions** (Skill 1.A) and **Collect Data** (Skills 2.A–2.E: sampling methods, bias diagnosis, experimental designs, and hypothesis identification).

- **Question 2 (Multi-Focus on Practices 3 & 4):** Evaluates your ability to **Analyze Data** (Skills 3.A–3.E: distributions, summary statistics, predicted responses) and **Interpret Results** (Skills 4.A–4.D: comparative descriptions with context and claim justifications).

- **Question 3 (Statistical Inference):** A comprehensive inferential investigation requiring you to conduct a full hypothesis test (**PHANTOM**) or construct a confidence interval (**PANIC**), verify all necessary conditions, perform calculations, and interpret in context.

- **Question 4 (Multi-Focus Synthesis on Practices 2, 3, & 4):** An extended, multi-practice investigative task integrating experimental/observational design, graphical/analytical calculations, and inferential conclusions under uncertainty.

2. The Official 5-Unit Curriculum & Course Skills Framework

The AP Statistics curriculum is structured around 5 foundational units and 4 core practices:

Unit	Curriculum Domain	MCQ Exam Weighting
Unit 1	Exploring One-Variable Data and Collecting Data	**20% – 30%**
Unit 2	Probability, Random Variables, and Probability Distributions	**15% – 25%**
Unit 3	Inference for Categorical Data: Proportions	**15% – 25%**
Unit 4	Inference for Quantitative Data: Means	**10% – 20%**
Unit 5	Regression Analysis	**10% – 20%**

The 4 Statistical Course Skills & Practices

- **Practice 1: Formulate Questions (5%–10% Weight):** Determine valid investigative questions requiring statistical investigation (Skill 1.A).

- **Practice 2: Collect Data (20%–30% Weight):** Identify necessary information (2.A), justify ethical data collection methods (2.B), select appropriate inference methods (2.C), diagnose errors (2.D), and state hypotheses (2.E).

- **Practice 3: Analyze Data (25%–35% Weight):** Construct graphical displays (3.A), compute summary statistics (3.B), compute expected counts and probabilities (3.C), find parameters (3.D), and compute test statistics/intervals (3.E).
- **Practice 4: Interpret Results (25%–35% Weight):** Describe distributions and compare relative positions (4.A, 4.C), justify claims (4.B, 4.G), verify inference conditions (4.E), and interpret statistical conclusions in context (4.D, 4.F).

3. 2027 Digital Bluebook & Calculator Policy Mandate

- **Calculator Policy Update for 2027: Nongraphing scientific calculators are STRICTLY PROHIBITED**, even if they contain statistical functions.
- **Approved Calculators:** You must use either an approved handheld **Graphing Calculator with statistical capabilities** (such as the TI-84 Plus CE) OR the **Built-in Desmos Graphing Calculator** accessible directly within the Bluebook application.
- **Digital Notation Entry on QWERTY Keyboards:** On the digital Bluebook exam, statistical symbols can be entered cleanly as plain text without special equation editors:

- Sample Proportion: Type `p-hat` or `^p`
- Sample Mean: Type `x-bar`
- Population Parameters: Type `mu`, `sigma`, or `beta`
- Test Statistics: Type `z`, `t`, or `Chi-Square` (or `X^2`)
- Significance Levels: Type `alpha = 0.05`
- Confidence Intervals: Enter as `(lower, upper)` or `statistic +/- margin of error`

4. The Official AP Scoring Architecture & Composite Scale

Free-response questions are evaluated using a 4-point holistic rubric:

- **E — Essentially Correct (1.0 Pt):** Complete, accurate, and contextually grounded.
- **P — Partially Correct (0.5 Pt):** Correct approach but has minor slips, missing units, or incomplete context.
- **I — Incomplete (0.0 Pts):** Major conceptual misunderstanding or fails to address the prompt.

The Composite Conversion Formula

Your raw score converts into a standardized 100-point composite score:

$$\text{Composite Score} = \left(\frac{50}{42} \times \text{MCQ Raw}\right) + \left(3.125 \times \sum_{i=1}^{4} \text{FRQ}_i\right)$$

Typical historical composite cut scores for the 5-point AP scale:

AP Score	Composite Score Range (Approx.)	College Board Recommendation
5	**70 – 100 Points**	Extremely Well Qualified (College Credit & AP)
4	**57 – 69 Points**	Well Qualified
3	**44 – 56 Points**	Qualified (Standard Passing Threshold)
2	**33 – 43 Points**	Possibly Qualified
1	**00 – 32 Points**	No Recommendation

Notice: A composite score of approximately 70 out of 100 (70%) consistently earns the top score of 5! Rigorous attention to context and condition verification will secure your 5.

5. The 10 Deadliest Exam Traps (Avoided by Score-5 Students)

1. **Trap 1: Forgetting Context in FRQs.** Writing *"The distribution is skewed right with median 14"* earns a P instead of an E. Always specify the variable and measurement units: *"The distribution of hospital recovery times is skewed right with a median of 14 days."*

2. **Trap 2: Claiming to "Prove" the Null Hypothesis.** Never state *"We accept H_0"* or *"We have proven the true mean is 50."* When p-value $> \alpha$, the only correct statistical statement is: *"We fail to reject H_0. There is insufficient statistical evidence that..."*

3. **Trap 3: Using Sample Statistics in Hypotheses.** Writing $H_0 : \bar{x} = 100$ or $H_0 : \hat{p} = 0.5$ results in an automatic deduction. Hypotheses represent unknown **population parameters**: $H_0 : \mu = 100$ or $H_0 : p = 0.5$.

4. **Trap 4: Subtracting Standard Deviations When Combining Random Variables.** For independent variables, variances ALWAYS add: $\sigma_{X-Y} = \sqrt{\sigma_X^2 + \sigma_Y^2}$. Standard deviations never subtract!

5. **Trap 5: Confusing Stratified Sampling with Cluster Sampling.**

   - *Stratified:* Divide population into **homogeneous** groups (strata); take an SRS from **every** group. (Some from all).

   - *Cluster:* Divide population into **heterogeneous** groups (clusters); randomly select **some clusters** and survey **everyone** inside them. (All from some).

6. **Trap 6: Calculator Syntax As Sole Work.** Writing `normalcdf(10, 9999, 12, 2)` on an FRQ earns an automatic P or I. You must draw or state the boundary, mean, and standard deviation explicitly: $z = \frac{10-12}{2} = -1.00$, $P(X \geq 10) = P(Z \geq -1.00) = 0.8413$.

7. **Trap 7: Using $\hat{p}$ Instead of p_0 in the Hypothesis Test Standard Error.** For a one-proportion z-test, the standard error MUST be computed under the assumption that H_0 is true: $\text{SE}_0 = \sqrt{\frac{p_0(1-p_0)}{n}}$. Using $\hat{p}$ in the denominator is an immediate penalty.

8. **Trap 8: Checking Observed Counts Instead of Expected Counts in Chi-Square.** The Large Counts condition requires all **expected counts** to be ≥ 5. Observed counts can be 1, 2, or even 0.

9. **Trap 9: Extrapolation in Linear Regression.** Never use an LSRL to predict response values for x-values outside the domain of the collected data. Always note the hazard of extrapolation.

10. **Trap 10: Omission of Comparison Words in Comparative Distributions.** Writing *"Class A median is 80. Class B median is 70"* does not earn credit for comparison. You MUST use explicit comparative language: *"The median exam score for Class A (80 points) is strictly greater than the median for Class B (70 points)."*

Chapter 1

Unit 1: Exploring One-Variable Data

1.1 Unit Overview & Exam Weight

Unit 1 accounts for **15% to 23%** of the total score on the official AP Statistics Exam, making it one of the highest-yield foundational units in the entire curriculum. Mastery of variable classification, distribution descriptions, summary statistics, and normal distributions forms the indispensable grammar for every inference procedure tested in Units 6 through 9.

Key Unit Vocabulary

Individual:
An object or person described by a set of data (e.g., students, patients, vehicles).

Variable:
Any characteristic of an individual that can take different values for different individuals.

Categorical Variable:

Places an individual into one of several non-numerical groups or categories (e.g., eye color, grade level, zip code).

Quantitative Variable:

Takes numerical values for which arithmetic operations such as addition or averaging make sense (e.g., weight, reaction time, exam score).

Discrete Quantitative Variable:

A quantitative variable whose possible values can be counted (e.g., number of siblings, goals scored).

Continuous Quantitative Variable:

A quantitative variable that can take any value in an interval on the real number line (e.g., height, temperature, running speed).

Distribution:

The pattern of variation of a variable; tells us what values the variable takes and how frequently it takes those values.

Resistance:

A statistical measure is resistant if extreme observations (outliers or strong skewness) have little to no influence on its value. The median and IQR are resistant; the mean, range, and standard deviation are non-resistant.

Percentile:

The percentage of values in a distribution that fall strictly below a given value.

Z-Score (Standardized Value):

Measures the directed distance between an individual data value and the mean in units of standard deviation: $z = (x - \mu)/\sigma$.

1.2 Core Concept Lessons

Lesson 1.1: Variable Classification & The Identifier Trap

Foundational Principle: Not every variable expressed as a number is quantitative! A zip code (e.g., 90210), a social security number, or a jersey number (#23) uses numbers strictly as categorical labels. Taking the average of zip codes has no mathematical meaning.

- **Categorical:** Look for categories or groupings. Operations like averaging are meaningless.
- **Quantitative:** Look for counts or measurements with units (e.g., inches, seconds, dollars). Calculating the mean makes conceptual sense.

Lesson 1.2: Visualizing Categorical Data & Simpson's Paradox

Categorical distributions are displayed using **Bar Charts, Segmented Bar Charts, and Mosaic Plots**.

- **Bar Chart:** Bars must have distinct spaces between them to emphasize discrete categories. Heights represent frequencies or relative frequencies.
- **Mosaic Plot:** A segmented bar chart where the width of each bar is proportional to the size of the corresponding

group, allowing visual assessment of association.

- **Simpson's Paradox:** An apparent association between two categorical variables can reverse or disappear when data from several groups are combined into a single aggregate table.

Lesson 1.3: Visualizing Quantitative Data

Quantitative data must be displayed using charts that preserve or bin numerical scale:

- **Dotplots:** Ideal for small to moderate datasets ($n < 50$). Every individual observation is plotted as a single dot above a continuous number line. Shows exact data values, clusters, gaps, and outliers.
- **Stemplots (Stem-and-Leaf Displays):** Separates each value into a stem and a leaf. Always include an explicit **KEY** (e.g., $4 \mid 2 = 42$ mph).
- **Histograms:** Bins contiguous quantitative data into equal-width intervals along the horizontal axis. Unlike bar charts, the vertical bars in a histogram touch each other.

Lesson 1.4: Describing Distributions with C-U-S-S

Whenever an AP exam prompt asks you to **"describe"** or **"compare"** distributions, you MUST write complete sentences addressing all four pillars in context:

1. **C — Center:** State the numerical median (for skewed data or outliers) or mean (for roughly symmetric, non-

outlier data). Always include units!

2. **U — Unusual Features:** Explicitly state any outliers, clusters, or notable gaps. If no outliers exist, explicitly state: *"There are no apparent outliers or gaps."*

3. **S — Spread:** State the numerical IQR and range (for skewed data) or standard deviation and variance (for symmetric data).

4. **S — Shape:** State whether the distribution is **Skewed Left** (tail extends to the left), **Skewed Right** (tail extends to the right), **Roughly Symmetric**, **Bimodal**, or **Uniform**.

Exam Trap Alert: Comparative Language on FRQs

When comparing two or more distributions on the exam, you **cannot** simply list summary statistics for each group separately! You MUST use explicit comparative adjectives such as *"greater than"*, *"less than"*, *"more variable than"*, or *"approximately equal to"*.

- **Incorrect (Score: Incomplete):** "Group A has a median of 45. Group B has a median of 30."

- **Correct (Score: Complete):** "The median exam score for Group A (45 points) is **significantly greater than** the median score for Group B (30 points)."

Lesson 1.5: Measures of Center — Mean vs. Median

$$\text{Sample Mean: } \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = \frac{x_1 + x_2 + \cdots + x_n}{n} \tag{1.1}$$

The **median** (M) is the midpoint of the ordered array:

- If n is odd, M is the exact middle observation at position $(n+1)/2$.
- If n is even, M is the average of the two middle observations at positions $n/2$ and $(n/2)+1$.

The Skewness Relationship:

- **Symmetric:** Mean $\approx$ Median
- **Skewed Right:** Mean $>$ Median (the extreme right tail pulls the non-resistant mean upward)
- **Skewed Left:** Mean $<$ Median (the extreme low tail pulls the mean downward)

Lesson 1.6: Measures of Spread — IQR and Standard Deviation

$$\text{Interquartile Range: IQR} = Q_3 - Q_1 \tag{1.2}$$

Q_1 is the median of the lower half of data; Q_3 is the median of the upper half of data.

$$\text{Sample Standard Deviation: } s_x = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2} \tag{1.3}$$

Interpretation of Standard Deviation: *"The standard deviation of s_x measures the typical distance that individual observations in this dataset deviate from the sample mean $\bar{x}$."*

Lesson 1.7: The $1.5 \times$ IQR Rule for Outliers

To mathematically identify outliers, establish two boundary fences:

$$\text{Lower Fence} = Q_1 - 1.5(\text{IQR}), \qquad \text{Upper Fence} = Q_3 + 1.5(\text{IQR}) \tag{1.4}$$

Any individual observation x satisfying:

$$x < \text{Lower Fence} \quad \text{OR} \quad x > \text{Upper Fence} \tag{1.5}$$

is classified as a statistical outlier. In a **Modified Boxplot**, whiskers extend only to the smallest and largest observations inside the fences; outliers are plotted as isolated dots.

Lesson 1.8: Linear Transformations of Datasets

Suppose a dataset undergoes a linear transformation $y_i = a + bx_i$, where a and b are constants:

- **Adding/Subtracting a Constant (a):** Adds a to all measures of **center and position** (Mean, Median, Quartiles, Min, Max). Measures of **spread** (Standard

Deviation, Variance, Range, IQR) remain **strictly unchanged**!

- **Multiplying/Dividing by a Constant (b):** Multiplies measures of center and position by b. Multiplies measures of spread by $|b|$. Variances multiply by b^2.
- **Shape:** Linear transformations **never** change the shape of a distribution!

Lesson 1.9: Standardized Scores (Z-Scores) & Normal Distribution

$$z = \frac{x-\mu}{\sigma} \quad \text{(Population)}, \qquad z = \frac{x-\bar{x}}{s_x} \quad \text{(Sample)} \tag{1.6}$$

A z-score indicates how many standard deviations an observation falls above ($z > 0$) or below ($z < 0$) the mean. **The Empirical 68–95–99.7 Rule:** For approximately normal distributions $N(\mu, \sigma)$:

- Approx. **68%** of observations fall within $\mu \pm 1\sigma$.
- Approx. **95%** of observations fall within $\mu \pm 2\sigma$.
- Approx. **99.7%** of observations fall within $\mu \pm 3\sigma$.

1.3 Worked Step-by-Step Examples

Example 1.1: Comprehensive C-U-S-S Analysis & Outlier Testing

Problem: A sports scientist records the recovery heart rates (in beats per minute) of 15 elite cross-country runners 5 minutes post-race:

$$68,\ 72,\ 74,\ 75,\ 76,\ 78,\ 79,\ 80,\ 81,\ 82,\ 85,\ 88,\ 91,\ 94,\ 118$$

(a) Determine the 5-number summary and construct the outlier boundary fences.
(b) Identify any outliers and justify mathematically.
(c) Describe the distribution using the C-U-S-S framework in complete context.

Solution:

- **Part (a):** $n = 15$. Sorted observations:
  Min $= 68$ bpm, Max $= 118$ bpm.
  Median position: $(15 + 1)/2 = 8^{\text{th}}$ value $\implies M = 80$ bpm.
  Lower half (first 7 values): $Q_1 = 4^{\text{th}}$ value $= 75$ bpm.
  Upper half (last 7 values): $Q_3 = 4^{\text{th}}$ value $= 88$ bpm.
  5-Number Summary: $\{68,\ 75,\ 80,\ 88,\ 118\}$ bpm.
  $\text{IQR} = Q_3 - Q_1 = 88 - 75 = 13$ bpm.
  Lower Fence $= 75 - 1.5(13) = 75 - 19.5 = 55.5$ bpm.
  Upper Fence $= 88 + 1.5(13) = 88 + 19.5 = 107.5$ bpm.

- **Part (b):** Because 118 bpm $>$ 107.5 bpm, the observation 118 bpm is a confirmed high outlier. There are no low outliers because Min $= 68 > 55.5$.

- **Part (c):** *"The distribution of recovery heart rates among these 15 elite runners is skewed to the right, with an isolated* *high outlier at 118 bpm. The center of the distribution is best represented by a median recovery rate of 80 bpm. The spread is characterized by an Interquartile Range (IQR) of 13 bpm and an overall range of 50 bpm (from 68 bpm to 118 bpm)."*

Example 1.2: Normal Distribution Forward Probability & Percentile

Problem: The birth weights of full-term newborn infants in a hospital system are approximately normally distributed with a mean of $\mu = 3,450$ grams and a standard deviation of $\sigma = 480$ grams. (a) What proportion of full-term newborns weigh more than $4,000$ grams?
(b) An infant is classified as "low birth weight" if its weight falls below the 10^{th} percentile. What is the maximum cutoff weight for this classification?

Solution:

- **Part (a):** Let $X \sim N(3450, 480)$. Standardize $x = 4000$:

$$z = \frac{4000 - 3450}{480} = \frac{550}{480} \approx 1.15$$

$P(X > 4000) = P(Z > 1.15) = 1 - 0.8749 = 0.1251$ (**12.51%**).

- **Part (b):** For 10^{th} percentile, $z^* \approx -1.282$ via standard normal table.

$x^* = \mu + z^*\sigma = 3450 + (-1.282)(480) = 3450 - 615.36 = \mathbf{2,834.64}$ gram

1.4 TI-84 Plus CE Calculator Playbook

TI-84 Playbook: Summary Statistics & Modified Boxplots

1. Press `[STAT]` → Select `1:Edit...` → Enter data into `L1`.
2. Press `[STAT]` → `CALC` → Select `1:1-Var Stats` → Calculate.
3. To plot: Press `[2nd] [Y=]` (`STAT PLOT`) → Select `Plot1` → Turn `ON`.
4. Under `Type:`, select the first boxplot icon (with isolated outlier points).
5. Press `[ZOOM]` → Select `9:ZoomStat`.

1.5 Comprehensive Practice Problem Set

Part I: 20 Multiple-Choice Questions

1. A survey records: GPA, Gender, Miles driven per week, and Zip code. How many variables are quantitative?

   (A) 0

   (B) 1

   (C) 2

   (D) 3

   (E) 4

Answer: (C). GPA and Miles driven are quantitative. Gender and Zip code are categorical.

2. A distribution of employee salaries is strongly skewed to the right. Which relationship is true?

   (A) Mean $<$ Median

   (B) Mean $\approx$ Median

   (C) Mean $>$ Median

   (D) SD $=$ IQR

   (E) Mean $= 0$

   Answer: (C). Right skewness pulls the non-resistant mean above the median.

3. A dataset has $Q_1 = 28$ and $Q_3 = 44$. An outlier must fall:

   (A) Below 4 or above 68

   (B) Below 12 or above 60

   (C) Below 20 or above 50

   (D) Exactly at 44

   (E) Between 28 and 44

   Answer: (A). IQR $= 16$. Lower Fence $= 28 - 24 = 4$. Upper Fence $= 44 + 24 = 68$.

4. If every observation is multiplied by 3 and increased by 10, how do the mean and standard deviation change?

   (A) Mean $\times 3 + 10$; SD $\times 3 + 10$

   (B) Mean $\times 3 + 10$; SD $\times 3$

(C) Mean +10; SD ×3

(D) Both ×3

(E) Neither changes

Answer: (B). Center shifts by addition and multiplication; spread scales only by multiplication.

5. A runner's time in a 10K race corresponds to $z = -1.80$. This indicates:

   (A) The runner finished 1.8 minutes faster than average.

   (B) The runner's time was 1.80 standard deviations below the mean.

   (C) The runner was in the bottom 1.8% of competitors.

   (D) The runner was 1.80 standard deviations above the mean.

   (E) The time was negative.

   Answer: (B). $z = -1.80$ means 1.80 standard deviations below the mean.

6. Adult male heights follow $N(70 \text{ in}, 3 \text{ in})$. Approximately what percentage are between 64 and 76 inches?

   (A) 50%

   (B) 68%

   (C) 95%

   (D) 99.7%

   (E) 100%

   Answer: (C). Within $\mu \pm 2\sigma$ is approximately 95% by the Empirical Rule.

7. Which summary statistic is resistant to outliers?

   (A) Sample Mean

   (B) Range

   (C) Standard Deviation

   (D) Interquartile Range (IQR)

   (E) Variance

   Answer: (D). IQR is based on percentiles, making it resistant.

8. If scores are transformed by $y = 1.2x - 5$ and original variance was 25, the new variance is:

   (A) 25

   (B) 30

   (C) 36

   (D) 20

   (E) 28.8

   Answer: (C). Variance scales by $b^2 = 1.2^2 = 1.44$. $25 \times 1.44 = 36$.

9. On an exam, Alice scored 82 ($\mu = 76, \sigma = 4$). Bob scored 88 ($\mu = 82, \sigma = 5$). Who performed better relatively?

   (A) Alice ($z = 1.50$ vs Bob's $z = 1.20$)

   (B) Bob

   (C) Both equal

   (D) Cannot determine

(E) Alice, because s was smaller

Answer: (A). Alice: $z = (82 - 76)/4 = 1.50$. Bob: $z = (88 - 82)/5 = 1.20$.

10. Test scores follow $N(50, 10)$. What percentage exceed 70?

(A) 2.5%

(B) 5%

(C) 16%

(D) 32%

(E) 0.15%

Answer: (A). $z = (70 - 50)/10 = 2.0$. Tail area = 2.5%.

11. The 75$^{\text{th}}$ percentile of $N(100, 15)$ is closest to:

(A) 110.1

(B) 115.0

(C) 107.5

(D) 112.3

(E) 120.0

Answer: (A). $z \approx 0.674$. $x = 100 + 0.674(15) = 110.1$.

12. In a boxplot, if the right whisker is much longer than the left whisker, the shape is:

(A) Skewed left

(B) Symmetric

(C) Skewed right

(D) Uniform

(E) Normal

Answer: (C). Elongated upper values reflect positive (rightward) skewness.

13. In a standard normal distribution, what is the mean and standard deviation?

(A) $\mu = 1, \sigma = 0$

(B) $\mu = 0, \sigma = 1$

(C) $\mu = 1, \sigma = 1$

(D) $\mu = 0, \sigma = 0$

(E) Variable

Answer: (B). By definition, $Z \sim N(0, 1)$.

14. If the minimum value in a dataset decreases by 10, which statistic remains unchanged?

(A) Mean

(B) Range

(C) Standard Deviation

(D) Median

(E) Minimum

Answer: (D). The median is resistant to changes in extreme values.

15. A distribution of incomes has Mean = $\$70,000$ and Median = $\$52,000$. This indicates:

(A) Symmetric

(B) Skewed left

(C) Skewed right

(D) Uniform

(E) Bell-shaped

Answer: (C). Mean > Median indicates right skewness.

16. The area under a standard normal curve between $z = -1.0$ and $z = +1.5$ is:

(A) 0.6826

(B) 0.7745

(C) 0.8185

(D) 0.8664

(E) 0.9332

Answer: (B). $0.9332 - 0.1587 = 0.7745$.

17. A 5-number summary is $10, 20, 35, 50, 95$. The upper outlier fence is:

(A) 50

(B) 80

(C) 95

(D) 100

(E) 110

Answer: (C). IQR $= 30$. Upper Fence $= 50 + 1.5(30) = 95$.

18. Back-to-back stemplots are most useful for:

(A) Displaying two categorical variables

(B) Comparing two quantitative distributions

(C) Scatterplots

(D) Time series

(E) Normal probability plots

Answer: (B). Enables direct comparison of shapes, centers, and spreads on the same scale.

19. If $P(Z < z) = 0.20$, what is the approximate value of z?

(A) -0.84

(B) $+0.84$

(C) -0.20

(D) $+0.20$

(E) -1.28

Answer: (A). By table lookup, $z \approx -0.84$.

20. A dotplot showing two distinct separated peaks with few observations in between is described as:

(A) Uniform

(B) Unimodal

(C) Bimodal

(D) Skewed left

(E) Normal

Answer: (C). Two major peaks represent a bimodal distribution.

Part II: Free-Response Practice Problem

FRQ 1.1: Comparing Class Distributions

Problem: Two classes take a 100-point statistics exam: Class A ($n = 25$) has $\text{Med} = 82, \text{IQR} = 18$, skewed left. Class B ($n = 25$) has $\text{Med} = 74, \text{IQR} = 10$, roughly symmetric. Neither class has outliers. Compare the distributions.

Grader-Approved Score-4 Solution: *Center:* The median exam score for Class A (82 points) is greater than Class B (74 points).

Spread: The variability in scores is greater for Class A than Class B, as Class A has an IQR of 18 points compared to Class B's IQR of 10 points.

Shape: Class A is skewed to the left, whereas Class B is approximately symmetric.

Unusual Features: Neither distribution exhibits any outliers or notable gaps.

FRQ 1.2: Standardized Scores, Outliers, and Transformations

Problem: A regional 10K road race records finish times (in minutes) for a random sample of 28 adult runners. The 5-number summary of finish times is:

$$\text{Min} = 38.0, \quad Q_1 = 45.0, \quad \text{Median} = 52.0, \quad Q_3 = 58.0, \quad \text{Max} = 82.0$$

The sample mean is $\bar{x} = 53.5$ minutes and the standard deviation is $s = 9.2$ minutes.

(a) Determine the outlier boundary fences using the $1.5 \times$ IQR rule. Is the maximum finish time of 82.0 minutes a confirmed outlier? Justify your answer.

(b) A competitor, Alice, finished the race in 42.0 minutes. Calculate Alice's z-score and interpret this value in context.

(c) Due to an official timing malfunction at the starting gun, race judges subtract 2.0 minutes from every runner's recorded time, and then convert all times from minutes to seconds by multiplying by 60. Describe how this linear transformation affects the mean, standard deviation, and shape of the finish time distribution.

Grader-Approved Score-4 Solution & Rubric:

- **Part (a):** $\text{IQR} = Q_3 - Q_1 = 58.0 - 45.0 = 13.0$ minutes.
  Lower Fence $= Q_1 - 1.5(\text{IQR}) = 45.0 - 1.5(13.0) = 45.0 - 19.5 = 25.5$ minutes.
  Upper Fence $= Q_3 + 1.5(\text{IQR}) = 58.0 + 1.5(13.0) = 58.0 + 19.5 = 77.5$ minutes.
  Because the maximum time 82.0 minutes > 77.5 minutes, 82.0 minutes is a confirmed high outlier.

- **Part (b):** $z = \frac{x-\bar{x}}{s} = \frac{42.0-53.5}{9.2} = \frac{-11.5}{9.2} \approx -1.25$.
  Interpretation: Alice's race time was approximately 1.25 standard deviations below the average finish time of the sample of runners.

- **Part (c):** Let $Y = 60(X - 2) = 60X - 120$ seconds.
  Center (Mean): $\bar{y} = 60(53.5 - 2) = 60(51.5) = 3090$ seconds. Center shifts by both subtracting 2 minutes and multiplying by 60.

Spread (SD): $s_y = 60(s_x) = 60(9.2) = 552$ seconds. Spread scales strictly by multiplication by $|60|$; subtracting 2 has zero effect on spread.
Shape: The shape of the distribution remains unchanged (still skewed right due to the high outlier).

Unit 1 Top 5 AP Exam Pitfalls to Avoid

1. **No Context Deduction:** Stating "the distribution is skewed right" without mentioning the variable (e.g., "race finish times") loses credit.

2. **Comparing Distributions Without Comparative Words:** Always use words like *"strictly greater than"*, *"less variable"*, rather than simply listing numbers.

3. **Confusing Mean and Median in Skewed Data:** Remember the mean is pulled toward the long tail; the median is resistant.

4. **Failing to Box Outliers on Boxplots:** Whiskers must stop at the adjacent non-outlier value, not at the fences!

5. **Adding Constants to Standard Deviation:** Adding or subtracting a constant shifts all data equally and leaves standard deviation completely unchanged.

1.6 Unit 1 Master Summary

Unit 1 Essential Review Checklist

- **C-U-S-S:** Center, Unusual Features, Spread, Shape in complete context.
- **Outliers:** $Q_1 - 1.5(\text{IQR})$ and $Q_3 + 1.5(\text{IQR})$.
- **Resistance:** Median & IQR resist extreme values; Mean & SD do not.
- **Linear Transformation ($y = a + bx$):** Center changes by $a + bx$; Spread scales by $|b|$; Shape is unchanged.
- **Standardized Score:** $z = (x - \mu)/\sigma$.

Chapter 2

Unit 2: Exploring Two-Variable Data

2.1 Unit Overview & Exam Weight

Unit 2 accounts for **5% to 7%** of the AP Statistics Exam. Despite its modest percentage weight, regression analysis, correlation interpretation, residual evaluation, and computer output reading appear on virtually every single AP Exam, frequently as a high-scoring Free-Response Question (FRQ).

Key Unit Vocabulary

Explanatory Variable (x):
The independent variable hypothesized to explain or predict changes in the response variable.

Response Variable (y):
The dependent variable whose values are predicted or measured as an outcome.

Scatterplot:

A two-dimensional plot displaying the relationship between two quantitative variables measured on the same individuals.

Correlation Coefficient (r):

Measures the strength and direction of the linear relationship between two quantitative variables ($-1 \leq r \leq 1$).

Least-Squares Regression Line (LSRL):

The unique line that minimizes the sum of squared vertical residuals: $\min \sum (y - \hat{y})^2$.

Residual (e):

The directed difference between the observed value and the predicted value: Residual $= y - \hat{y}$.

Coefficient of Determination (r^2):

The percentage of total variability in y explained by the linear relationship with x.

Extrapolation:

Predicting y using x-values outside the domain of the collected data. Highly unreliable!

High Leverage Point:

An observation with an extreme x-value relative to other observations.

Influential Point:

An observation whose removal substantially changes the slope, y-intercept, or correlation.

2.2 Core Concept Lessons

Lesson 2.1: Describing Scatterplots with D-O-F-S

Whenever an exam question asks you to describe the relationship shown in a scatterplot, write complete sentences addressing all four components in context:

1. **D — Direction:** State whether the relationship is **Positive** (as x increases, y tends to increase) or **Negative** (as x increases, y tends to decrease).
2. **O — Outliers:** Identify any individual points that deviate substantially from the overall linear trend.
3. **F — Form:** State whether the pattern is **Linear** or **Non-linear** (curved).
4. **S — Strength:** Characterize the scatter as **Strong**, **Moderate**, or **Weak** based on how closely points cluster around a central trend line.

Lesson 2.2: The Correlation Coefficient (r)

$$r = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{x_i-\bar{x}}{s_x}\right)\left(\frac{y_i-\bar{y}}{s_y}\right) \tag{2.1}$$

Essential Properties of r:

- r is strictly bounded: $-1 \leq r \leq 1$.
- r has **NO units of measurement**. Standardizing x and y makes r invariant to changes in units (e.g., converting inches to centimeters does not change r).

- Switching x and y leaves r completely unchanged!
- r measures **LINEAR relationships only**! A scatterplot can have a perfect quadratic relationship ($y = x^2$) and still have $r = 0$.
- r is **non-resistant**; an isolated outlier can dramatically alter its value.
- **Correlation does NOT imply causation!** An observed association may be driven by confounding variables.

Lesson 2.3: The Least-Squares Regression Line (LSRL)

$$\hat{y} = b_0 + b_1 x \quad \text{or} \quad \hat{y} = a + bx \tag{2.2}$$

$$\text{Slope: } b_1 = r\left(\frac{s_y}{s_x}\right), \qquad y\text{-Intercept: } b_0 = \bar{y} - b_1\bar{x} \tag{2.3}$$

Key Geometric Properties:

- The LSRL **always** passes through the point of means: $(\bar{x}, \bar{y})$.
- The sum of the vertical residuals is always mathematically equal to zero: $\sum(y - \hat{y}) = 0$.
- The sign of the slope b_1 is identical to the sign of the correlation coefficient r.

Score-4 Interpretation Templates: Slope & y-Intercept

Slope (b_1): *"For each additional 1 [unit of x] increase in [explanatory variable], the predicted [response variable] increases/decreases by approximately $|b_1|$ [units of y]." (Note: You MUST include the word "predicted" or "estimated" to earn full credit!)*

y-Intercept (b_0): *"When the [explanatory variable] is 0 [units of x], the predicted [response variable] is b_0 [units of y]." (Note: Comment if $x = 0$ makes physical sense in the real-world context).*

Lesson 2.4: Residuals & Residual Plot Analysis

$$\text{Residual } e = y - \hat{y} = \text{Actual } y - \text{Predicted } y \qquad (2.4)$$

The Linear Appropriateness Test: A linear regression model is appropriate if and only if the residual plot exhibits a **random scatter of points around the zero line with roughly constant variance and no curved pattern**.

- If the residual plot exhibits a curved (U-shaped or inverted U) pattern, the relationship between x and y is non-linear, and a linear model is **inappropriate**!
- A positive residual ($y > \hat{y}$) means the actual value was *above* prediction (underestimate).
- A negative residual ($y < \hat{y}$) means the actual value was *below* prediction (overestimate).

Lesson 2.5: Coefficient of Determination (r^2) & Residual SD (s)

Score-4 Template: r^2 and s Interpretation

Coefficient of Determination (r^2): *"Approximately* [$r^2 \times 100\%$] *of the variability in [response variable y] is accounted for by the linear relationship with [explanatory variable x]."*

Standard Deviation of Residuals (s):

$$s = \sqrt{\frac{\sum(y - \hat{y})^2}{n - 2}} \tag{2.5}$$

"When using the least-squares regression line to predict [response variable y] from [explanatory variable x], our predictions will typically be off by approximately s [units of y]."

Lesson 2.6: High Leverage, Outliers, & Influential Points

- **High Leverage Point:** Has an x-value substantially higher or lower than the mean $\bar{x}$. Acts as an anchor point; if it aligns with the overall trend, it inflates r without altering slope.

- **Outlier in Regression:** An observation with a very large residual (far above or below the regression line).

- **Influential Point:** An observation whose removal markedly changes the regression slope, y-intercept, or correlation. High leverage points that deviate from the linear trend are typically highly influential on slope!

Lesson 2.7: Reading Regression Computer Output

Computer printouts (from Minitab, R, or JMP) always follow this standardized grid:

Predictor	**Coef**	**SE Coef**	**T**	**P**
Constant	b_0	$\text{SE}(b_0)$	t	p
Variable (x)	b_1	$\text{SE}(b_1)$	t	p

$S = s$ (Standard deviation of residuals), R-sq $= r^2$. To obtain r from R-sq, compute $\sqrt{r^2}$ and **match the sign of the slope (b_1)!**

2.3 Worked Step-by-Step Examples

Example 2.1: Interpreting Computer Output & Calculating Residuals

Problem: A biologist investigates the relationship between ambient water temperature (x, in °C) and the larval development time of a species of fish (y, in days). Regression output for 12 tanks is displayed below:

Predictor	**Coef**	**SE Coef**	**T**	**P**
Constant	48.650	2.140	22.73	0.000
Temp	-1.420	0.095	-14.95	0.000

$S = 1.15$ days, R-sq $= 95.7\%$, R-sq(adj) $= 95.3\%$.

(a) Write the equation of the least-squares regression line.

(b) Interpret the slope of the regression line in context.

(c) Calculate and interpret the correlation coefficient r.

(d) One tank at 18°C had an observed larval development time of 21.5 days. Calculate the residual.

Solution:

- **Part (a):** $\widehat{\text{Development Time}} = 48.650 - 1.420(\text{Temperature})$.

- **Part (b):** *"For each additional* $1°C$ *increase in ambient water temperature, the predicted larval development time decreases by approximately* 1.420 *days."*

- **Part (c):** $r = -\sqrt{0.957} \approx -0.978$. (We assign the negative sign because the slope $b_1 = -1.420$ is negative). *"There is a very strong, negative linear relationship between water temperature and larval development time."*

- **Part (d):** Predicted development time for $x = 18$:

$$\hat{y} = 48.650 - 1.420(18) = 48.650 - 25.56 = 23.09 \text{ days}$$

Residual $e = y - \hat{y} = 21.5 - 23.09 = -1.59$ days.

"The actual development time was 1.59 *days shorter than predicted by the linear regression model."*

2.4 TI-84 Plus CE Calculator Playbook

TI-84 Playbook: Linear Regression & Residual Plots

1. **Diagnostic ON:** Press `[2nd]` `[0]` (`CATALOG`) → Scroll to `DiagnosticOn` → Press `[ENTER]` twice. (Crucial to display r and r^2!).
2. Enter x-data into `L1` and y-data into `L2`.
3. Press `[STAT]` → `CALC` → Select `8:LinReg(a+bx)`.
4. Set `Xlist: L1`, `Ylist: L2`, `Store RegEQ: Y1` (press `[VARS]` → `Y-VARS` → `1:Function` → `1:Y1`). Press `Calculate`.
5. **To plot residuals:** In `STAT PLOT`, set `Ylist: RESID` (press `[2nd]` `[STAT]` → select `RESID`). Press `[ZOOM]` `[9:ZoomStat]`.

2.5 Comprehensive Practice Problem Set

Part I: 20 Multiple-Choice Questions

1. If the correlation between two variables x and y is $r = 0.85$, what is the correlation if x is converted from feet to meters (1 ft $= 0.3048$ m)?

   (A) 0.85×0.3048

   (B) $0.85/0.3048$

   (C) 0.85

   (D) -0.85

(E) 0.7225

Answer: (C). Changing measurement units does not alter the correlation coefficient r.

2. A regression line has equation $\hat{y} = 12 - 3.5x$. What is the predicted change in y for each 2-unit increase in x?

   (A) Decreases by 3.5

   (B) Decreases by 7.0

   (C) Increases by 7.0

   (D) Increases by 12

   (E) Decreases by 12

   Answer: (B). $2 \times (-3.5) = -7.0$.

3. Which of the following indicates that a linear model is appropriate?

   (A) The correlation r is positive

   (B) The residual plot shows a clear curved pattern

   (C) The residual plot displays random scatter around the horizontal zero line

   (D) r^2 is exactly 0.50

   (E) The slope is greater than 1

   Answer: (C). Random scatter without systematic curvature confirms appropriateness.

4. An observation has an actual value of $y = 45$ and a predicted value of $\hat{y} = 52$. The residual is:

(A) -7

(B) $+7$

(C) 45

(D) 52

(E) 0.865

Answer: (A). Residual $= y - \hat{y} = 45 - 52 = -7$.

5. If $r^2 = 0.64$ and the slope of the regression line is negative, what is r?

   (A) 0.64

   (B) 0.80

   (C) -0.80

   (D) -0.64

   (E) 0.4096

   Answer: (C). $r = -\sqrt{0.64} = -0.80$.

6. The point of means $(\bar{x}, \bar{y})$:

   (A) Always lies on the least-squares regression line

   (B) Lies on the line only if $r = 1$

   (C) Is always an outlier

   (D) Has a non-zero residual

   (E) Determines the direction of curvature

   Answer: (A). Mathematically, $\bar{y} = b_0 + b_1\bar{x}$.

7. What does $r^2 = 0.81$ mean in the context of predicting fuel efficiency (y) from car weight (x)?

(A) 81% of cars have high fuel efficiency.

(B) 81% of the variability in fuel efficiency is accounted for by the linear model with car weight.

(C) The correlation is 0.81.

(D) The slope of the line is 0.81.

(E) Predictions will be correct 81% of the time.

Answer: (B). Standard definition of the coefficient of determination.

8. An influential point in regression is best described as an observation that:

(A) Has a residual of zero.

(B) Substantially alters the slope, intercept, or correlation when removed.

(C) Is always an error in data collection.

(D) Falls exactly on the mean of x.

(E) Has a positive z-score.

Answer: (B). Defining property of an influential observation.

9. Using a model $\hat{y} = 5 + 2x$ established for $x \in [10, 50]$ to predict y when $x = 120$ is an example of:

(A) Interpolation

(B) Extrapolation

(C) Randomization

(D) Confounding

(E) Residual analysis

Answer: (B). Predicting beyond the range of explanatory data is extrapolation.

10. If $s_x = 4, s_y = 6$, and $r = 0.75$, what is the slope b_1 of the LSRL?

(A) 0.75

(B) 1.125

(C) 0.50

(D) 1.50

(E) 0.888

Answer: (B). $b_1 = r(s_y/s_x) = 0.75(6/4) = 0.75(1.5) = 1.125$.

11. Which of the following correlations indicates the strongest linear relationship?

(A) $r = 0.78$

(B) $r = 0.00$

(C) $r = -0.89$

(D) $r = 0.85$

(E) $r = -0.12$

Answer: (C). Strength is determined by $|r|$; $|-0.89| = 0.89$ is the largest magnitude.

12. If the standard deviation of residuals is $s = 3.2$ cm, this means:

(A) All residuals are less than 3.2 cm.

(B) Predictions of y typically deviate from actual values by about 3.2 cm.

(C) The slope is 3.2.

(D) $r = 0.32$.

(E) The mean residual is 3.2 cm.

Answer: (B). Standard interpretation of s.

13. If the residuals for a model sum to a non-zero value, what is the most likely explanation?

(A) The relationship is non-linear.

(B) Rounding error in calculations.

(C) The line used is not the least-squares regression line.

(D) r was negative.

(E) There are outliers.

Answer: (C). The true LSRL strictly satisfies $\sum e = 0$; any other line will not.

14. High leverage points in a scatterplot always have:

(A) Large residuals

(B) Unusual x-values

(C) Negative slopes

(D) $r > 0.90$

(E) Low variance

Answer: (B). Leverage is governed purely by the distance of x from $\bar{x}$.

15. When two variables have a strong non-linear relationship of the form $y = ab^x$, what transformation linearizes the data?

    (A) Plotting y vs. x^2

    (B) Plotting $\log(y)$ vs. x

    (C) Plotting $\log(y)$ vs. $\log(x)$

    (D) Plotting $1/y$ vs. $1/x$

    (E) Plotting $\sqrt{y}$ vs. $\sqrt{x}$

    Answer: (B). An exponential model is linearized by taking the logarithm of y only ($\log y = \log a + (\log b)x$).

16. A power model $y = ax^b$ is linearized by plotting:

    (A) $\log(y)$ vs. x

    (B) $\log(y)$ vs. $\log(x)$

    (C) y vs. $\log(x)$

    (D) y^2 vs. x

    (E) $\sqrt{y}$ vs. x

    Answer: (B). Power models linearize when both variables are transformed logarithmically.

17. In a regression analysis, the explanatory variable is speed (mph) and response is stopping distance (ft). The slope is 3.5. What are the units of the slope?

    (A) mph

    (B) ft

    (C) ft per mph

    (D) mph per ft

(E) Dimensionless

Answer: (C). Slope units are always [y-units]/[x-units] = ft per mph.

18. If $r = 0$, what does this conclude about the relationship between x and y?

    (A) There is no relationship of any kind.

    (B) There is no linear relationship between x and y.

    (C) All residuals are zero.

    (D) The slope is 1.

    (E) The data is non-random.

    Answer: (B). A strong curved relationship can still have $r = 0$; r detects only linear trends.

19. A researcher observes a strong positive correlation ($r = 0.92$) between ice cream sales and drowning deaths. The most plausible explanation is:

    (A) Eating ice cream causes drowning.

    (B) Drowning causes people to buy ice cream.

    (C) Both variables are influenced by a common confounding variable: outdoor summer temperature.

    (D) The correlation was miscalculated.

    (E) Extrapolation error.

    Answer: (C). Classic lurking/confounding variable scenario.

20. A regression line has $\hat{y} = 10 + 2x$. If an observation at $x = 5$ has an actual value of $y = 23$, the point lies:

(A) Exactly on the line

(B) 3 units below the line

(C) 3 units above the line

(D) 23 units above the line

(E) 10 units below the line

Answer: (C). $\hat{y} = 10 + 2(5) = 20$. Actual $y = 23 \implies$ Residual $= 23 - 20 = +3$.

Part II: Free-Response Practice Problem

FRQ 2.1: Regression Modeling & Residual Evaluation

Problem: An environmental engineer investigates the relationship between wind speed (x, in mph) and electricity generation (y, in kilowatts) of a wind turbine. The regression line is $\widehat{\text{Power}} = -15.4 + 4.8(\text{Wind})$. $r^2 = 0.88, s = 6.2$ kW.
(a) Interpret the slope 4.8 in context.
(b) Interpret $r^2 = 0.88$ in context.
(c) At a wind speed of 20 mph, the turbine produced 85 kW. Calculate the residual and state whether the model overpredicted or underpredicted power.

Grader-Approved Score-4 Solution:

- **Part (a):** *"For each additional 1 mph increase in wind speed, the predicted electricity generation increases by approximately 4.8 kilowatts."*

- **Part (b):** *"Approximately 88% of the variability in electricity generation is accounted for by the linear relationship with wind speed."*
- **Part (c):** Predicted: $\hat{y} = -15.4 + 4.8(20) = -15.4 + 96.0 = 80.6$ kW.
  Residual $= y - \hat{y} = 85 - 80.6 = +\mathbf{4.4}$ kW.
  Because the residual is positive, the linear regression model **underpredicted** the actual electricity generation by 4.4 kW.

FRQ 2.2: Regression Output, Residuals, and Outliers vs. High Leverage

Problem: A real estate analyst investigates the relationship between living area (x, in hundreds of square feet) and selling price (y, in thousands of dollars) for 24 recently sold suburban homes. A computer regression package outputs:

Predictor	Coef	SE Coef	T	P
Constant	45.20	16.40	2.76	0.011
Area	12.80	1.60	8.00	0.000

$S = 18.50$ (\$1,000s), R-sq $= 74.4\%$.

(a) State the equation of the least-squares regression line and predict the selling price of a home with 2,500 square feet ($x = 25.0$).

(b) A home with 2,500 square feet sold for \$390,000 ($y = 390.0$). Calculate and interpret the residual.

(c) Distinguish between an outlier in regression and a point with high leverage. If a 6,000-sq-ft mansion is added to the dataset, would it be classified as high leverage? Explain.

(d) Explain why the analyst should not use this model to predict the price of a 7,500-square-foot commercial estate.

Grader-Approved Score-4 Solution & Rubric:

- **Part (a):** $\widehat{\text{Price}} = 45.20 + 12.80(\text{Area})$.
  For $x = 25.0$: $\hat{y} = 45.20 + 12.80(25.0) = 45.20 + 320.00 = 365.20$ thousand dollars ($365,200).

- **Part (b):** Residual $= y - \hat{y} = 390.0 - 365.20 = +24.80$ thousand dollars ($24,800).
  Interpretation: The actual selling price of the home was $24,800 higher than the price predicted by the regression line based on its living area.

- **Part (c):** An outlier has an unusually large residual (far above or below the regression line). A point with high leverage has an extreme x-value (far from $\bar{x}$) with the potential to strongly steer the slope. A 6,000-sq-ft home ($x = 60.0$) is far beyond the typical sample range, giving it high leverage.

- **Part (d):** Predicting at 7,500 square feet represents dangerous **extrapolation**. We have no evidence that the linear relationship continues beyond the observed data range.

Unit 2 Top 5 AP Exam Pitfalls to Avoid

1. **Implying Causation from Correlation:** r measures linear association, never cause and effect!
2. **Claiming r^2 Means the Model is Correct:** r^2 can be high even if the underlying data is curved. Always inspect the residual plot!
3. **Confusing Residual Direction:** Residual is Observed minus Predicted $(y - \hat{y})$, never $\hat{y} - y$.
4. **Omitting Units in Slope Interpretation:** Always state "predicted increase in [units of y] per 1 [unit of x]".
5. **Extrapolation Danger:** Never make predictions outside the domain of observed x-values.

2.6 Unit 2 Master Summary

Unit 2 Essential Review Checklist

- **D-O-F-S:** Direction, Outliers, Form, Strength for scatterplots.
- **LSRL Formula:** $\hat{y} = b_0 + b_1 x$, where $b_1 = r(s_y/s_x)$ and $b_0 = \bar{y} - b_1\bar{x}$.
- **Residual:** $e = y - \hat{y}$ (Actual minus Predicted).
- **Residual Plot Appropriateness:** Random scatter around zero confirms a linear model is appropriate; any curved pattern indicates a non-linear relationship.

- **Interpretation Keywords:** Always use ”predicted” or ”estimated” for slope interpretations.

Chapter 3

Unit 3: Collecting Data

3.1 Unit Overview & Exam Weight

Unit 3 represents **12% to 15%** of the AP Statistics Exam. It is heavily tested on both the Multiple-Choice section and Section II Free-Response Questions. This unit focuses on the foundational rules of statistical data collection: distinguishing between sampling methods, diagnosing bias, structuring valid experiments, and establishing the exact scope of statistical inference.

Key Unit Vocabulary

Population:
The entire group of individuals about which we desire information.

Sample:
A subset of individuals selected from the population, used to collect data.

Census:
An attempt to collect data from every individual in the entire

population.

Observational Study:

Observes individuals and measures variables of interest without attempting to influence responses. Can demonstrate association, but **CANNOT establish causation**!

Experiment:

Deliberately imposes treatments on experimental units to observe responses. A properly designed randomized experiment is the **ONLY** valid method to establish cause-and-effect!

Confounding:

Occurs when two variables are associated in such a way that their separate effects on a response variable cannot be distinguished.

Simple Random Sample (SRS):

A sample of size n chosen in such a way that every group of n individuals in the population has an equal chance of being selected.

Stratified Random Sample:

Dividing the population into homogeneous groups (strata) of similar characteristics, then taking a separate SRS within every stratum.

Cluster Sample:

Dividing the population into heterogeneous, naturally occurring groups (clusters), selecting a random sample of whole clusters, and surveying all members within selected clusters.

Random Assignment:

Using chance to assign experimental units to treatment groups, balancing the effects of unmeasured confounding variables across all groups.

3.2 Core Concept Lessons

Lesson 3.1: Probability Sampling Methods

- **Simple Random Sample (SRS):** The gold standard. Assign each individual in the sampling frame a distinct integer from 1 to N. Use a random number generator to select n unique integers without replacement.
- **Stratified Random Sampling:** *"Some from all."* Form homogeneous strata based on a variable known to influence the response (e.g., gender, grade level). Take an SRS from each stratum and combine them. Reduces sampling variability!
- **Cluster Sampling:** *"All from some."* Form heterogeneous clusters that mimic the overall population (e.g., homerooms, city blocks). Randomly select entire clusters and survey every individual within selected clusters. Highly cost-effective and logistically practical.
- **Systematic Sampling:** Select every k^{th} individual from an ordered list after a random starting point between 1 and k.

Lesson 3.2: Diagnosing Sampling Bias & Pitfalls

Bias occurs when a sampling design consistently favors certain outcomes:

- **Convenience Sampling:** Choosing individuals who are easiest to reach (e.g., mall intercepts). Severely unrepresentative.
- **Voluntary Response Bias:** Allowing individuals to choose whether to participate (e.g., online polls, call-in shows). Over-represents individuals with strong, typically negative opinions!
- **Undercoverage:** Occurs when some groups in the population are left out of the process of choosing the sample (e.g., telephone surveys omitting people without phones).
- **Nonresponse Bias:** Occurs when a selected individual cannot be contacted or refuses to participate. Note: Nonresponse is NOT the same as voluntary response!
- **Response Bias / Wording of Questions:** Misleading, confusing, or leading questions that influence respondents to answer untruthfully.

Lesson 3.3: The 4 Principles of Experimental Design

Every well-designed randomized experiment must incorporate:

1. **Comparison:** Compare two or more treatments to prevent confounding from external temporal factors.
2. **Random Assignment:** Use chance process to assign experimental units to treatments. This creates roughly equivalent groups by balancing lurking variables.
3. **Replication:** Apply each treatment to a sufficient num-

ber of experimental units so differences in response can be distinguished from chance variation.

4. **Control:** Keep other external variables identical for all groups to prevent confounding.

Lesson 3.4: Experimental Design Frameworks

- **Completely Randomized Design:** All experimental units are assigned to treatment groups strictly by chance.
- **Randomized Block Design:** Form blocks of units known to be similar with respect to a variable expected to affect response (e.g., age group). Within each block, randomly assign units to treatments. *"Block what you can control, randomize what you cannot!"*
- **Matched Pairs Design:** A specialized block design comparing two treatments. Either each subject receives both treatments in random order (serving as their own control), or subjects are paired based on a matching characteristic and randomly assigned to separate treatments.

Lesson 3.5: Placebos, Blinding, & The Placebo Effect

- **Placebo:** A dummy or sham treatment that has no active therapeutic ingredient, used to control for psychological response.
- **Placebo Effect:** The measurable psychological response of a patient to a dummy treatment.
- **Single-Blind:** Either the subjects OR the evaluators

measuring response do not know which treatment was administered.

- **Double-Blind:** Neither the subjects NOR the medical personnel interacting with and evaluating them know which treatment was assigned.

Lesson 3.6: The Scope of Statistical Inference Matrix

This 2×2 grid is tested on almost every AP Statistics exam:

	Were Individuals Randomly Assigned?	**No Random Assignment?**
Random Selection?	Inference about **Population** AND **Causation**	Inference about **Population** only (NO Causation)
No Random Selection?	Inference about **Causation** only (NO Population)	**NO Inference** about Population or Causation!

3.3 Worked Step-by-Step Examples

Example 3.1: Designing a Randomized Block Experiment

Problem: An agricultural researcher wants to test the effect of three different fertilizers (A, B, and C) on corn yield. The researcher has 18 plots of land available: 9 plots on a sunny hillside and 9 plots in a shady valley. (a) Identify the experimental units, factor, levels, and response variable.
(b) Explain why a randomized block design is preferable to a completely randomized design.
(c) Describe how you would carry out the random assignment within blocks.

Solution:

- **Part (a):** Experimental units: The 18 agricultural plots. Factor: Fertilizer type. Levels: 3 levels (A, B, C). Response variable: Corn yield (e.g., bushels per acre).

- **Part (b):** Sun exposure and moisture differences between the sunny hillside and shady valley will likely affect corn yield independently of the fertilizer. A randomized block design groups plots into two blocks of 9 plots (Block 1: Hillside, Block 2: Valley), isolating sunlight variation and reducing unexplained variability in yield.
- **Part (c):** Within the 9 hillside plots, assign each plot an integer from 1 to 9. Use a random number generator to select 3 unique integers without replacement; assign these 3 plots to Fertilizer A. Generate 3 more unique integers from the remaining numbers; assign these to Fertilizer B. Assign the final 3 plots to Fertilizer C. Repeat this exact randomization procedure independently for the 9 valley plots.

3.4 TI-84 Plus CE Calculator Playbook

TI-84 Playbook: Random Number Simulation (randIntNoRep)

1. Press `[MATH]` → Arrow right to `PRB` (Probability).
2. **For sampling without replacement:** Select `8:randIntNoRep(`.
3. Syntax: `randIntNoRep(lower, upper, n)`. Example: To select 5 unique students from a roster of 30: `randIntNoRep(1, 30, 5)`.
4. Press `[ENTER]`. The calculator returns a list of 5 distinct

integers with zero repeats!

3.5 Comprehensive Practice Problem Set

Part I: 20 Multiple-Choice Questions

1. What is the primary purpose of random assignment in an experiment?

   (A) To ensure the sample is representative of the entire population.

   (B) To create treatment groups that are roughly equivalent by balancing lurking variables.

   (C) To eliminate the need for a control group.

   (D) To guarantee double-blinding.

   (E) To reduce sample size requirements.

   Answer: (B). Random assignment balances unmeasured confounding variables across treatment groups.

2. A school principal surveys all students in 5 randomly selected homerooms out of 40 homerooms. What sampling method was used?

   (A) Simple Random Sample

   (B) Stratified Random Sample

   (C) Cluster Sample

   (D) Systematic Sample

   (E) Convenience Sample

Answer: (C). Entire naturally occurring groups (homerooms) were sampled ("all from some").

3. A researcher selects 50 freshman, 50 sophomores, 50 juniors, and 50 seniors from a high school. This is an example of:

   (A) Cluster sampling

   (B) Stratified random sampling

   (C) Systematic sampling

   (D) Simple random sampling

   (E) Voluntary response

   Answer: (B). Homogeneous strata (grade levels) were sampled ("some from all").

4. Which of the following is required to establish cause-and-effect?

   (A) A large observational study

   (B) A high correlation coefficient ($r > 0.95$)

   (C) Random assignment of treatments in a well-designed experiment

   (D) A voluntary response survey

   (E) A stratified random sample

   Answer: (C). Only randomized experiments can prove causation.

5. An online news site asks readers to vote on whether taxes should be increased. 82% say "No." This result is most vulnerable to:

(A) Nonresponse bias

(B) Voluntary response bias

(C) Undercoverage

(D) High standard deviation

(E) Matched pairs bias

Answer: (B). Self-selected participation over-represents individuals with strong opinions.

6. In a medical trial, neither the patients nor the examining doctors know who received the real drug versus the placebo. This study is:

   (A) Completely confounded

   (B) Single-blind

   (C) Double-blind

   (D) Retrospective

   (E) Observational

   Answer: (C). Blinding both subjects and evaluators is double-blind.

7. What is the main reason for blocking in an experiment?

   (A) To reduce variability within treatment groups caused by a known confounding variable.

   (B) To eliminate the need for random assignment.

   (C) To ensure everyone gets the active treatment.

   (D) To increase the number of treatments.

   (E) To eliminate the placebo effect.

Answer: (A). Blocking controls for known sources of variation, making treatment comparisons clearer.

8. A survey about gun control is conducted by law enforcement officers in full uniform. The responses are most likely subject to:

   (A) Undercoverage

   (B) Nonresponse bias

   (C) Response bias

   (D) Voluntary response bias

   (E) Stratification error

   Answer: (C). The interviewer's presence influences respondents to provide socially desirable answers.

9. An experiment assigns runners to test two brands of shoes. Each runner wears Brand A on one foot and Brand B on the other, determined by a coin flip. This is:

   (A) A completely randomized design

   (B) A matched pairs design

   (C) An observational study

   (D) A stratified design

   (E) A cluster design

   Answer: (B). Subjects act as their own control in a matched pairs design.

10. If volunteers are used in an experiment that includes random assignment to treatments, what can be concluded from statistically significant results?

(A) Causation for the volunteers, but results cannot be generalized to the broader population.

(B) Causation for the entire population.

(C) Association only.

(D) No valid conclusions can be drawn.

(E) The experiment was biased.

Answer: (A). Random assignment permits causal inference, but lack of random selection prevents generalization to the population.

11. Which of the following is an example of undercoverage?

(A) A telephone survey conducted by calling landline numbers during the day.

(B) People refusing to answer a mail-in survey.

(C) People lying about their income on an IRS audit.

(D) An interviewer misreading a survey question.

(E) Assigning treatments using coin flips.

Answer: (A). Households without landlines (or adults away at work) are systematically excluded from the sampling frame.

12. What is the control group's role in an experiment?

(A) To provide a baseline for comparison against the active treatments.

(B) To double the sample size.

(C) To eliminate the need for random selection.

(D) To prevent any placebo effect from occurring.

(E) To guarantee a positive correlation.

Answer: (A). Controls isolate whether changes in response are truly caused by the treatment rather than outside variables.

13. Selecting every 15^{th} customer entering a retail store after a random starting point is:

    (A) Simple Random Sampling

    (B) Systematic Sampling

    (C) Stratified Sampling

    (D) Cluster Sampling

    (E) Convenience Sampling

    Answer: (B). Defining procedure of systematic sampling.

14. Why is an observational study incapable of demonstrating cause-and-effect?

    (A) Because the sample sizes are too small.

    (B) Because treatments are not deliberately imposed and confounding variables cannot be controlled.

    (C) Because they never use random samples.

    (D) Because observational studies use only categorical variables.

    (E) Because the standard error is too large.

    Answer: (B). Confounding between the explanatory variable and lurking factors prevents causal attribution.

15. A researcher tests 4 levels of nitrogen and 3 levels of watering frequency on plant growth. How many treatments are in this experiment?

    (A) 4

    (B) 3

    (C) 7

    (D) 12

    (E) 64

    Answer: (D). Treatments equal the product of the factor levels: $4 \times 3 = 12$.

16. What is the difference between nonresponse bias and voluntary response bias?

    (A) In nonresponse, individuals are selected by the researcher but cannot be reached or refuse; in voluntary response, individuals self-select to participate.

    (B) They are identical terms.

    (C) Nonresponse occurs only in experiments.

    (D) Voluntary response occurs only in censuses.

    (E) Nonresponse always produces positive bias.

    Answer: (A). Key distinction frequently tested on the AP exam.

17. In an experiment evaluating an anti-anxiety drug, subjects receiving the sugar pill report feeling less anxious. This is:

    (A) Confounding

(B) The placebo effect

(C) Response bias

(D) Observer bias

(E) Blocking error

Answer: (B). Psychological improvement resulting from expectation of treatment.

18. A company wants to survey its 500 employees. Which method guarantees an SRS?

(A) Number employees 1 to 500 and use a random number generator to pick 50 unique numbers.

(B) Survey the first 50 employees who arrive at the office.

(C) Pick 10 employees from each of the 5 departments.

(D) Post a survey link on the company intranet and take the first 50 responses.

(E) Survey all 50 employees in the marketing department.

Answer: (A). Every combination of 50 employees has an equal probability of selection.

19. In an experiment comparing two diets, dogs are grouped by breed into blocks before assigning diets. Why?

(A) Because breed is expected to affect metabolism and weight gain.

(B) Because breed is the response variable.

(C) To eliminate the need for replication.

(D) To make the study double-blind.

(E) To ensure all dogs lose weight.

Answer: (A). Blocking reduces within-treatment variation caused by breed differences.

20. A study concludes that people who drink coffee daily have lower rates of cardiovascular disease. Can we conclude coffee prevents cardiovascular disease?

(A) Yes, because the sample size was large.

(B) No, because this was an observational study and other lifestyle factors could confound the relationship.

(C) Yes, if $r > 0.8$.

(D) No, because coffee is a categorical variable.

(E) Yes, if the sample was an SRS.

Answer: (B). Observational studies show association, not causation.

Part II: Free-Response Practice Problem

FRQ 3.1: Designing a Valid Medical Experiment

Problem: A pharmaceutical firm develops a new medication to reduce migraine duration. 60 adult migraine sufferers (30 men, 30 women) volunteer to participate. (a) Describe how you would implement a randomized block design for this study, blocking by gender.
(b) Explain why a placebo and double-blinding should be used in this experiment.

Grader-Approved Score-4 Solution:

- **Part (a):** Because gender differences in hormone levels may affect migraine duration independently of the drug, gender serves as a blocking variable. Form two blocks: 30 men and 30 women. Within the block of 30 men, assign each subject an integer from 1 to 30. Use a random number generator to select 15 unique integers without replacement; assign these 15 men to the new medication. Assign the remaining 15 men to an identical-looking placebo. Repeat this exact randomization procedure independently for the block of 30 women. Compare the reduction in migraine duration between the treatment and placebo groups within each gender block and overall.

- **Part (b):** A placebo (inactive pill identical in appearance, taste, and size) is essential to control for the placebo effect, ensuring that measured improvements are attributable to the active chemical compound rather than the psychological expectation of relief. Double-blinding (neither the subjects nor the physicians recording migraine outcomes know who receives the active drug) prevents both patient expectation bias and physician evaluation bias.

FRQ 3.2: Matched Pairs vs. Completely Randomized Design with Blocking

Problem: An athletic footwear company designs a new synthetic rubber sole material intended to reduce shoe wear. Thirty collegiate cross-country runners volunteer to participate in a 10-week testing trial.

(a) Describe how you would implement a matched pairs experimental design where each runner wears both sole materials simultaneously (one on the left foot, one on the right foot). Explain the role of randomization in your design.

(b) Explain the primary statistical advantage of this matched pairs design over a completely randomized design where 15 runners wear only the new sole and 15 runners wear only the standard sole.

(c) Can the results of this experiment be generalized to all casual weekend joggers? Explain your reasoning.

Grader-Approved Score-4 Solution & Rubric:

- **Part (a):** Number the 30 runners from 01 to 30. For each runner, flip a fair coin: if Heads, the runner wears the new synthetic sole on their left shoe and the standard sole on their right shoe; if Tails, the runner wears the standard sole on their left shoe and the new sole on their right shoe. Randomizing which foot receives which sole controls for natural asymmetry in runner stride, dominant foot wear patterns, and biomechanical imbalances.
- **Part (b):** The matched pairs design uses each runner as their own control, eliminating between-runner variability (such as runner weight, weekly mileage, running surface, and cadence). This substantially reduces experimental error variance, making it much easier to detect true differences between sole materials.
- **Part (c):** No. The subjects were collegiate cross-country runners who volunteered. Volunteers are not a random

sample, and elite collegiate runners have vastly different biomechanics, speeds, and training volumes compared to casual weekend joggers.

Unit 3 Top 5 AP Exam Pitfalls to Avoid

1. **Confusing Random Sampling with Random Assignment:** Random sampling allows generalization to the population; random assignment allows causal inference (cause-and-effect).

2. **Confusing Stratified Sampling with Blocking:** Stratified sampling is a survey technique for observational data; blocking is an experimental design technique.

3. **Thinking Large Samples Eliminate Bias:** A huge sample size cannot overcome voluntary response or undercoverage bias!

4. **Failing to Explain How to Randomize:** On FRQs, always explain the random mechanism (numbering, random number table without replacement, coin flips).

5. **Confusing Confounding with Common Response:** Confounding occurs when the treatment effect cannot be separated from an extraneous variable.

3.6 Unit 3 Master Summary

Unit 3 Essential Review Checklist

- **Causation vs. Association:** Experiments with random assignment show causation; observational studies show association only.
- **4 Principles of Experiments:** Comparison, Random Assignment, Replication, Control.
- **Blocking:** Group by a variable known to affect the response variable before random assignment within blocks.
- **Scope of Inference:** Random selection $\implies$ generalize to population. Random assignment $\implies$ causal inference.

Chapter 4

Unit 4: Probability, Random Variables, & Probability Distributions

4.1 Unit Overview & Exam Weight

Unit 4 accounts for **10% to 20%** of the AP Statistics Exam. Probability theory provides the vital mathematical bridge connecting descriptive data exploration (Units 1–3) with statistical inference (Units 6–9). Mastery of random variable transformations, independence, Binomial models, and Geometric distributions is essential for scoring a 5.

Key Unit Vocabulary

Law of Large Numbers:
As the number of repetitions of a chance experiment increases, the relative frequency of an event approaches its true theoretical probability.

Sample Space (S):
The set of all possible outcomes of a random phenomenon.

Disjoint (Mutually Exclusive) Events:

Events that cannot occur simultaneously: $P(A \cap B) = 0$.

Independent Events:

Two events A and B are independent if knowing that one event has occurred does not change the probability that the other event occurs: $P(A \mid B) = P(A)$ and $P(B \mid A) = P(B)$.

Conditional Probability:

The probability that one event happens given that another event is already known to have occurred: $P(A \mid B) = P(A \cap B)/P(B)$.

Discrete Random Variable (X):

Takes a fixed set of countable values with gaps between them, defined by a probability distribution where $\sum P(x_i) = 1$.

Expected Value ($E(X) = \mu_X$):

The long-run average outcome of a random variable over many repeated trials: $\mu_X = \sum x_i P(x_i)$.

Variance (Var$(X) = \sigma_X^2$):

The average squared deviation from the mean: $\sigma_X^2 = \sum (x_i - \mu_X)^2 P(x_i)$.

Binomial Setting ($B - I - N - S$):

Binary outcomes, Independent trials, Number of trials n fixed, Same probability of success p for each trial.

Geometric Setting ($B - I - T - S$):

Binary outcomes, Independent trials, Trials continue until the first success, Same probability of success p.

4.2 Core Concept Lessons

Lesson 4.1: Fundamental Probability Axioms & The Independence Test

- **Complement Rule:** $P(A^c) = 1 - P(A)$.
- **General Addition Rule:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- **Conditional Probability Formula:**

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(A \cap B) = P(B) \cdot P(A \mid B) \tag{4.1}$$

- **The Official AP Independence Verification Test:** Two events A and B are independent if and only if any of the following equivalence conditions hold:

$$P(A \mid B) = P(A) \quad \text{OR} \quad P(B \mid A) = P(B) \quad \text{OR} \quad P(A \cap B) = P(A) \tag{4.2}$$

Exam Trap Alert: Disjoint vs. Independent Events

Disjoint (mutually exclusive) events and Independent events are **NEVER** the same thing! In fact, two events with non-zero probabilities can **never** be both disjoint and independent!

- If A and B are **disjoint**, they cannot happen together ($P(A \cap B) = 0$). If A happens, the probability that B happens drops to exactly 0! Thus, knowing A occurred completely changes the probability of B, proving disjoint events are **heavily dependent**!

- On FRQs, never write *"Events A and B are independent because they cannot happen at the same time."* That earns an immediate **Incomplete (I)** score!

Lesson 4.2: Discrete Random Variables & Linear Transformations

$$\mu_X = E(X) = \sum_{i=1}^{k} x_i P(x_i), \qquad \sigma_X^2 = \sum_{i=1}^{k} (x_i - \mu_X)^2 P(x_i), \qquad \sigma_X = \sqrt{\;} \tag{4.3}$$

Linear Transformation of a Single Random Variable ($Y = a + bX$):

$$\mu_{a+bX} = a + b\mu_X, \qquad \sigma_{a+bX}^2 = b^2\sigma_X^2, \qquad \sigma_{a+bX} = |b|\sigma_X \tag{4.4}$$

Adding a constant shifts the center but leaves spread completely untouched.

Lesson 4.3: Linear Combinations of Independent Random Variables

Let X and Y be two **independent** random variables:

$$\mu_{X+Y} = \mu_X + \mu_Y, \qquad \mu_{X-Y} = \mu_X - \mu_Y \tag{4.5}$$

$$\sigma_{X+Y}^2 = \sigma_X^2 + \sigma_Y^2, \qquad \sigma_{X-Y}^2 = \sigma_X^2 + \sigma_Y^2 \tag{4.6}$$

$$\sigma_{X\pm Y} = \sqrt{\sigma_X^2 + \sigma_Y^2} \tag{4.7}$$

The Universal Rule of Variances: *"Standard deviations NEVER add directly! Variances ALWAYS add, even when* *subtracting two random variables!"*

Lesson 4.4: The Binomial Probability Distribution (B-I-N-S)

A random variable $X \sim B(n, p)$ represents the number of successes in n independent trials:

1. **B — Binary:** Each trial has exactly two outcomes: Success (p) or Failure ($1 - p$).
2. **I — Independent:** Trials are mutually independent.
3. **N — Number:** The number of trials n is fixed in advance.
4. **S — Same Probability:** Probability of success p is constant for each trial.

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} \tag{4.8}$$

$$\mu_X = np, \qquad \sigma_X = \sqrt{np(1-p)} \tag{4.9}$$

10% Condition for Binomial Sampling: When sampling without replacement from a finite population of size N, trials are approximately independent if $n \leq 0.10N$.

Lesson 4.5: The Geometric Probability Distribution (B-I-T-S)

A random variable $Y \sim G(p)$ counts the number of trials required until the **first success** occurs:

$$P(Y = k) = (1-p)^{k-1} p \tag{4.10}$$

$$\mu_Y = E(Y) = \frac{1}{p}, \qquad \sigma_Y = \frac{\sqrt{1-p}}{p} \tag{4.11}$$

$$P(Y > k) = (1-p)^k \quad \text{(the probability of failing on the first } k \text{ consecut} \tag{4.12}$$

4.3 Worked Step-by-Step Examples

Example 4.1: Combining Independent Random Variables

Problem: A packaging company ships boxes containing a glass vase and protective packing peanuts.

- The weight of the empty shipping box follows $X \sim N(12 \text{ oz}, 1.5 \text{ oz})$.
- The weight of the glass vase follows $Y \sim N(28 \text{ oz}, 2.0 \text{ oz})$.
- The weight of the packing peanuts follows $W \sim N(4 \text{ oz}, 0.5 \text{ oz})$.

Assuming the weights are independent, what is the probability that a randomly assembled total package exceeds 48 oz?

Solution: Let $T = X+Y+W$ = total package weight. Because the linear combination of independent normal random variables is itself normal:

$$\mu_T = \mu_X + \mu_Y + \mu_W = 12 + 28 + 4 = 44.0 \text{ oz}$$

$$\sigma_T^2 = \sigma_X^2 + \sigma_Y^2 + \sigma_W^2 = 1.5^2 + 2.0^2 + 0.5^2 = 2.25 + 4.00 + 0.25 = 6.50 \text{ oz}^2$$

$$\sigma_T = \sqrt{6.50} \approx 2.55 \text{ oz}$$

Thus, $T \sim N(44.0, 2.55)$. We seek $P(T > 48)$:

$$z = \frac{48 - 44.0}{2.55} = \frac{4.0}{2.55} \approx 1.57$$

$P(T > 48) = P(Z > 1.57) = 1 - 0.9418 = \mathbf{0.0582}$ (approximately **5.82%**).

4.4 TI-84 Plus CE Calculator Playbook

TI-84 Playbook: Binomial and Geometric Distributions

- **Single Binomial Probability** $P(X = k)$**:** Press `[2nd]` `[VARS]` (`DISTR`) → Select `A:binompdf(`. Enter n, p, k.
- **Cumulative Binomial Probability** $P(X \leq k)$**:** Press `[2nd]` `[VARS]` → Select `B:binomcdf(`. Enter n, p, k.
- **Geometric Probability** $P(Y = k)$**:** Press `[2nd]` `[VARS]` → Select `E:geometpdf(`. Enter p, k.
- **Cumulative Geometric Probability** $P(Y \leq k)$**:** Press `[2nd]` `[VARS]` → Select `F:geometcdf(`. Enter p, k.

4.5 Comprehensive Practice Problem Set

Part I: 20 Multiple-Choice Questions

1. If $P(A) = 0.40, P(B) = 0.50$, and $P(A \cap B) = 0.20$, are events A and B independent?

   (A) Yes, because $P(A) \cdot P(B) = 0.40 \times 0.50 = 0.20 = P(A \cap$

B).

(B) No, because $P(A \cap B) \neq 0$.

(C) Yes, because $P(A) + P(B) \neq 1$.

(D) No, because $P(A) \neq P(B)$.

(E) Cannot be determined.

Answer: (A). The multiplication rule $P(A \cap B) = P(A)P(B)$ proves independence.

2. Two events C and D have $P(C) = 0.6$ and $P(D) = 0.3$. If C and D are mutually exclusive, what is $P(C \cup D)$?

(A) 0.18

(B) 0.90

(C) 0.72

(D) 0.30

(E) 0.00

Answer: (B). For disjoint events, $P(C \cup D) = P(C) + P(D) = 0.6 + 0.3 = 0.90$.

3. A random variable X has $\mu_X = 10, \sigma_X = 3$. If $Y = 4X - 5$, what are the mean and standard deviation of Y?

(A) $\mu_Y = 35, \sigma_Y = 12$

(B) $\mu_Y = 35, \sigma_Y = 7$

(C) $\mu_Y = 40, \sigma_Y = 12$

(D) $\mu_Y = 35, \sigma_Y = 48$

(E) $\mu_Y = 40, \sigma_Y = 3$

Answer: (A). $\mu_Y = 4(10) - 5 = 35$. $\sigma_Y = |4|(3) = 12$.

4. Independent variables X and Y have $\sigma_X = 4$ and $\sigma_Y = 3$. What is the standard deviation of $D = X - Y$?

   (A) 1

   (B) 5

   (C) 7

   (D) 25

   (E) $\sqrt{7}$

   Answer: (B). Variances add: $\sigma_D = \sqrt{\sigma_X^2 + \sigma_Y^2} = \sqrt{16+9} = \sqrt{25} = 5$.

5. 20% of cereal boxes contain a prize. A student buys 5 boxes. What is the probability of getting exactly 2 prizes?

   (A) 0.040

   (B) 0.2048

   (C) 0.3277

   (D) 0.0512

   (E) 0.1638

   Answer: (B). $P(X = 2) = \binom{5}{2}(0.2)^2(0.8)^3 = 10(0.04)(0.512) = 0.2048$.

6. In a binomial distribution with $n = 100$ and $p = 0.30$, what are the mean and standard deviation?

   (A) $\mu = 30, \sigma = 21$

   (B) $\mu = 30, \sigma = 4.58$

(C) $\mu = 30, \sigma = 2.10$

(D) $\mu = 50, \sigma = 5.00$

(E) $\mu = 100, \sigma = 30$

Answer: (B). $\mu = np = 30$. $\sigma = \sqrt{np(1-p)} = \sqrt{100(0.3)(0.7)} = \sqrt{21} \approx 4.58$.

7. A basketball player makes 75% of her free throws. What is the probability that her first miss occurs on her 4^{th} attempt?

(A) $(0.75)^3(0.25)$

(B) $(0.75)(0.25)^3$

(C) $\binom{4}{1}(0.75)^3(0.25)$

(D) $(0.25)^4$

(E) $(0.75)^4$

Answer: (A). Geometric distribution: Success = Miss ($p = 0.25$). Trials $= 4 \implies (0.75)^3(0.25) \approx 0.1055$.

8. What is the expected number of rolls of a fair 6-sided die until the first 6 appears?

(A) 1

(B) 3

(C) 3.5

(D) 6

(E) 36

Answer: (D). For a geometric distribution, $\mu = 1/p = 1/(1/6) = 6$.

9. A probability distribution for a discrete random variable X requires which two conditions?

   (A) $0 \leq P(x_i) \leq 1$ and $\sum P(x_i) = 1$

   (B) $\mu = 0$ and $\sigma = 1$

   (C) $P(x_i)$ is symmetric and bell-shaped

   (D) $n \geq 30$ and $np \geq 10$

   (E) X takes only positive values

   Answer: (A). Foundational axioms for valid probability distributions.

10. If $P(A \mid B) = 0.70$ and $P(B) = 0.50$, what is $P(A \cap B)$?

    (A) 0.20

    (B) 0.35

    (C) 1.20

    (D) 0.714

    (E) 0.25

    Answer: (B). $P(A \cap B) = P(B) \cdot P(A \mid B) = 0.50 \times 0.70 = 0.35$.

11. Which of the following is NOT a requirement for a binomial distribution?

    (A) Fixed number of trials n

    (B) Exactly two outcomes per trial

    (C) Trials continue until the first success

    (D) Independent trials

(E) Constant probability of success p

Answer: (C). Waiting for the first success defines a geometric distribution, not binomial.

12. X and Y are independent random variables with $\mu_X = 50, \mu_Y = 20$. What is μ_{X-2Y}?

    (A) 10

    (B) 30

    (C) 50

    (D) 90

    (E) -10

    Answer: (A). $\mu = 50 - 2(20) = 10$.

13. In problem 12, if $\sigma_X = 5$ and $\sigma_Y = 3$, what is the standard deviation of $X - 2Y$?

    (A) -1

    (B) $\sqrt{25 + 36} = \sqrt{61} \approx 7.81$

    (C) $\sqrt{25 - 36}$

    (D) 11

    (E) 16

    Answer: (B). $\sigma^2 = \sigma_X^2 + 2^2\sigma_Y^2 = 25 + 4(9) = 25 + 36 = 61 \implies \sigma = \sqrt{61}$.

14. The probability of winning a carnival game is $p = 0.05$. What is the probability that you fail on the first 10 plays?

    (A) $(0.05)^{10}$

(B) $(0.95)^{10}$

(C) $1-(0.95)^{10}$

(D) $10(0.05)$

(E) $(0.95)(0.05)^9$

Answer: (B). $P(Y>10)=(1-p)^{10}=(0.95)^{10}\approx 0.5987$.

15. Two fair coins are tossed. Let X = number of heads. What is $\text{Var}(X)$?

(A) 0.25

(B) 0.50

(C) 0.75

(D) 1.00

(E) 2.00

Answer: (B). Binomial with $n=2, p=0.5 \implies \sigma^2 = np(1-p)=2(0.5)(0.5)=0.50$.

16. If $P(A)=0.7, P(B)=0.4$, and $P(A\cup B)=0.85$, what is $P(A\cap B)$?

(A) 0.15

(B) 0.25

(C) 0.28

(D) 0.55

(E) 0.00

Answer: (B). $P(A\cap B)=P(A)+P(B)-P(A\cup B)=0.7+0.4-0.85=0.25$.

17. When sampling from a population of size $N = 500$, what is the maximum sample size n allowed to use the binomial formula safely without replacement?

    (A) 10

    (B) 30

    (C) 50

    (D) 100

    (E) 250

    Answer: (C). By the 10% condition, $n \leq 0.10(500) = 50$.

18. Which of the following statements about independent events A and B is always true?

    (A) $P(A \cup B) = 1$

    (B) $P(A \mid B) = P(A)$

    (C) $P(A \cap B) = 0$

    (D) $P(A) = P(B)$

    (E) $P(A) + P(B) = 1$

    Answer: (B). Definition of statistical independence.

19. A discrete random variable X takes values 1, 2, 3 with probabilities 0.2, 0.5, 0.3. What is $E(X)$?

    (A) 2.0

    (B) 2.1

    (C) 2.3

    (D) 2.5

(E) 3.0

Answer: (B). $E(X) = 1(0.2)+2(0.5)+3(0.3) = 0.2+1.0+0.9 = 2.1$.

20. A binomial distribution with $n = 50$ and $p = 0.40$ can be approximated by a normal distribution because:

(A) $n \geq 30$

(B) $np = 20 \geq 10$ and $n(1-p) = 30 \geq 10$

(C) $p = 0.40 < 0.50$

(D) $n \leq 0.10N$

(E) The trials are dependent

Answer: (B). The Large Counts condition ($np \geq 10$ and $n(1-p) \geq 10$) justifies normality.

Part II: Free-Response Practice Problem

FRQ 4.1: Binomial Probability & Random Variable Combinations

Problem: An airline knows that 5% of passengers who purchase tickets do not show up for their flight. A flight operates on an aircraft with 100 seats, but the airline sells 102 tickets. (a) Define the random variable and check whether the conditions for a binomial distribution are met.
(b) Calculate the probability that more passengers show up than there are seats available.
(c) The airline charges \$200 per ticket sold. If more than 100 passengers show up, each bumped passenger is paid \$500 in compensation. What is the expected net revenue for this

flight?

Grader-Approved Score-4 Solution:

- **Part (a):** Let X = number of passengers who show up. $X \sim B(102, 0.95)$.

  Binary: Passenger shows up (Success, $p = 0.95$) or does not show up (Failure, $1 - p = 0.05$).

  Independent: Passenger decisions are assumed independent.

  Number: Fixed $n = 102$ tickets.

  Same Probability: $p = 0.95$ constant for each passenger.

- **Part (b):** More passengers than seats means $X = 101$ or $X = 102$.

  $P(X = 101) = \binom{102}{101}(0.95)^{101}(0.05)^1 = 102(0.00569)(0.05) \approx 0.0290$.

  $P(X = 102) = \binom{102}{102}(0.95)^{102}(0.05)^0 = (0.95)^{102} \approx 0.0054$.

  $P(X > 100) = 0.0290 + 0.0054 = \mathbf{0.0344}$ (approx. **3.44%**).

- **Part (c):** Guaranteed revenue from 102 tickets $= 102 \times \$200 = \$20,400$.

  If 101 show up (prob 0.0290), 1 passenger bumped $\implies$ Cost $= \$500$.

  If 102 show up (prob 0.0054), 2 passengers bumped $\implies$ Cost $= \$1,000$.

  Expected compensation cost $= 0.0290(\$500) + 0.0054(\$1,000) = \$14.50 + \$5.40 = \$19.90$.

  Expected Net Revenue $= \$20,400 - \$19.90 = \$\mathbf{20,380.10}$.

FRQ 4.2: Linear Combinations of Random Variables & Binomial Distributions

Problem: A specialty roaster fills coffee bags using two automated dispensers. Dispenser 1 dispense volume (X) follows $N(\mu_X = 16.2 \text{ oz}, \sigma_X = 0.5 \text{ oz})$. Dispenser 2 dispense volume (Y) follows $N(\mu_Y = 16.2 \text{ oz}, \sigma_Y = 0.5 \text{ oz})$. Assume the dispensers operate independently.

(a) Find the mean and standard deviation of the total coffee volume dispensed when a two-bag combo is filled ($T = X + Y$).

(b) Find the probability that the absolute difference between the two bags $|X - Y|$ exceeds 1.0 ounce.

(c) An automated sealing line inspects filled bags. The probability that any single bag has a seal defect is $p = 0.04$. In an independent random batch of 50 bags, calculate the probability that at least 2 bags have seal defects.

Grader-Approved Score-4 Solution & Rubric:

- **Part (a):** $\mu_T = \mu_X + \mu_Y = 16.2 + 16.2 = 32.4$ oz.
  Because X and Y are independent, variances add: $\sigma_T = \sqrt{\sigma_X^2 + \sigma_Y^2} = \sqrt{0.5^2 + 0.5^2} = \sqrt{0.25 + 0.25} = \sqrt{0.50} \approx 0.707$ oz.

- **Part (b):** Let $D = X - Y$. $\mu_D = 16.2 - 16.2 = 0$ oz.
  $\sigma_D = \sqrt{\sigma_X^2 + \sigma_Y^2} = \sqrt{0.5^2 + 0.5^2} \approx 0.707$ oz. (Variances ALWAYS add!)
  $D \sim N(0, 0.707)$. $z = \frac{1.0-0}{0.707} \approx 1.414$.

$P(|D| > 1.0) = P(Z > 1.414) + P(Z < -1.414) = 2 \times 0.0786 = \mathbf{0.1573}$.

- **Part (c):** Let W be the number of defective bags. $W \sim B(n = 50, p = 0.04)$.
  $P(W \geq 2) = 1 - [P(W = 0) + P(W = 1)]$.
  $P(W = 0) = (0.96)^{50} \approx 0.1299$.
  $P(W = 1) = \binom{50}{1}(0.04)^1(0.96)^{49} = 50(0.04)(0.1353) \approx 0.2706$.
  $P(W \geq 2) = 1-(0.1299+0.2706) = 1-0.4005 = \mathbf{0.5995}$.

Unit 4 Top 5 AP Exam Pitfalls to Avoid

1. **Subtracting Variances:** Never subtract variances! When subtracting random variables ($X - Y$), variances still add: $\sigma^2_{X-Y} = \sigma^2_X + \sigma^2_Y$.
2. **Confusing Binomial with Geometric:** Binomial counts successes in fixed n trials; Geometric counts trials until first success.
3. **Forgetting to State Independence Before Multiplying:** $P(A \cap B) = P(A)P(B)$ only applies when events are independent.
4. **Confusing Disjoint (Mutually Exclusive) with Independent:** Disjoint events can NEVER be independent (if one occurs, the other cannot!).
5. **Multiplying Standard Deviations by Negative Numbers:** Linear transformation $\sigma_{a+bx} = |b|\sigma_x$. Standard deviations are always non-negative.

4.6 Unit 4 Master Summary

Unit 4 Essential Review Checklist

- **Independence Test:** $P(A \mid B) = P(A)$ or $P(A \cap B) = P(A)P(B)$.
- **Combining RVs:** $\mu_{X \pm Y} = \mu_X \pm \mu_Y$; $\sigma^2_{X \pm Y} = \sigma^2_X + \sigma^2_Y$ (variances always add!).
- **Binomial ($BINS$):** $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$, $\mu = np$, $\sigma = \sqrt{np(1-p)}$.
- **Geometric ($BITS$):** $P(Y = k) = (1-p)^{k-1} p$, $\mu = 1/p$.

Chapter 5

Unit 5: Sampling Distributions & The Central Limit Theorem

5.1 Unit Overview & Exam Weight

Unit 5 accounts for **7% to 12%** of the AP Statistics Exam. It is the crucial conceptual bridge between probability models and formal statistical inference. Understanding the sampling distributions of proportions ($\hat{p}$) and means ($\bar{x}$) is required to justify every confidence interval and significance test in Units 6 through 9.

Key Unit Vocabulary

Parameter:
A numerical value describing a characteristic of an entire population. Fixed, but typically unknown (μ, σ, p).

Statistic:
A numerical value computed from sample data, used to estimate a population parameter ($\bar{x}, s_x, \hat{p}$).

Sampling Distribution:

The theoretical probability distribution of all possible values of a statistic taken across all possible random samples of the exact same size n from the same population.

Sampling Variability:

The fundamental reality that different random samples produce different sample statistics.

Unbiased Estimator:

A statistic whose sampling distribution has a mean equal to the true value of the population parameter being estimated ($E(\hat{p}) = p$ and $E(\bar{x}) = \mu$).

Central Limit Theorem (CLT):

For a sufficiently large random sample ($n \geq 30$), the sampling distribution of the sample mean $\bar{x}$ is approximately normal, regardless of the underlying shape of the population distribution!

10% Condition:

When sampling without replacement, the sample size must not exceed 10% of the population ($n \leq 0.10N$) to satisfy the independence assumption in standard error formulas.

Large Counts Condition:

For proportions, both $np \geq 10$ and $n(1 - p) \geq 10$ must be satisfied to ensure the sampling distribution of $\hat{p}$ is approximately normal.

5.2 Core Concept Lessons

Lesson 5.1: The Three Distributions Distinguished

To avoid disastrous confusion on the AP exam, strictly distinguish between:

1. **Population Distribution:** Gives the values of the variable for all individuals in the entire population. Fixed mean μ and spread σ.
2. **Sample Distribution:** Gives the values of the variable for the n individuals in a single selected sample. Summary statistics are $\bar{x}$ and s_x.
3. **Sampling Distribution:** The distribution of values taken by the statistic (e.g., $\bar{x}$ or $\hat{p}$) across **all possible samples of size** n. As n increases, its spread shrinks!

Lesson 5.2: Sampling Distribution of a Sample Proportion ($\hat{p}$)

Let $\hat{p} = X/n$ be the sample proportion of successes in an SRS of size n from a population with true proportion p:

$$\text{Center: } \mu_{\hat{p}} = p \quad (\hat{p} \text{ is an unbiased estimator of } p) \tag{5.1}$$

$$\text{Spread: } \sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} \quad (\text{Requires 10\% condition: } n \leq 0.10N) \tag{5.2}$$

Shape (Normality Condition): The sampling distribution of $\hat{p}$ is approximately normal if and only if the **Large**

Counts Condition is satisfied:

$$np \geq 10 \quad \text{AND} \quad n(1-p) \geq 10 \tag{5.3}$$

Lesson 5.3: Sampling Distribution of a Sample Mean ($\bar{x}$)

Let $\bar{x}$ be the sample mean of an SRS of size n drawn from a population with mean μ and standard deviation σ:

$$\text{Center: } \mu_{\bar{x}} = \mu \quad (\bar{x} \text{ is an unbiased estimator of } \mu) \tag{5.4}$$

$$\text{Spread: } \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \quad (\text{Requires 10\% condition: } n \leq 0.10N) \tag{5.5}$$

Shape (Normality Condition):

- If the population distribution is **Normal**, the sampling distribution of $\bar{x}$ is **Normal for any sample size** n.
- If the population distribution is **NOT Normal (skewed or multimodal)**, the **Central Limit Theorem (CLT)** guarantees that the sampling distribution of $\bar{x}$ is approximately normal provided the sample size is sufficiently large:

$$n \geq 30 \tag{5.6}$$

Exam Trap Alert: Central Limit Theorem vs. Large Counts

- The **Central Limit Theorem applies ONLY to sample means ($\bar{x}$)**!
- Never write *"The sampling distribution of $\hat{p}$ is normal by*

the CLT." That is mathematically false and loses credit. Proportions are justified normal by **Large Counts** ($np \geq 10, n(1-p) \geq 10$).

- Furthermore, the CLT says the **sampling distribution** becomes normal as n grows; it does NOT say the population or the raw sample becomes normal!

5.3 Worked Step-by-Step Examples

Example 5.1: Sampling Distribution of Proportions in Quality Control

Problem: A semiconductor foundry produces microchips with a defect rate of $p = 0.08$. An inspector tests a random sample of $n = 250$ microchips from a production run of 10,000 chips.
(a) Confirm that the sampling distribution of $\hat{p}$ is approximately normal.
(b) Calculate the mean and standard deviation of $\hat{p}$.
(c) What is the probability that the sample contains more than 10% defective chips?

Solution:

- **Part (a):** *Random:* Stated random sample.
  10% Condition: $n = 250 \leq 0.10(10,000) = 1,000$. Met.
  Large Counts: $np = 250(0.08) = 20 \geq 10$ and $n(1-p) = 250(0.92) = 230 \geq 10$. Both are at least 10, so the sampling distribution of $\hat{p}$ is approximately normal.

- **Part (b):** $\mu_{\hat{p}} = p = \mathbf{0.08}$.
  $\sigma_{\hat{p}} = \sqrt{\frac{0.08(0.92)}{250}} = \sqrt{\frac{0.0736}{250}} = \sqrt{0.0002944} \approx \mathbf{0.01716}$.

- **Part (c):** We seek $P(\hat{p} > 0.10)$. Standardize $\hat{p} = 0.10$:

$$z = \frac{0.10 - 0.08}{0.01716} = \frac{0.02}{0.01716} \approx 1.17$$

$P(\hat{p} > 0.10) = P(Z > 1.17) = 1 - P(Z \leq 1.17) = 1 - 0.8790 = \mathbf{0.1210}$ (approx. **12.1%**).

Example 5.2: Applying the Central Limit Theorem to Heavily Skewed Data

Problem: The claim processing times at an insurance company are heavily right-skewed with a mean of $\mu = 14.5$ days and a standard deviation of $\sigma = 6.0$ days. A manager reviews an SRS of $n = 36$ claims. What is the probability that the average processing time $\bar{x}$ for these 36 claims is under 13.0 days?

Solution:

- **Condition Check:** Although the population distribution is heavily skewed, the sample size is $n = 36 \geq 30$. By the Central Limit Theorem, the sampling distribution of the sample mean $\bar{x}$ is approximately normal!

- **Parameters of $\bar{x}$:** $\mu_{\bar{x}} = \mu = 14.5$ days.
$\sigma_{\bar{x}} = \sigma/\sqrt{n} = 6.0/\sqrt{36} = 6.0/6 = 1.0$ day.

- **Probability Calculation:** $P(\bar{x} < 13.0)$:

$$z = \frac{13.0 - 14.5}{1.0} = \frac{-1.5}{1.0} = -1.50$$

$P(\bar{x} < 13.0) = P(Z < -1.50) = \mathbf{0.0668}$ (approx. **6.68%**).

5.4 TI-84 Plus CE Calculator Playbook

TI-84 Playbook: Sampling Distribution Probabilities

- **For Proportions $\hat{p}$:** Press `[2nd] [VARS]` $\rightarrow$ `2:normalcdf(`. Enter `lower`, `upper`, $\mu = p$, $\sigma = \sqrt{p(1-p)/n}$.
- **For Means $\bar{x}$:** Press `[2nd] [VARS]` $\rightarrow$ `2:normalcdf(`. Enter `lower`, `upper`, $\mu = \mu_{\text{pop}}$, $\sigma = \sigma_{\text{pop}}/\sqrt{n}$.

5.5 Comprehensive Practice Problem Set

Part I: 20 Multiple-Choice Questions

1. Which of the following is a statistic?

   (A) The true proportion p of all voters

   (B) The population mean μ

   (C) The sample proportion $\hat{p}$

   (D) The population standard deviation σ

   (E) The census parameter

   Answer: (C). $\hat{p}$ is computed from sample data.

2. As sample size n increases from 100 to 400, the standard deviation of $\hat{p}$:

   (A) Multiplies by 4

   (B) Multiplies by 2

(C) Is cut in half

(D) Is divided by 4

(E) Remains unchanged

Answer: (C). $\sigma_{\hat{p}} \propto 1/\sqrt{n}$. $\sqrt{400}/\sqrt{100} = 20/10 = 2 \implies$ divided by 2.

3. A population distribution is heavily skewed to the left with $\mu = 80, \sigma = 12$. A random sample of size $n = 64$ is selected. The sampling distribution of $\bar{x}$ is:

(A) Heavily skewed left with $\mu = 80, \sigma = 12$

(B) Approximately normal with $\mu = 80, \sigma = 1.5$

(C) Approximately normal with $\mu = 80, \sigma = 12$

(D) Uniform with $\mu = 80, \sigma = 1.5$

(E) Cannot be determined because the population is skewed

Answer: (B). By the CLT ($n = 64 \geq 30$), the sampling distribution is approximately normal. $\sigma_{\bar{x}} = 12/\sqrt{64} = 1.5$.

4. The Central Limit Theorem states that:

(A) As n grows, the sample data looks more normal.

(B) As n grows, the population becomes normal.

(C) As n grows, the sampling distribution of $\bar{x}$ approaches a normal distribution.

(D) As n grows, the standard error increases.

(E) Proportions are always normal.

Answer: (C). Standard definition of the CLT.

5. Which condition justifies using $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$?

   (A) $np \geq 10$ and $n(1-p) \geq 10$

   (B) $n \geq 30$

   (C) $n \leq 0.10N$ (10% Condition)

   (D) Population is normal

   (E) $p = 0.50$

   Answer: (C). The 10% condition satisfies the independence requirement for standard error formulas.

6. An unbiased estimator is a statistic that:

   (A) Has a standard deviation of zero.

   (B) Has a sampling distribution whose mean equals the true parameter.

   (C) Never produces an error.

   (D) Is calculated from a census.

   (E) Has a sample size greater than 100.

   Answer: (B). Defining condition of unbiasedness.

7. If $p = 0.60$ and $n = 100$, what is $P(\hat{p} \leq 0.50)$?

   (A) 0.0207

   (B) 0.0500

   (C) 0.1000

   (D) 0.4793

   (E) 0.9793

Answer: (A). $\sigma = \sqrt{0.6(0.4)/100} = 0.04899$. $z = (0.50 - 0.60)/0.04899 = -2.04$. $P(Z < -2.04) = 0.0207$.

8. A population has $\mu = 50, \sigma = 10$. For an SRS of size $n = 25$, what is $P(\bar{x} > 54)$ if the population is normal?

   (A) 0.0228

   (B) 0.0500

   (C) 0.1587

   (D) 0.3446

   (E) 0.4500

   Answer: (A). $\sigma_{\bar{x}} = 10/\sqrt{25} = 2.0$. $z = (54 - 50)/2 = 2.00$. $P(Z > 2.00) = 0.0228$.

9. Which of the following estimators of a population mean μ is biased?

   (A) Sample mean $\bar{x}$

   (B) Sample median (for symmetric populations)

   (C) $\bar{x} + 2$

   (D) Midrange (for symmetric populations)

   (E) None of the above

   Answer: (C). Adding a constant of 2 guarantees the expected value is $\mu + 2 \neq \mu$.

10. For a population with $p = 0.02$, what is the minimum sample size required to satisfy the Large Counts condition?

    (A) 30

    (B) 100

(C) 250

(D) 500

(E) 1000

Answer: (D). $np \geq 10 \implies n(0.02) \geq 10 \implies n \geq 10/0.02 = 500$.

11. Why does the standard deviation of $\bar{x}$ decrease as n increases?

    (A) Larger samples average out extreme individual values, clustering sample means closer to μ.

    (B) The population standard deviation shrinks.

    (C) The degrees of freedom decrease.

    (D) The sampling frame becomes biased.

    (E) μ increases.

    Answer: (A). Averaging over larger sample sizes dampens extreme individual deviations.

12. A fair coin is tossed n times. What is the standard deviation of the proportion of heads if $n = 400$?

    (A) 0.05

    (B) 0.025

    (C) 0.0125

    (D) 0.25

    (E) 0.0025

    Answer: (B). $\sigma = \sqrt{0.5(0.5)/400} = \sqrt{0.25/400} = 0.5/20 = 0.025$.

13. If the shape of a population is uniform between 0 and 10, the sampling distribution of $\bar{x}$ for $n = 50$ will be:

    (A) Uniform

    (B) Skewed right

    (C) Approximately normal

    (D) Bimodal

    (E) Exponential

    Answer: (C). By the Central Limit Theorem ($n = 50 \geq 30$).

14. The spread of a sampling distribution is also commonly called the:

    (A) Parameter

    (B) Standard error of the statistic

    (C) Bias

    (D) Confidence level

    (E) P-value

    Answer: (B). The standard deviation of an estimator is the standard error.

15. If $n = 16$ from a normal population with $\sigma = 8$, the standard deviation of $\bar{x}$ is:

    (A) 8

    (B) 4

    (C) 2

    (D) 0.5

(E) 1

Answer: (C). $\sigma_{\bar{x}} = 8/\sqrt{16} = 8/4 = 2$.

16. A study surveys 100 students from a university of 20,000. Is the 10% condition satisfied?

(A) Yes, because $100 \leq 0.10(20,000) = 2,000$.

(B) No, because $100 < 30$.

(C) Yes, because $100 \geq 30$.

(D) No, because the university is too large.

(E) Yes, by the Central Limit Theorem.

Answer: (A). $100 \leq 2,000$.

17. What happens to the mean of a sampling distribution as the sample size quadruples?

(A) Quadruples

(B) Doubles

(C) Is cut in half

(D) Remains unchanged

(E) Approaches zero

Answer: (D). The mean $\mu_{\bar{x}} = \mu$ is invariant to sample size.

18. An estimator that consistently overestimates the true parameter is said to have:

(A) High variability

(B) Positive bias

(C) Low degrees of freedom

(D) Negative variance

(E) Normality

Answer: (B). Systematically exceeding the parameter reflects positive bias.

19. If $P(\bar{x} < 100) = 0.50$, what is the population mean μ?

(A) 0

(B) 50

(C) 100

(D) 200

(E) Cannot be determined

Answer: (C). Because normal distributions are symmetric around their mean, 50% falls below $\mu \implies \mu = 100$.

20. Which of the following conditions is required to apply the Central Limit Theorem?

(A) The population must be normal.

(B) The sample size must be sufficiently large ($n \geq 30$).

(C) $np \geq 10$.

(D) The data must be categorical.

(E) The variance must be 1.

Answer: (B). Core criterion for the CLT.

Part II: Free-Response Practice Problem

FRQ 5.1: Central Limit Theorem & Sampling Distribution of Means

Problem: An automated coffee machine dispenses coffee with a mean volume of $\mu = 10.0$ oz and a standard deviation of $\sigma = 0.8$ oz. The distribution of volumes for individual cups is skewed to the left. (a) Can we calculate the probability that a single randomly selected cup has fewer than 9.8 oz of coffee? Explain.
(b) A tray holds an SRS of $n = 36$ cups. Describe the sampling distribution of the sample mean volume $\bar{x}$ (shape, center, spread).
(c) What is the probability that the average volume of these 36 cups is less than 9.8 oz?

Grader-Approved Score-4 Solution:

- **Part (a):** No. Because the population distribution of individual cup volumes is skewed to the left and $n = 1$, we cannot assume normality for a single cup, making normal probability calculations invalid.

- **Part (b):** *Shape:* Approximately Normal because $n = 36 \geq 30$, satisfying the Central Limit Theorem.
  Center: $\mu_{\bar{x}} = \mu = \mathbf{10.0}$ oz.
  Spread: Because $n = 36 \leq 0.10(\text{all cups produced})$, $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{0.8}{\sqrt{36}} = \frac{0.8}{6} \approx \mathbf{0.1333}$ oz.

- **Part (c):** Standardize $\bar{x} = 9.8$:

$$z = \frac{9.8 - 10.0}{0.1333} = \frac{-0.2}{0.1333} = -1.50$$

$P(\bar{x} < 9.8) = P(Z < -1.50) =$ **0.0668** (approx. **6.68%**).

FRQ 5.2: Sampling Distribution of Means with Central Limit Theorem

Problem: A commercial cargo airline transports standard shipping crates. The crate weights have a distribution with mean $\mu = 125$ lbs and standard deviation $\sigma = 40$ lbs. The distribution of individual crate weights is heavily skewed to the right with occasional very heavy crates.

(a) Can you calculate the probability that a single randomly chosen crate weighs more than 135 lbs? Explain.

(b) A cargo aircraft is loaded with an SRS of $n = 64$ crates. Describe the sampling distribution of the sample mean crate weight $\bar{x}$, including shape, center, and spread.

(c) The maximum allowable average weight per crate for safe takeoff is 135 lbs. What is the probability that the sample mean weight of the 64 crates exceeds 135 lbs?

(d) If the airline increased the sample size to $n = 100$ crates, how would the probability of exceeding the 135-lb safety threshold change? Explain.

Grader-Approved Score-4 Solution & Rubric:

- **Part (a):** No. We are told the population of individual crate weights is heavily skewed to the right. Because the population is not normal and $n = 1 < 30$, we cannot use the normal distribution to compute probabilities for a single crate.

- **Part (b):** *Shape:* Approximately normal by the Central Limit Theorem because the sample size $n = 64 \geq 30$.
  Center: $\mu_{\bar{x}} = \mu = 125$ lbs.
  Spread: $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{40}{\sqrt{64}} = \frac{40}{8} = 5.0$ lbs (assuming 64 crates is less than 10% of all crates).

- **Part (c):** $z = \frac{135-125}{5.0} = \frac{10}{5.0} = 2.00$.
  $P(\bar{x} > 135) = P(Z > 2.00) = 1 - 0.9772 = \mathbf{0.0228}$ (approx. 2.28%).

- **Part (d):** For $n = 100$, $\sigma_{\bar{x}} = \frac{40}{\sqrt{100}} = 4.0$ lbs. $z = \frac{135-125}{4.0} = 2.50$, so $P(Z > 2.50) = 0.0062$. Increasing n reduces the standard error of the sample mean, concentrating the sampling distribution closer to $\mu = 125$ and decreasing the probability of exceeding 135 lbs.

Unit 5 Top 5 AP Exam Pitfalls to Avoid

1. **Claiming the CLT Makes the Population Normal:** The Central Limit Theorem only normalizes the **sampling distribution of** $\bar{x}$, NEVER the population or sample!

2. **Applying CLT to Proportions:** CLT applies to sample means $\bar{x}$. For proportions $\hat{p}$, normality is justified by the Large Counts condition ($np \geq 10$ and $n(1-p) \geq 10$).

3. **Forgetting to Divide by $\sqrt{n}$:** The standard deviation of $\bar{x}$ is $\sigma/\sqrt{n}$, not σ!

4. **Confusing the 10% Condition with Large Counts:** 10% condition ($n \leq 0.10N$) checks independence; Large Counts checks normality.

5. **Confusing Parameter with Statistic:** μ and p are fixed population parameters; $\bar{x}$ and $\hat{p}$ are variable statistics.

5.6 Unit 5 Master Summary

Unit 5 Essential Review Checklist

- **Proportions ($\hat{p}$):** $\mu = p$, $\sigma = \sqrt{p(1-p)/n}$. Normal if $np \geq 10$ and $n(1-p) \geq 10$.
- **Means ($\bar{x}$):** $\mu = \mu$, $\sigma = \sigma/\sqrt{n}$. Normal if pop is normal OR $n \geq 30$ (CLT).
- **10% Condition ($n \leq 0.10N$):** Required for standard deviation formulas when sampling without replacement.
- **CLT Distinction:** CLT applies ONLY to means ($\bar{x}$), never to proportions!

Chapter 6

Unit 6: Inference for Categorical Data: Proportions

6.1 Unit Overview & Exam Weight

Unit 6 accounts for **12% to 15%** of the AP Statistics Exam. This unit marks the formal beginning of statistical inference. You will master the two primary pillars of inference: constructing and interpreting **Confidence Intervals** (estimating population proportions) and conducting **Significance Tests** (evaluating hypotheses). You will also master error analysis (Type I vs. Type II errors) and statistical power.

Key Unit Vocabulary

Point Estimate:

A single value calculated from sample data that serves as the best guess for an unknown population parameter ($\hat{p}$ for p).

Confidence Interval:

An interval of plausible values for a parameter, structured as: Point Estimate ± Margin of Error.

Margin of Error:

The maximum expected difference between the true parameter and point estimate at a specified confidence level: $\text{ME} = z^* \cdot \text{SE}$.

Critical Value (z^*):

A multiplier based on the desired confidence level from the standard normal curve (e.g., $z^* = 1.96$ for 95% confidence, $z^* = 2.576$ for 99%).

Null Hypothesis (H_0):

The default claim of ”no difference,” ”no change,” or ”status quo” ($H_0 : p = p_0$). Assumed true throughout calculations until disproven.

Alternative Hypothesis (H_a):

The research claim we seek convincing evidence for ($p > p_0$, $p < p_0$, or $p \neq p_0$).

P-Value:

The probability of obtaining a test statistic as extreme as or more extreme than the observed sample result, assuming the null hypothesis H_0 is true.

Type I Error (α):

Rejecting the null hypothesis H_0 when H_0 is actually true (a ”false alarm”). $P(\text{Type I Error}) = \alpha$.

Type II Error (β):

Failing to reject H_0 when H_0 is actually false (a ”missed detection”).

Power of a Test $(1-\beta)$:

The probability of correctly rejecting a false null hypothesis.

6.2 Core Concept Lessons

Lesson 6.1: The P-A-N-I-C Framework for Confidence Intervals

Whenever constructing an AP confidence interval on an FRQ, follow the 5-step P-A-N-I-C process:

1. **P — Parameter:** State the population parameter in complete context: *"Let p = the true proportion of all [individuals in population] who [characteristic]."*
2. **A — Assumptions / Conditions:** Verify the 3 mandatory conditions:
   - *Random:* Data gathered via an SRS or random sample stated in the prompt.
   - *10% Condition:* $n \leq 0.10N$ (sampling without replacement).
   - *Large Counts:* $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$. (Show actual numbers!).
3. **N — Name the Interval:** State: *"One-sample z-interval for a population proportion p."*
4. **I — Interval Calculation:**

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \tag{6.1}$$

5. **C — Conclusion in Context:** Use the official AP sentence frame: *"We are [C%] confident that the true proportion of [parameter in context] is between [Lower Bound] and [Upper Bound]."*

Exam Trap Alert: Confidence Interval vs. Confidence Level

- **Interpreting the Interval:** *"We are 95% confident that the true proportion of voters who support Candidate X is between 0.52 and 0.58."*
- **Interpreting the Level (95%):** *"If we were to take many repeated random samples of the same size from this population and construct a 95% confidence interval from each sample, approximately 95% of the resulting intervals would successfully capture the true population proportion p."*
- **NEVER say:** *"There is a 95% probability that the true proportion is in this interval."* (Once calculated, the true parameter is either in the interval or it is not; probability applies to the method over repeated sampling, not a fixed interval!).

Lesson 6.2: The P-H-A-N-T-O-M Framework for Significance Tests

Follow the 7-step P-H-A-N-T-O-M protocol for testing claims:

1. **P — Parameter:** Define p in context.
2. **H — Hypotheses:** $H_0 : p = p_0$ versus $H_a : p > p_0, p <$

p_0, or $p \neq p_0$.

3. **A — Assumptions:** Random, 10%, and Large Counts: $np_0 \geq 10$ and $n(1-p_0) \geq 10$! (Notice: Use the hypothesized p_0 to check conditions, NOT $\hat{p}$!).

4. **N — Name the Test:** *"One-sample z-test for a population proportion p."*

5. **T — Test Statistic:**

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \tag{6.2}$$

6. **O — Obtain P-Value:** Compute $P(Z \geq z)$, $P(Z \leq z)$, or $2 \times P(Z \geq |z|)$.

7. **M — Make Decision & State Conclusion in Context:**

   - **If p-value $\leq \alpha$:** *"Because the p-value of [Value] is less than $\alpha = 0.05$, we reject H_0. There is convincing statistical evidence that [state H_a in context]."*
   - **If p-value $> \alpha$:** *"Because the p-value of [Value] is greater than $\alpha = 0.05$, we fail to reject H_0. There is not convincing statistical evidence that [state H_a in context]."*

Lesson 6.3: Error Analysis & Statistical Power

	H_0 is Actually True	**H_0 is Actually False**
Reject H_0	**Type I Error** (prob $= \alpha$)	**Correct Decision** (Power $= 1 - \beta$)
Fail to Reject H_0	**Correct Decision**	**Type II Error** (prob $= \beta$)

How to Increase the Power of a Test:

- **Increase sample size n:** Decreases standard error, sharpening distributions (most practical method!).
- **Increase significance level α:** Widens the rejection region, decreasing β and increasing power. (Tradeoff: Increases Type I error risk).
- **Increase effect size:** A larger true difference between the parameter and null value makes detection easier.

Lesson 6.4: Two-Sample Inference for Proportions ($p_1 - p_2$)

- **Two-Sample z-Interval for $p_1 - p_2$:**

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} \tag{6.3}$$

- **Two-Sample z-Test for $p_1 - p_2$ ($H_0 : p_1 - p_2 = 0$):** Because H_0 assumes $p_1 = p_2$, pool sample counts into a single combined proportion:

$$\hat{p}_C = \frac{X_1 + X_2}{n_1 + n_2} \tag{6.4}$$

$$z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}_C(1-\hat{p}_C)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \tag{6.5}$$

6.3 Worked Step-by-Step Examples

Example 6.1: Complete 1-Proportion Significance Test (PHANTOM)

Problem: A streaming service claims that 70% of its subscribers watch content on mobile devices. A competitor believes the true proportion is lower. In an SRS of $n = 400$ subscribers, 260 watch on mobile devices. Conduct a hypothesis test at $\alpha = 0.05$.

Solution:

- **P:** Let $p =$ the true proportion of all subscribers who watch on mobile
- **H:** $H_0 : p = 0.70$ versus $H_a : p < 0.70$.
- **A:** *Random:* Stated SRS of subscribers.
  10% Condition: $n = 400 \leq 0.10(\text{all subscribers})$ (millions of users).
  Large Counts: $np_0 = 400(0.70) = 280 \geq 10$ and $n(1 - p_0) = 400(0.30) = 120 \geq 10$. Both ≥ 10.
- **N:** One-sample z-test for a population proportion p.
- **T:** $\hat{p} = 260/400 = 0.65$.

$$z = \frac{0.65 - 0.70}{\sqrt{\frac{0.70(0.30)}{400}}} = \frac{-0.05}{\sqrt{0.000525}} = \frac{-0.05}{0.02291} \approx -2.18$$

- **O:** p-value $= P(Z \leq -2.18) = \mathbf{0.0146}$.
- **M:** Because the p-value (0.0146) is less than $\alpha = 0.05$, we reject the null hypothesis H_0. There is convincing statistical evidence that the true proportion of subscribers who watch on mobile devices is strictly less than 70%.

6.4 TI-84 Plus CE Calculator Playbook

TI-84 Playbook: Proportions Inference Tests and Intervals

- **1-Prop z-Test:** Press `[STAT]` → `TESTS` → `5:1-PropZTest`. Enter p_0, x, n, and direction of H_a.
- **1-Prop z-Interval:** Press `[STAT]` → `TESTS` → `A:1-PropZInt`. Enter x, n, C-Level.
- **2-Prop z-Test:** Press `[STAT]` → `TESTS` → `6:2-PropZTest`. Enter x_1, n_1, x_2, n_2.
- **2-Prop z-Interval:** Press `[STAT]` → `TESTS` → `B:2-PropZInt`. Enter x_1, n_1, x_2, n_2, C-Level.

6.5 Comprehensive Practice Problem Set

Part I: 20 Multiple-Choice Questions

1. What does it mean to be "95% confident" in the context of a confidence interval?

   (A) 95% of the data values fall within the interval.

   (B) There is a 95% probability that the true parameter lies in this specific interval.

   (C) In repeated random sampling, approximately 95% of all intervals constructed will capture the true population parameter.

(D) 95% of future sample means will fall within this interval.

(E) The test has a 5% margin of error.

Answer: (C). Standard definition of confidence level.

2. To reduce the margin of error of a confidence interval by a factor of 3 without changing the confidence level, the sample size must:

(A) Triple

(B) Multiply by 6

(C) Multiply by 9

(D) Multiply by $\sqrt{3}$

(E) Be cut in third

Answer: (C). ME $\propto 1/\sqrt{n}$. To cut ME by 3, n must multiply by $3^2 = 9$.

3. In an SRS of $n = 100$ students, 40 play an instrument. What is the standard error of $\hat{p}$?

(A) $\sqrt{0.4(0.6)/100} = 0.04899$

(B) $0.4/100 = 0.004$

(C) $\sqrt{100(0.4)(0.6)} = 4.899$

(D) 1.96×0.04899

(E) 0.40

Answer: (A). $\text{SE}(\hat{p}) = \sqrt{\hat{p}(1-\hat{p})/n} = \sqrt{0.24/100} \approx 0.049$.

4. A 95% confidence interval for p is $(0.42, 0.54)$. What is the point estimate $\hat{p}$ and margin of error?

(A) $\hat{p} = 0.48, \mathrm{ME} = 0.06$

(B) $\hat{p} = 0.48, \mathrm{ME} = 0.12$

(C) $\hat{p} = 0.50, \mathrm{ME} = 0.04$

(D) $\hat{p} = 0.42, \mathrm{ME} = 0.54$

(E) $\hat{p} = 0.45, \mathrm{ME} = 0.09$

Answer: (A). Center $= (0.42 + 0.54)/2 = 0.48$. ME $= 0.54 - 0.48 = 0.06$.

5. In testing $H_0 : p = 0.50$ vs. $H_a : p > 0.50$, the test statistic is $z = 1.96$. At $\alpha = 0.05$, the decision is:

(A) Reject H_0 because p-value $= 0.025 \leq 0.05$.

(B) Fail to reject H_0 because $z > 0$.

(C) Accept H_a with 100% certainty.

(D) Reject H_a.

(E) Retest with a larger sample.

Answer: (A). For $z = 1.96$, one-sided p-value $= 0.025 < 0.05 \implies$ Reject H_0.

6. A Type I error is committed when:

(A) We reject H_0 when H_0 is actually true.

(B) We fail to reject H_0 when H_0 is false.

(C) The sample size is too small.

(D) The null hypothesis is false.

(E) The confidence interval includes zero.

Answer: (A). False alarm / rejecting a true null hypothesis.

7. Increasing the significance level α from 0.01 to 0.05:

   (A) Increases the power of the test and decreases the probability of a Type II error.

   (B) Decreases power.

   (C) Increases β.

   (D) Has no effect on Type II errors.

   (E) Decreases the probability of a Type I error.

   Answer: (A). A higher α widens the rejection zone, increasing power and reducing β.

8. A 99% confidence interval for $p_1 - p_2$ is $(-0.08, 0.04)$. Based on this interval, what can be concluded?

   (A) p_1 is definitely greater than p_2.

   (B) Because zero is included in the interval, there is no convincing evidence of a difference between p_1 and p_2 at $\alpha = 0.01$.

   (C) p_2 is significantly greater than p_1.

   (D) $p_1 = p_2 = 0$.

   (E) The sample sizes were unequal.

   Answer: (B). An interval containing 0 cannot rule out $p_1 = p_2$.

9. What is the pooled proportion $\hat{p}_C$ for $X_1 = 30, n_1 = 50$ and $X_2 = 45, n_2 = 50$?

   (A) 0.60

   (B) 0.75

(C) 0.75

(D) 0.75

(E) 0.75

Answer: (C). $\hat{p}_C = (30+45)/(50+50) = 75/100 = 0.75$.

10. When testing $H_0 : p = 0.40$ vs. $H_a : p \neq 0.40$, the test statistic is $z = -2.15$. The p-value is:

(A) 0.0158

(B) 0.0316

(C) 0.9842

(D) 0.0500

(E) 0.0079

Answer: (B). Two-sided test: $2 \times P(Z \leq -2.15) = 2(0.0158) = 0.0316$.

11. Which critical value z^* is used for a 90% confidence interval for a proportion?

(A) 1.282

(B) 1.645

(C) 1.960

(D) 2.326

(E) 2.576

Answer: (B). Middle 90% leaves 5% in each tail $\implies z^* = 1.645$.

12. If a test fails to reject H_0, which error could have potentially been made?

    (A) Type I Error

    (B) Type II Error

    (C) Confounding Error

    (D) Standard Error

    (E) No error is possible

    Answer: (B). Failing to reject when H_0 is actually false is a Type II error (β).

13. If an experimenter rejects H_0 at $\alpha = 0.05$, will they definitely reject H_0 at $\alpha = 0.01$?

    (A) Yes, always.

    (B) No, only if the p-value is also less than 0.01.

    (C) Yes, because 0.05 ¿ 0.01.

    (D) No, because power decreases.

    (E) Cannot be determined.

    Answer: (B). A result with $p = 0.03$ rejects at 0.05 but fails to reject at 0.01.

14. The power of a significance test against a specific alternative parameter value is 0.85. What is β?

    (A) 0.05

    (B) 0.15

    (C) 0.85

    (D) 0.90

(E) 0.01

Answer: (B). Power $= 1 - \beta \implies \beta = 1 - 0.85 = 0.15$.

15. In a poll of $n = 500$ voters, $\hat{p} = 0.52$. Can we claim Candidate A has a statistically significant lead over 50% at $\alpha = 0.05$?

   (A) Yes, because $0.52 > 0.50$.

   (B) No, because $z = (0.52 - 0.50)/\sqrt{0.5(0.5)/500} = 0.02/0.02236 \approx 0.89$, giving p-value $= 0.1867 > 0.05$.

   (C) Yes, because $500 \geq 30$.

   (D) Yes, by the Central Limit Theorem.

   (E) No, because 500 is too small.

   Answer: (B). $z = 0.89$ produces p-value $= 0.187 > 0.05$, failing to reject H_0.

16. Why do we use hypothesized p_0 instead of $\hat{p}$ when calculating standard error in a 1-proportion test?

   (A) Because p_0 is always larger.

   (B) Because hypothesis tests proceed under the formal assumption that H_0 is true.

   (C) To avoid calculating square roots.

   (D) Because $\hat{p}$ is biased.

   (E) It is required by the 10% condition.

   Answer: (B). Testing logic evaluates how likely the sample is assuming $H_0 : p = p_0$ is the truth.

17. A 95% confidence interval for p is calculated as $(0.18, 0.26)$. Which of the following two-sided hypotheses would be rejected at $\alpha = 0.05$?

    (A) $H_0 : p = 0.20$

    (B) $H_0 : p = 0.22$

    (C) $H_0 : p = 0.25$

    (D) $H_0 : p = 0.28$

    (E) $H_0 : p = 0.24$

    Answer: (D). Any value outside the 95% CI ($p = 0.28$) is rejected at $\alpha = 0.05$.

18. What is the minimum sample size required to estimate a population proportion within ± 0.03 with 95% confidence if no prior estimate of p is available?

    (A) 385

    (B) 752

    (C) 1,068

    (D) 1,500

    (E) 2,000

    Answer: (C). Conservative estimate uses $p^* = 0.50$: $1.96\sqrt{0.25/n} \leq 0.03 \implies n \geq (1.96/0.03)^2(0.25) \approx 1,067.1 \implies 1,068$.

19. In a 2-sample z-test for $p_1 - p_2$, why do we pool the proportions?

    (A) Because $n_1 = n_2$.

(B) Because the null hypothesis assumes $p_1 = p_2$, so both samples come from a population with a single common proportion.

(C) To reduce bias.

(D) Because the variables are continuous.

(E) It is required by the Central Limit Theorem.

Answer: (B). Under $H_0 : p_1 = p_2$, pooling provides the best estimate of the shared parameter.

20. If an AP exam question states "interpret the p-value in context," you should write:

(A) "The probability that H_0 is true."

(B) "The probability that H_a is true."

(C) "Assuming the null hypothesis is true, the probability of obtaining a sample proportion as extreme as or more extreme than observed purely by random chance."

(D) "The probability that we made an error."

(E) "The probability of getting an outlier."

Answer: (C). Standard statistical definition of p-value.

Part II: Free-Response Practice Problem

FRQ 6.1: Full PANIC Confidence Interval & Claim Evaluation

Problem: A consumer advocacy group surveys a random sample of 200 electric vehicle (EV) owners and finds that 156 are satisfied with their vehicle's battery range. (a) Construct and interpret a 95% confidence interval for the true proportion of all EV owners who are satisfied with battery range.
(b) An industry report claims that at least 85% of all EV owners are satisfied. Based on your confidence interval from part (a), what can you conclude about this claim?
Grader-Approved Score-4 Solution:

- **Part (a) [PANIC]:**
  Parameter: Let p = the true proportion of all EV owners satisfied with battery range.
  Assumptions/Conditions: 1. *Random:* Stated random sample of 200 owners. 2. *10% Condition:* $n = 200 \leq 0.10(\text{all EV owners})$ (hundreds of thousands of owners). 3. *Large Counts:* $n\hat{p} = 156 \geq 10$ and $n(1-\hat{p}) = 200 - 156 = 44 \geq 10$. Both ≥ 10.
  Name: One-sample z-interval for a population proportion p.
  Interval Calculation: $\hat{p} = 156/200 = 0.78$. For 95% confidence, $z^* = 1.96$.

$$\text{SE} = \sqrt{\frac{0.78(0.22)}{200}} = \sqrt{\frac{0.1716}{200}} \approx 0.02929$$

$$\text{ME} = 1.96(0.02929) \approx 0.0574$$

Interval $= 0.78 \pm 0.0574 = (\mathbf{0.7226}, \mathbf{0.8374})$ or **(72.26%, 83.74%)**.

Conclusion: "We are 95% confident that the true proportion of all EV owners satisfied with their battery range is between 72.26% and 83.74%."

- **Part (b):** Because the claimed value of 85% (0.85) lies entirely above and outside the 95% confidence interval of (0.7226, 0.8374), our interval provides convincing statistical evidence that the true satisfaction proportion is **strictly less than 85%**. The industry claim is not supported.

FRQ 6.2: Two-Proportion z-Interval & Error Analysis in Medicine

Problem: A pharmaceutical clinical trial compares a new allergy medication (Drug A) against a standard over-the-counter medication (Drug B). Two independent random samples of adult allergy sufferers are tested:

- Drug A: In an SRS of 250 patients, 215 report significant symptom relief.
- Drug B: In an SRS of 250 patients, 175 report significant symptom relief.

(a) Construct and interpret a 95% confidence interval for the difference in relief proportions ($p_A - p_B$).

(b) Based on your interval, is there convincing statistical evidence that Drug A provides a higher relief rate than Drug B?

(c) Suppose researchers conduct a formal hypothesis test of $H_0 : p_A = p_B$ vs. $H_a : p_A > p_B$. Describe a Type I error and a Type II error in the context of this study, and identify which error would be more serious for patient care.

Grader-Approved Score-4 Solution & Rubric:

- **Part (a):** $\hat{p}_A = \frac{215}{250} = 0.86$, $\hat{p}_B = \frac{175}{250} = 0.70$. $\hat{p}_A - \hat{p}_B = 0.86 - 0.70 = 0.16$.

  Conditions: Independent random samples; $250 \le 0.10N$ for both; all 4 counts $(215, 35, 175, 75)$ are ≥ 10.

  $\text{SE} = \sqrt{\frac{0.86(0.14)}{250} + \frac{0.70(0.30)}{250}} = \sqrt{0.0004816 + 0.000840} = \sqrt{0.0013216} \approx 0.03635$.

  For 95% confidence, $z^* = 1.960$.

  $\text{ME} = 1.960(0.03635) \approx 0.0712$.

  $\text{CI} = 0.16 \pm 0.0712 \implies (\mathbf{0.0888},\ \mathbf{0.2312})$.

  Interpretation: We are 95% confident that the true difference in the proportion of patients experiencing relief between Drug A and Drug B ($p_A - p_B$) is between 8.88% and 23.12%.

- **Part (b):** Yes. Because the entire 95% confidence interval is strictly positive and does not contain 0, there is convincing evidence at $\alpha = 0.05$ that Drug A has a higher relief rate than Drug B.

- **Part (c):** *Type I Error:* Concluding Drug A is more effective when in reality Drug A and Drug B have identical efficacy. (Consequence: Patients pay more for a new drug that offers no actual clinical advantage).

Type II Error: Failing to detect that Drug A is genuinely more effective. (Consequence: Patients miss out on a superior therapeutic treatment).

Unit 6 Top 5 AP Exam Pitfalls to Avoid

1. **Using Pooled $\hat{p}_c$ for Confidence Intervals:** Pooled proportions are used ONLY for hypothesis tests ($H_0 : p_1 = p_2$), NEVER for confidence intervals!

2. **Hypothesizing with Statistics:** Writing $H_0 : \hat{p}_1 = \hat{p}_2$ loses credit instantly. Always use parameters p_1 and p_2.

3. **Confusing Margin of Error with Standard Error:** $\text{ME} = z^* \times \text{SE}$. Standard error does not include the critical value.

4. **Claiming p-value is the Probability H_0 is True:** The p-value is $P(\text{data as extreme} \mid H_0 \text{ is true})$, NOT $P(H_0 \text{ is true})$.

5. **Power Confusion:** Power increases when sample size n increases, α increases, or the true parameter is further from the null value.

6.6 Unit 6 Master Summary

Unit 6 Essential Review Checklist

- **PANIC for Intervals:** Parameter, Assumptions, Name, Interval, Conclusion.
- **PHANTOM for Tests:** Parameter, Hypotheses, Assumptions, Name, Test Stat, Obtain p-value, Make decision.
- **Conditions for Proportions:** Random, 10% ($n \leq 0.10N$), Large Counts ($np \geq 10, n(1-p) \geq 10$).
- **Decision Rule:** If p-value $\leq \alpha \implies$ Reject H_0; If p-value $> \alpha \implies$ Fail to reject H_0.
- **Errors:** Type I = False Alarm (α); Type II = Miss (β); Power $= 1 - \beta$.

Chapter 7

Unit 7: Inference for Quantitative Data: Means

7.1 Unit Overview & Exam Weight

Unit 7 accounts for **10% to 18%** of the AP Statistics Exam. Whenever dealing with quantitative variables where the true population standard deviation σ is unknown (which is practically 100% of real-world scenarios), we use the t**-distribution** discovered by William Sealy Gosset (writing under the pseudonym "Student"). You will master 1-sample t-procedures, paired t-procedures, and 2-sample t-procedures.

Key Unit Vocabulary

Student's t-Distribution:

A family of bell-shaped, symmetric density curves defined by degrees of freedom (df). It has heavier tails and more probability in the extremes than the standard normal distribution to account for the extra uncertainty of estimating σ with s_x. As $df \to \infty$, the t-distribution approaches $N(0, 1)$.

Degrees of Freedom (df):

For a 1-sample t-distribution, $df = n - 1$.

Standard Error of the Mean:

$\text{SE}(\bar{x}) = \frac{s_x}{\sqrt{n}}$.

1-Sample t-Interval:

An interval estimating an unknown population mean μ: $\bar{x} \pm t^* \frac{s_x}{\sqrt{n}}$.

1-Sample t-Test:

A hypothesis test evaluating claims about a population mean μ: $t = \frac{\bar{x} - \mu_0}{s_x/\sqrt{n}}$.

Paired Data (Matched Pairs):

Quantitative data resulting from paired observations on the same individuals (e.g., pre-test vs. post-test) or pairs of similar individuals. Analyzed by taking differences ($d = x_1 - x_2$) and running 1-sample t-procedures on the differences ($\mu_d, \bar{x}_d, s_d$).

Two-Sample t-Procedures:

Inference comparing two independent population means $\mu_1 - \mu_2$. Never pool variances for means!

Robustness:

An inference procedure is robust if the stated confidence level or p-value remains approximately accurate even when a condition (such as normality) is moderately violated. The t-procedures are robust against non-normality when $n \geq 30$.

7.2 Core Concept Lessons

Lesson 7.1: The t-Distribution vs. Standard Normal (z)

Why can we NOT use z when working with sample means? Because the true population standard deviation σ is almost never known! When we replace σ with the sample standard deviation s_x, we introduce additional sampling variability. To compensate for this extra variation, the t-distribution spreads out wider, with fatter tails.

- Bell-shaped and symmetric about 0.
- Area under curve = 1.
- Controlled by degrees of freedom: $df = n - 1$.
- As df increases, s_x becomes a more precise estimate of σ, and the t-distribution converges to the standard normal Z curve!

Lesson 7.2: One-Sample t-Interval for a Population Mean μ

Confidence Interval Formula:

$$\bar{x} \pm t^* \left(\frac{s_x}{\sqrt{n}} \right) \tag{7.1}$$

where t^* is the critical value with $df = n - 1$. **Mandatory Conditions Check:**

1. **Random:** Data comes from an SRS or randomized experiment.

2. **10% Condition:** $n \leq 0.10N$ when sampling without replacement.

3. **Normal / Large Sample Condition:**

   - If the population is stated to be **Normal**, the condition is met.
   - If $n \geq 30$, the Central Limit Theorem guarantees the sampling distribution is approximately normal.
   - If $n < 30$, you MUST graph the sample data (dotplot, stemplot, or boxplot) and confirm there are **no strong skewness or outliers**!

Lesson 7.3: Paired t-Procedures (Matched Pairs Design)

Critical Distinction: When two sets of quantitative measurements are collected on the **same individuals** (e.g., before-treatment and after-treatment) or matched pairs, they are NOT independent!

- Compute individual differences for each pair: $d_i = x_{1i} - x_{2i}$.
- Treat the list of differences as a **single sample** of size n with sample mean difference $\bar{x}_d$ and sample standard deviation of differences s_d.
- Hypotheses: $H_0 : \mu_d = 0$ versus $H_a : \mu_d > 0, \mu_d < 0$, or $\mu_d \neq 0$.

- Test Statistic:

$$t = \frac{\bar{x}_d - 0}{s_d/\sqrt{n}}, \qquad df = n - 1 \tag{7.2}$$

- Confidence Interval: $\bar{x}_d \pm t^* \frac{s_d}{\sqrt{n}}$.

Lesson 7.4: Two-Sample t-Procedures ($\mu_1 - \mu_2$)

Comparing the means of two independent groups:

- **Confidence Interval:**

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \tag{7.3}$$

- **Test Statistic ($H_0 : \mu_1 - \mu_2 = 0$):**

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \tag{7.4}$$

- **Degrees of Freedom:** Use technology (Welch-Satterthwaite approximation) or the conservative formula: $df = \min(n_1 - 1, n_2 - 1)$.

- **Important Rule: NEVER POOL** variances for two-sample t-procedures on the AP exam! Always select `Pooled: NO` on your TI-84.

7.3 Worked Step-by-Step Examples

Example 7.1: One-Sample t-Test for a Mean (PHANTOM)

Problem: A consumer agency tests whether a brand of AAA batteries lasts less than the advertised mean lifetime of 30 hours. A random sample of $n = 16$ batteries has a mean lifetime of $\bar{x} = 28.5$ hours with a standard deviation of $s = 2.4$ hours. A dotplot of the 16 values shows no outliers or strong skewness. Conduct a test at $\alpha = 0.05$.

Solution:

- **P:** Let $\mu =$ the true mean lifetime of all AAA batteries of this brand
- **H:** $H_0 : \mu = 30$ versus $H_a : \mu < 30$.
- **A:** 1. *Random:* Stated random sample of 16 batteries. 2. *10% Condition:* $n = 16 \leq 0.10(\text{all batteries produced})$. 3. *Normal/Large Sample:* Although $n = 16 < 30$, the problem states the dotplot of sample data shows no outliers or strong skewness. Therefore, t-procedures are safe to use.
- **N:** One-sample t-test for a population mean μ.
- **T:** $df = n - 1 = 16 - 1 = 15$.

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{28.5 - 30}{2.4/\sqrt{16}} = \frac{-1.5}{0.60} = -2.50$$

- **O:** p-value $= P(t_{15} \leq -2.50) = \mathbf{0.0123}$.
- **M:** Because the p-value (0.0123) is less than $\alpha = 0.05$, we reject H_0. There is convincing statistical evidence that the true mean lifetime of this battery brand is strictly less than 30 hours.

7.4 TI-84 Plus CE Calculator Playbook

TI-84 Playbook: Inference for Means

- **1-Sample t-Test:** Press `[STAT]` → `TESTS` → `2:T-Test`. Select `Inpt: Stats` (or `Data` if raw list in `L1`). Enter $\mu_0, \bar{x}, s_x, n$, and direction of H_a.
- **1-Sample t-Interval:** Press `[STAT]` → `TESTS` → `8:TInterval`. Enter stats, C-Level.
- **Paired t-Test:** Enter before-data in `L1`, after-data in `L2`. In `L3`, define `L1 - L2`. Run `2:T-Test` using `Data: L3` with $\mu_0 = 0$.
- **2-Sample t-Test:** Press `[STAT]` → `TESTS` → `4:2-SampTTest`. Always set `Pooled: NO`!
- **2-Sample t-Interval:** Press `[STAT]` → `TESTS` → `0:2-SampTInt`. Always set `Pooled: NO`!

7.5 Comprehensive Practice Problem Set

Part I: 20 Multiple-Choice Questions

1. Why do we use the t-distribution instead of the standard normal distribution when constructing a confidence interval for a population mean?

   (A) Because the sample size is always small.

   (B) Because the population standard deviation σ is unknown and estimated by s.

(C) Because the population is always non-normal.

(D) Because the mean is non-resistant.

(E) To avoid calculating degrees of freedom.

Answer: (B). Estimating σ with s introduces additional variability, requiring t.

2. As degrees of freedom increase, the t-distribution:

(A) Becomes more skewed.

(B) Approaches the standard normal distribution Z.

(C) Becomes wider with fatter tails.

(D) Has a mean greater than 0.

(E) Has a standard deviation of 0.

Answer: (B). As $df \to \infty$, $t \to Z$.

3. A random sample of $n = 25$ students has $\bar{x} = 75, s = 10$. For a 95% confidence interval for μ, what is df and t^*?

(A) $df = 25, t^* = 1.960$

(B) $df = 24, t^* = 2.064$

(C) $df = 24, t^* = 1.960$

(D) $df = 25, t^* = 2.060$

(E) $df = 23, t^* = 2.500$

Answer: (B). $df = n-1 = 24$. From the t-table, $t^* = 2.064$.

4. A matched pairs study measures cholesterol before and after a diet program for 20 patients. What degrees of freedom should be used for the hypothesis test?

(A) 38

(B) 39

(C) 19

(D) 20

(E) 18

Answer: (C). Paired data is analyzed as a single sample of $n = 20$ differences $\implies df = 20 - 1 = 19$.

5. If $n = 10$ and the sample data displays a severe outlier, is it appropriate to conduct a 1-sample t-test?

   (A) Yes, because t-procedures are always robust.

   (B) No, because for $n < 30$, the presence of an outlier violates the normality condition.

   (C) Yes, if $\alpha = 0.01$.

   (D) Yes, by the Central Limit Theorem.

   (E) No, because $10 < 30$ is never allowed.

   Answer: (B). For small samples ($n < 30$), outliers invalidate t-procedures.

6. A 95% confidence interval for $\mu_1 - \mu_2$ is $(-4.5, -1.2)$. What can be concluded?

   (A) $\mu_1 = \mu_2$.

   (B) There is convincing evidence that $\mu_1 < \mu_2$ because the entire interval is negative (strictly below 0).

   (C) There is no significant difference between the two means.

   (D) $\mu_1 > \mu_2$.

(E) The test was two-sided.

Answer: (B). All plausible values for $\mu_1 - \mu_2$ are negative $\implies \mu_1 < \mu_2$.

7. In a two-sample t-test, why do AP Statistics standards mandate "Never Pool"?

   (A) Pooling requires assuming the two population variances are equal ($\sigma_1^2 = \sigma_2^2$), an assumption rarely justified in practice.

   (B) Pooling decreases power.

   (C) Calculators cannot compute pooled tests.

   (D) Pooling is allowed only for proportions.

   (E) Pooling violates the 10% condition.

   Answer: (A). Unequal variance t-tests (Welch) are far more robust and realistic.

8. If a 90% confidence interval for μ is $(45.2, 54.8)$, what was the sample mean $\bar{x}$?

   (A) 45.2

   (B) 50.0

   (C) 54.8

   (D) 4.8

   (E) 9.6

   Answer: (B). Center of interval: $(45.2+54.8)/2 = 100/2 = 50.0$.

9. A test of $H_0 : \mu = 100$ vs. $H_a : \mu \neq 100$ produces $t = 2.45$ with $df = 15$. The p-value is:

   (A) $P(t_{15} > 2.45)$

   (B) $2 \times P(t_{15} > 2.45)$

   (C) $P(t_{15} < 2.45)$

   (D) 0.05

   (E) 0.0245

   Answer: (B). Two-sided alternative requires doubling the tail probability.

10. If the margin of error for a 1-sample t-interval is ME $= 3.5$, increasing the sample size will:

    (A) Increase the margin of error

    (B) Decrease the margin of error

    (C) Have no effect on ME

    (D) Increase the sample mean

    (E) Make t^* larger

    Answer: (B). Larger n decreases standard error $s/\sqrt{n}$, reducing ME.

11. Which of the following is an example of a matched pairs design?

    (A) Comparing test scores of 30 girls and 30 boys.

    (B) Comparing blood pressure measurements of 50 patients before and after taking a medication.

(C) Comparing corn yields from 10 plots in Texas and 10 plots in Iowa.

(D) Surveying 100 randomly chosen voters.

(E) Measuring weights of 40 dogs of different breeds.

Answer: (B). Pre- and post-treatment measurements on the same individual constitute matched pairs.

12. In a 1-sample t-test with $n = 20$, the test statistic is $t = -1.73$. For $H_a : \mu < \mu_0$, the p-value falls in which range?

(A) $p < 0.01$

(B) $0.025 < p < 0.05$

(C) $0.05 < p < 0.10$

(D) $p > 0.10$

(E) Exactly 0.05

Answer: (C). For $df = 19$, $t_{0.05} = 1.729 \implies p \approx 0.050$. Specifically, p-value $\approx 0.0499 \approx 0.05$.

13. If an investigator knows the true population standard deviation $\sigma = 15$, which procedure should be used to estimate μ?

(A) 1-sample t-interval

(B) 1-sample z-interval

(C) 2-sample t-test

(D) Chi-square test

(E) Regression test

Answer: (B). Knowing σ allows using z instead of t.

14. What is the conservative degrees of freedom for a two-sample t-test with $n_1 = 15$ and $n_2 = 22$?

 (A) 37

 (B) 35

 (C) 14

 (D) 21

 (E) 15

 Answer: (C). Conservative $df = \min(n_1 - 1, n_2 - 1) = \min(14, 21) = 14$.

15. When constructing a confidence interval for μ, the critical value t^* depends on:

 (A) Only the sample mean $\bar{x}$

 (B) The confidence level and degrees of freedom $(n - 1)$

 (C) The population size N

 (D) The margin of error

 (E) The p-value

 Answer: (B). t^* is determined exclusively by the desired confidence level and df.

16. If zero is contained within a 95% confidence interval for μ_d in a paired experiment, what decision would be made for $H_0 : \mu_d = 0$ at $\alpha = 0.05$?

 (A) Reject H_0

 (B) Fail to reject H_0

 (C) Accept H_a

(D) Increase sample size

(E) Conclude a Type I error was made

Answer: (B). An interval containing 0 means the hypothesized value 0 is plausible, so we fail to reject H_0.

17. In a sample of $n = 35$, a histogram shows slight right skewness. Can a t-test be conducted safely?

(A) No, data must be perfectly symmetric.

(B) Yes, because $n = 35 \geq 30$, the Central Limit Theorem ensures robustness.

(C) No, because s cannot be calculated.

(D) Yes, only if $\mu = 0$.

(E) No, z must be used instead.

Answer: (B). t-procedures are robust to moderate skewness when $n \geq 30$.

18. Standard error of the difference between two independent sample means is:

(A) $\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

(B) $\frac{s_1}{n_1} + \frac{s_2}{n_2}$

(C) $\sqrt{s_1^2 + s_2^2}$

(D) $\frac{s_1+s_2}{\sqrt{n_1+n_2}}$

(E) $s_1 - s_2$

Answer: (A). Standard formula for $\mathrm{SE}(\bar{x}_1 - \bar{x}_2)$.

19. If $n = 100, \bar{x} = 50, s = 10$, what is the margin of error for a 95% confidence interval for μ ($t^* \approx 1.984$)?

(A) 1.984

(B) 0.1984

(C) 19.84

(D) 1.00

(E) 0.50

Answer: (A). ME $= t^*(s/\sqrt{n}) = 1.984(10/\sqrt{100}) = 1.984(1.0) =$ 1.984.

20. Which of the following is true regarding t-distributions?

(A) They are skewed to the right.

(B) They have a larger variance than the standard normal distribution.

(C) They can take only positive values.

(D) They do not depend on sample size.

(E) They are used exclusively for categorical variables.

Answer: (B). Fatter tails mean larger variance (Var $=$ $df/(df-2) > 1$).

Part II: Free-Response Practice Problem

FRQ 7.1: Paired t-Test for Matched Pairs

Problem: A fitness coach tests whether a 6-week plyometric training program increases vertical jump height in basketball players. 12 players have their vertical jumps measured (in inches) before and after the program. The mean difference is $\bar{x}_d = \text{After} - \text{Before} = +2.4$ inches with a sample standard deviation of differences $s_d = 1.8$ inches. A boxplot of the differences shows no outliers. Conduct a hypothesis test at $\alpha = 0.05$.

Grader-Approved Score-4 Solution:

- **P:** Let μ_d = the true mean difference (After − Before) in vertical jump height for all players.

- **H:** $H_0 : \mu_d = 0$ versus $H_a : \mu_d > 0$.

- **A:** 1. *Random:* Stated representative sample of players. 2. *10% Condition:* $n = 12 \leq 0.10$(all basketball players). 3. *Normal Condition:* $n = 12 < 30$, but the boxplot of differences displays no outliers or extreme skewness. Safe to proceed with t-distribution.

- **N:** Paired t-test for a mean difference μ_d.

- **T:** $df = 12 - 1 = 11$.

$$t = \frac{\bar{x}_d - 0}{s_d/\sqrt{n}} = \frac{2.4 - 0}{1.8/\sqrt{12}} = \frac{2.4}{0.5196} \approx 4.62$$

- **O:** p-value $= P(t_{11} \geq 4.62) = \mathbf{0.00036}$.

- **M:** Because the p-value (0.00036) is strictly less than $\alpha = 0.05$, we reject H_0. There is convincing statistical evidence that the 6-week training program produces a true mean increase in vertical jump height.

FRQ 7.2: Paired t-Test for Matched Pairs vs. Two-Sample t-Test

Problem: An educational psychologist evaluates a spatial reasoning software module. Twelve high school seniors take a standardized 50-point spatial visualization test before and after completing a 4-week computer course. The summary statistics of the test score differences ($d = \text{Post} - \text{Pre}$) are:

$$n = 12, \quad \bar{x}_d = 6.50 \text{ points}, \quad s_d = 2.40 \text{ points}$$

A normal probability plot of the differences displays a strong linear pattern with no outliers.

(a) Explain why a paired t-test is appropriate for this data rather than a two-sample independent t-test.

(b) State the appropriate null and alternative hypotheses to test whether the software course significantly improves spatial visualization scores.

(c) Check all required conditions for conducting the paired t-test.

(d) Calculate the test statistic and p-value, and state your conclusion at the $\alpha = 0.01$ significance level.

Grader-Approved Score-4 Solution & Rubric:

- **Part (a):** A paired t-test is appropriate because the data consist of matched pairs of measurements taken on the exact same subjects (pre-test and post-test scores for each student). The two sets of scores are inherently dependent, not two independent random samples.

- **Part (b):** Let μ_d be the true mean difference in spatial visualization scores (Post − Pre) for all students who complete the course.
  $H_0 : \mu_d = 0 \quad \text{vs.} \quad H_a : \mu_d > 0.$

- **Part (c):** *Random:* The 12 students are treated as representative of the target student population.
  10% Condition: $n = 12 \leq 0.10$(All high school seniors).
  Normal/Large Sample: $n = 12 < 30$, but the normal probability plot of differences is linear with no outliers, making the normality assumption plausible.

- **Part (d):** $\text{SE}(\bar{x}_d) = \frac{s_d}{\sqrt{n}} = \frac{2.40}{\sqrt{12}} \approx \frac{2.40}{3.4641} \approx 0.6928$ points.
  $t = \frac{\bar{x}_d - 0}{\text{SE}(\bar{x}_d)} = \frac{6.50}{0.6928} \approx 9.38.$
  Degrees of freedom: $df = n - 1 = 12 - 1 = 11.$
  p-value: $P(t_{11} \geq 9.38) < 0.00001.$
  Conclusion: Because p-value < 0.01, we reject H_0. There is overwhelming statistical evidence that the spatial visualization software module significantly improves student test scores.

Unit 7 Top 5 AP Exam Pitfalls to Avoid

1. **Running a Two-Sample t-Test on Paired Data:** If subjects are matched or measured twice, you MUST run a paired t-test on differences ($df = n - 1$), NOT a two-sample test!

2. **Selecting Pooled on the TI-84:** On the AP Statistics Exam, NEVER pool variances for a two-sample t-test. Always select `Pooled: NO`!

3. **Degrees of Freedom for Two-Sample t:** Conservative $df = \min(n_1 - 1, n_2 - 1)$ is always accepted if TI-84 Satterthwaite df is not used.

4. **Checking Normality When $n < 30$:** When $n < 30$, you MUST check the plot of sample data for outliers and severe skewness.

5. **Interpreting p-Value as Strength of Effect:** A tiny p-value means strong evidence against H_0; it does NOT mean the effect size is large!

7.6 Unit 7 Master Summary

Unit 7 Essential Review Checklist

- **1-Sample t:** $\bar{x} \pm t^*(s/\sqrt{n})$, $df = n - 1$.
- **Normality Check for Means:** Normal pop OR $n \geq 30$ (CLT) OR graph with no strong skewness/outliers.
- **Paired t:** Run 1-sample t on the list of differences ($d = x_1 - x_2$).
- **2-Sample t:** Independent groups. Never pool variances!

Chapter 8

Unit 8: Inference for Categorical Data: Chi-Square Tests

8.1 Unit Overview & AP Exam Weight

Unit 8 accounts for **2% to 5%** of the AP Statistics Exam multiple-choice section and appears on approximately **50% of all released Free-Response Questions**. While covering fewer overall CED topics than Units 6 or 7, mastering the distinctions among the three Chi-Square tests is essential for securing a Score 5.

Key Vocabulary Terms

Chi-Square (χ^2) Distribution:

A continuous, strictly positive probability distribution that is skewed to the right, parameterized entirely by its degrees of freedom (df). As degrees of freedom increase, the distribution shifts rightward and becomes progressively more symmetric.

Observed Count (O):

The actual raw count of individuals or outcomes observed in each category from the sample data.

Expected Count (E):

The theoretical frequency calculated under the mathematical assumption that the null hypothesis H_0 is true in the underlying population.

Chi-Square Goodness-of-Fit Test:

Evaluates whether a single sample of categorical data fits a hypothesized population distribution across k categories ($df = k - 1$).

Chi-Square Test for Homogeneity:

Evaluates whether the distribution of a single categorical variable is identical across two or more independent populations or treatment groups ($df = (r - 1)(c - 1)$).

Chi-Square Test for Independence:

Evaluates whether two categorical variables measured on a single random sample from one population are statistically independent ($df = (r - 1)(c - 1)$).

Two-Way Contingency Table:

A rectangular data grid displaying categorical frequencies classified simultaneously by row and column variables.

Large Counts Condition (χ^2):

The requirement that **all expected cell counts must be at least 5** ($E_i \geq 5$) to ensure the discrete multinomial distribution is adequately approximated by the continuous Chi-Square curve.

8.2 Core Concept Lessons

Lesson 8.1: Distinguishing the Three Chi-Square Tests

AP graders rigorously penalize students who misidentify the specific Chi-Square test being conducted. Use the following decision matrix:

Test Name	Variables	Samples / Populations	Degrees of Freedom
Goodness-of-Fit (GOF)	1 Categorical	1 Sample	$df = k - 1$ (k = number of categories)
Homogeneity	1 Categorical	2+ Independent Samples	$df = (r - 1)(c - 1)$
Independence / Association	2 Categorical	1 Single Random Sample	$df = (r - 1)(c - 1)$

Key Grader Clue for Homogeneity vs. Independence: Look closely at the data collection protocol described in the problem stem:

- If the stem states: *"Independent random samples of 100 men and 100 women were surveyed..."* → **Homogeneity** (2 distinct populations).
- If the stem states: *"A single random sample of 200 adults was selected and each person was asked their gender and preference..."* → **Independence** (1 population, 2 variables recorded per subject).

Lesson 8.2: Formulating Null and Alternative Hypotheses

- **Goodness-of-Fit:**

H_0 : The distribution of [categorical variable] matches the specified

H_a : At least one of the true categorical proportions differs from the

- **Homogeneity:**

H_0 : The distribution of [categorical variable] is the same across all

H_a : The distribution of [categorical variable] is not the same across

- **Independence:**

H_0 : There is no association between [Variable 1] and [Variable 2] in

H_a : There is an association between [Variable 1] and [Variable 2] in

Lesson 8.3: Calculating Expected Counts & Test Statistic

- **Goodness-of-Fit Expected Counts:**

$$E_i = n \times p_i \tag{8.1}$$

where n is the sample size and p_i is the hypothesized proportion for category i.

- **Two-Way Table Expected Counts (Homogeneity & Independence):**

$$E_{ij} = \frac{(\text{Row } i \text{ Total}) \times (\text{Column } j \text{ Total})}{\text{Table Grand Total } n} \tag{8.2}$$

- **Chi-Square Test Statistic Formula:**

$$\chi^2 = \sum_{\text{all cells}} \frac{(O-E)^2}{E} = \frac{(O_1-E_1)^2}{E_1} + \frac{(O_2-E_2)^2}{E_2} + \cdots + \frac{(O_m-E_m)^2}{E_m} \tag{8.3}$$

Each individual term $\frac{(O-E)^2}{E}$ is called a **cell component**. The cell with the largest component contributed the greatest discrepancy from the null model.

Exam Trap: Expected vs. Observed Counts Condition

The Large Counts condition for Chi-Square tests requires that **all expected cell counts must be at least 5** ($E_i \geq 5$).

- AP students frequently and erroneously examine the **observed counts** O instead of the expected counts E.
- An observed cell count can legally be 0, 1, 2, or 3 without violating the condition!
- On an FRQ, you must **explicitly write down the matrix or table of expected counts** to earn full credit ("Essentially Correct") for checking the Large Counts condition.

8.3 Worked Step-by-Step Examples

Example 8.1: Chi-Square Goodness-of-Fit Test

Problem: A confectionery manufacturer claims that its candy bags contain the following color distribution: 30% Brown, 20% Yellow, 20% Red, 10% Orange, 10% Green, and 10% Blue. A student purchases a random 200-piece candy bag and counts: 70 Brown, 35 Yellow, 45 Red, 15 Orange, 15 Green, and 20 Blue. Carry out a goodness-of-fit test at $\alpha = 0.05$.

Solution:

- **Step 1 (Hypotheses):** $H_0 : p_{\text{Brown}} = 0.30, p_{\text{Yellow}} = 0.20, p_{\text{Red}} = 0.20, p_{\text{Orange}} = 0.10, p_{\text{Green}} = 0.10, p_{\text{Blue}} = 0.10.$
  H_a : At least one of the true color proportions differs from the claimed distribution.

- **Step 2 (Conditions):** 1. *Random:* Stated as a random candy bag from the manufacturer. (Met)
  2. *10% Condition:* $n = 200 \leq 0.10$(Total candies produced by compar
  (Met)
  3. *Large Counts:* Compute expected counts $E_i = 200 \times p_i$: $E_{\text{Brown}} = 200(0.30) = 60$, $E_{\text{Yellow}} = 200(0.20) = 40$, $E_{\text{Red}} = 200(0.20) = 40$, $E_{\text{Orange}} = 200(0.10) = 20$, $E_{\text{Green}} = 200(0.10) = 20$, $E_{\text{Blue}} = 200(0.10) = 20$. All expected counts are ≥ 5 (minimum is 20). (Met)

- **Step 3 (Mechanics):**

  $$\chi^2 = \frac{(70-60)^2}{60} + \frac{(35-40)^2}{40} + \frac{(45-40)^2}{40} + \frac{(15-20)^2}{20} + \frac{(15-20)^2}{20} + \frac{(2$$

  $$\chi^2 = \frac{100}{60} + \frac{25}{40} + \frac{25}{40} + \frac{25}{20} + \frac{25}{20} + 0 = 1.667 + 0.625 + 0.625 + 1.250 + 1.250 + 0$$

  Degrees of freedom: $df = k - 1 = 6 - 1 = 5$.
  p-value: Using $\chi^2\text{cdf}(5.417, 10^{99}, 5) \approx 0.3671$.

- **Step 4 (Conclusion):** Because the p-value (0.3671) is greater than $\alpha = 0.05$, we fail to reject H_0. There is insufficient statistical evidence to conclude that the candy color distribution differs from the manufacturer's claimed proportions.

Example 8.2: Chi-Square Test for Independence

Problem: A market research firm surveys an SRS of 400 vehicle owners to examine whether vehicle body style (Sedan, SUV, Truck) is independent of buyer age bracket (< 35, 35–55, > 55). The sample counts are:

Age Group	**Sedan**	**SUV**	**Truck**	**Total**
< 35	60	50	30	140
35–55	50	80	30	160
> 55	50	30	20	100
Total	160	160	80	400

(a) Find the expected number of buyers aged < 35 who purchase an SUV under H_0.
(b) Calculate the Chi-Square test statistic, degrees of freedom, and state the conclusion at $\alpha = 0.01$.

Solution:

- **Part (a):** $E_{12} = \frac{\text{(Row 1 Total)}\times\text{(Col 2 Total)}}{\text{Grand Total}} = \frac{140\times160}{400} = \frac{22{,}400}{400} = \mathbf{56.0}$ buyers.

- **Part (b):** Expected counts matrix:

Age	**Sedan**	**SUV**	**Truck**
< 35	$140(160)/400 = 56$	$140(160)/400 = 56$	$140(80)/400 = 28$
35–55	$160(160)/400 = 64$	$160(160)/400 = 64$	$160(80)/400 = 32$
> 55	$100(160)/400 = 40$	$100(160)/400 = 40$	$100(80)/400 = 20$

All expected counts are ≥ 20, satisfying the Large Counts condition.

$$\chi^2 = \sum \frac{(O-E)^2}{E} = \frac{(60-56)^2}{56} + \frac{(50-56)^2}{56} + \frac{(30-28)^2}{28} + \frac{(50-64)^2}{64}$$

$$+ \frac{(30-32)^2}{32} + \frac{(50-40)^2}{40} + \frac{(30-40)^2}{40} + \frac{(20-20)^2}{20}$$

$$\chi^2 = 0.286 + 0.643 + 0.143 + 3.063 + 4.000 + 0.125 + 2.500 + 2.500 + 0 = \mathbf{13.2}$$

Degrees of freedom: $df = (r-1)(c-1) = (3-1)(3-1) = 2 \times 2 = \mathbf{4}$.

p-value: $\chi^2\text{cdf}(13.26, 10^{99}, 4) \approx \mathbf{0.0101}$.

At $\alpha = 0.01$, since p-value $= 0.0101 > 0.01$, we fail to reject H_0 (barely). There is insufficient evidence at the 1% level of an association between age group and vehicle preference.

8.4 TI-84 CE Calculator Procedures for Chi-Square

TI-84 Playbook: Chi-Square Tests

- **Chi-Square Goodness-of-Fit Test (χ^2`GOF-Test`):** 1. Press `[STAT]` → `1:Edit...`. Enter observed counts into `L1` and expected counts into `L2`.
  2. Press `[STAT]` → `TESTS` → `D:`χ^2`GOF-Test`.
  3. Enter `Observed: L1`, `Expected: L2`, `df: k-1`.
  4. Select `Calculate`. Displays χ^2, p-value, df, and the individual cell contributions in list `CNTRB`.

- **Two-Way Table Test (χ^2`-Test`):** 1. Press `[2nd]` `[x⁻¹]` (MATRIX) → Arrow to `EDIT` → Select `[A]`.
  2. Enter table dimensions ($r \times c$) and input raw observed frequencies.

3. Press `[STAT]` $\rightarrow$ `TESTS` $\rightarrow$ `C:`χ^2`-Test`.
4. Specify `Observed: [A]`, `Expected: [B]` $\rightarrow$ Select `Calculate`.
5. The calculator automatically calculates all expected counts and saves them into matrix `[B]`. Press `[2nd]` `[MATRIX]` $\rightarrow$ `[B]` to view and verify $E \geq 5$!

8.5 Comprehensive Practice Problem Set

Part I: 20 Multiple-Choice Questions with Explanations

1. A standard 6-sided die is rolled 120 times to test if it is fair. What are the degrees of freedom for the goodness-of-fit test?

   (A) 5

   (B) 6

   (C) 119

   (D) 120

   (E) 1

   Answer: (A). For a goodness-of-fit test, $df = k - 1$. With $k = 6$ possible faces, $df = 6 - 1 = 5$.

2. In a Chi-Square test of independence with a 3×4 contingency table, the degrees of freedom are:

   (A) 12

   (B) 11

   (C) 7

(D) 6

(E) 5

Answer: (D). $df = (r-1)(c-1) = (3-1)(4-1) = 2 \times 3 = 6$.

3. What is the minimum expected count required for every cell in a valid Chi-Square test?

(A) 1

(B) 5

(C) 10

(D) 30

(E) 50

Answer: (B). The Large Counts condition requires all expected cell counts to be at least 5 ($E \geq 5$).

4. Under the null hypothesis in a Chi-Square test, the test statistic χ^2 measures:

(A) The total sample size

(B) The overall discrepancy between observed and expected frequencies

(C) The variance of the population

(D) The probability that H_0 is true

(E) The correlation coefficient

Answer: (B). $\chi^2 = \sum \frac{(O-E)^2}{E}$ measures normalized deviations between observed and expected counts.

5. An observed cell count in a contingency table is 3. What does this indicate about the validity of the test?

(A) The test is invalid because observed counts must be at least 5.

(B) The test is invalid only if sample size is small.

(C) The test may still be valid because the condition applies to expected counts, not observed counts.

(D) The degrees of freedom must be reduced.

(E) A t-test must be used instead.

Answer: (C). The Large Counts condition applies strictly to expected counts ($E_i \geq 5$). Observed counts can be less than 5.

6. Which of the following describes the shape of a Chi-Square distribution with small degrees of freedom ($df = 3$)?

(A) Perfectly symmetric and bell-shaped

(B) Uniform

(C) Strongly skewed to the right

(D) Strongly skewed to the left

(E) Bimodal

Answer: (C). Chi-Square distributions are bounded below by 0 and are strongly right-skewed for small df.

7. A test of homogeneity evaluates whether:

(A) Two quantitative variables have zero slope

(B) Two independent samples have identical categorical distributions

(C) A single sample fits a normal curve

(D) Two means are equal

(E) Variances are pooled

Answer: (B). Homogeneity tests whether a single categorical distribution is identical across two or more separate populations.

8. In a 2×2 table, the row 1 total is 80, column 1 total is 50, and table grand total is 200. The expected count for cell $(1, 1)$ is:

(A) 20

(B) 40

(C) 50

(D) 80

(E) 130

Answer: (A). $E = \frac{80 \times 50}{200} = \frac{4000}{200} = 20$.

9. If the calculated $\chi^2 = 0$, what does this indicate?

(A) p-value is 0

(B) Observed counts perfectly match expected counts in every cell

(C) A calculation error occurred

(D) Degrees of freedom equal 0

(E) H_0 is rejected

Answer: (B). $\chi^2 = 0$ occurs if and only if $O_i = E_i$ for every single cell.

10. All Chi-Square tests are:

(A) Two-sided

(B) One-sided (upper-tailed)

(C) One-sided (lower-tailed)

(D) Directional tests

(E) Non-parametric t-tests

Answer: (B). Because deviations are squared, discrepancies make χ^2 larger, placing the rejection region exclusively in the upper right tail.

11. When the degrees of freedom of a Chi-Square distribution increase, the distribution:

(A) Becomes more skewed

(B) Becomes more bell-shaped and symmetric

(C) Shifts toward zero

(D) Becomes discrete

(E) Narrower with range 0 to 1

Answer: (B). As $df \to \infty$, the Chi-Square distribution approaches a normal distribution.

12. If a Chi-Square test yields $\chi^2 = 12.5$ with $df = 4$, and the critical value at $\alpha = 0.05$ is $\chi^* = 9.49$, the decision is:

(A) Fail to reject H_0

(B) Reject H_0

(C) Accept H_a as absolute proof

(D) Recalculate with $df = 3$

(E) Increase α

Answer: (B). Because the test statistic $\chi^2 = 12.5 > 9.49$, the result falls into the rejection region (p-value < 0.05).

13. A researcher selects 50 freshmen, 50 sophomores, 50 juniors, and 50 seniors, and asks their preferred dining option. Which test is appropriate?

    (A) Chi-Square Goodness-of-Fit

    (B) Chi-Square Test for Homogeneity

    (C) Chi-Square Test for Independence

    (D) Matched Pairs t-test

    (E) Two-sample z-test for proportions

    Answer: (B). Four separate, independent samples (by class standing) are compared on one categorical variable (dining choice).

14. A single SRS of 300 employees is surveyed regarding department and job satisfaction (Satisfied, Neutral, Dissatisfied). Which test is appropriate?

    (A) Goodness-of-Fit

    (B) Homogeneity

    (C) Independence

    (D) Two-sample t-test

    (E) Linear regression

    Answer: (C). One single sample categorized by two variables (department and satisfaction) calls for a test of independence.

15. What is the mean of a Chi-Square distribution with $df = 8$?

    (A) 0

    (B) 4

    (C) 8

    (D) 16

    (E) 64

    Answer: (C). The theoretical mean of any Chi-Square distribution equals its degrees of freedom ($\mu = df = 8$).

16. In a Goodness-of-Fit test with 4 categories, the observed counts are $25, 25, 25, 25$ and hypothesized proportions are equal. What is χ^2?

    (A) 0

    (B) 1

    (C) 4

    (D) 25

    (E) 100

    Answer: (A). Total $n = 100$. Expected count for each category is $100(0.25) = 25$. Since $O_i = E_i$ for all cells, $\chi^2 = 0$.

17. Which cell contributes most to the Chi-Square statistic?

    (A) Cell with largest expected count

    (B) Cell with largest observed count

    (C) Cell with largest individual component $(O - E)^2/E$

    (D) Cell with $O = E$

(E) The bottom-right total cell

Answer: (C). The component $(O - E)^2/E$ measures each individual cell's contribution to the total test statistic.

18. If an expected count in a 3×3 table is 3.8, what can the researcher do to satisfy conditions?

(A) Combine adjacent categories if meaningful to increase expected counts

(B) Discard the row

(C) Change the significance level

(D) Double the observed count

(E) Run a normal approximation

Answer: (A). Combining logically related categorical levels is a standard statistical remedy to ensure all $E_i \geq 5$.

19. Degrees of freedom for a Goodness-of-Fit test with 10 categories is:

(A) 10

(B) 9

(C) 8

(D) 1

(E) 100

Answer: (B). $df = k - 1 = 10 - 1 = 9$.

20. What happens to the critical value χ^* as the significance level α decreases from 0.05 to 0.01 for fixed df?

(A) Decreases

(B) Increases

(C) Remains constant

(D) Becomes negative

(E) Reaches 0

Answer: (B). A smaller α pushes the rejection threshold further into the right tail, increasing the critical value.

Part II: Score-4 Free-Response Problem with Complete Rubric

FRQ 8.1: Chi-Square Test for Homogeneity Across Age Brackets

Problem: An airline carrier surveys independent random samples of passengers across three route types (Domestic Short-Haul, Domestic Transcontinental, International) regarding on-board Wi-Fi satisfaction. The observed frequencies are:

Route Type	Satisfied	Neutral	Dissatisfied	Total
Short-Haul	65	45	40	150
Transcontinental	80	40	30	150
International	95	35	20	150
Total	240	120	90	450

Perform an appropriate hypothesis test at $\alpha = 0.05$ to determine if the satisfaction distribution differs across route types.

Grader-Approved Score-4 Solution:

Step 1: Hypotheses

H_0 : The distribution of Wi-Fi satisfaction is the same across all three route types.

H_a : The distribution of Wi-Fi satisfaction is not the same across all three route types.

Step 2: Conditions

- *Random:* Three independent random samples of passengers were surveyed. (Met)
- *10% Condition:* The 150 passengers in each sample represent less than 10% of all passengers flying each route type. (Met)
- *Large Counts:* Compute expected counts $E = \frac{\text{Row Total}\times\text{Col Total}}{450}$. Because all three row totals equal 150: $E_{\text{Satisfied}} = \frac{150\times240}{450} = 80$, $E_{\text{Neutral}} = \frac{150\times120}{450} = 40$, $E_{\text{Dissatisfied}} = \frac{150\times90}{450} = 30$. All expected counts are ≥ 30, easily exceeding 5. (Met)

Step 3: Mechanics

$$\chi^2 = \sum \frac{(O-E)^2}{E} = \frac{(65-80)^2}{80} + \frac{(45-40)^2}{40} + \frac{(40-30)^2}{30}$$

$$+\frac{(80-80)^2}{80}+\frac{(40-40)^2}{40}+\frac{(30-30)^2}{30}+\frac{(95-80)^2}{80}+\frac{(35-40)^2}{40}+\frac{(20-3}{30}$$

$$\chi^2 = \frac{225}{80} + \frac{25}{40} + \frac{100}{30} + 0 + 0 + 0 + \frac{225}{80} + \frac{25}{40} + \frac{100}{30}$$

$\chi^2 = 2.8125+0.625+3.3333+0+0+0+2.8125+0.625+3.3333 = \mathbf{13.54}$

$df = (3-1)(3-1) = 2 \times 2 = 4.$

p-value: $P(\chi^2_4 \geq 13.54) = 0.0089.$

Step 4: Conclusion

Because the p-value (0.0089) is less than $\alpha = 0.05$, we reject H_0. There is convincing statistical evidence that the distribution of Wi-Fi satisfaction differs significantly across the three airline route types.

8.6 Unit 8 Master Summary

Unit 8 Essential Review Checklist

- **Test Types:** GOF (1 sample, 1 var), Homogeneity (2+ samples, 1 var), Independence (1 sample, 2 vars).
- **Degrees of Freedom:** $df = k - 1$ for GOF; $df = (r - 1)(c - 1)$ for two-way tables.
- **Expected Counts:** Must show matrix or formula; ALL $E \geq 5$ (never check observed counts!).
- **Rejection Region:** Always upper right tail (p-value $= P(\chi^2 \geq \text{stat})$).

Chapter 9

Unit 9: Inference for Quantitative Data: Slopes

9.1 Unit Overview & AP Exam Weight

Unit 9 accounts for **2% to 5%** of the multiple-choice section of the AP Statistics Exam and appears frequently on Section II Free-Response Questions (often as part of Question 1 or Question 6 Investigative Task). It integrates two-variable regression analysis from Unit 2 with formal inference concepts (t-intervals and t-tests).

Key Vocabulary Terms

Population Regression Line:
$\mu_y = \alpha + \beta x$. Models the true mean response μ_y for any given value of explanatory variable x. Parameter β represents the true population slope.

Sample Regression Line (LSRL):
$\hat{y} = b_0 + b_1 x$. The estimated regression line calculated from

sample data, where b_1 is the point estimate of β and b_0 estimates α.

Standard Error of the Slope (SE(b_1)):

The estimated standard deviation of the sampling distribution of sample slopes b_1:

$$\mathrm{SE}(b_1) = \frac{s}{s_x\sqrt{n-1}} \tag{9.1}$$

where s is the standard deviation of the residuals and s_x is the sample standard deviation of x.

Standard Deviation of Residuals (s):

Estimates the standard deviation σ of the response variable y about the true population regression line:

$$s = \sqrt{\frac{\sum(y_i - \hat{y}_i)^2}{n-2}} = \sqrt{\frac{\sum e_i^2}{n-2}} \tag{9.2}$$

Degrees of Freedom for Regression:

$df = n - 2$, reflecting the estimation of two parameters (α and β).

LINER Conditions:

The five mandatory assumptions for valid regression inference: Linear, Independent, Normal residuals, Equal variance of residuals, Random sample/assignment.

9.2 Core Concept Lessons

Lesson 9.1: The L-I-N-E-R Conditions for Regression Inference

On an AP exam FRQ, you must state and verify all 5 LINER conditions:

1. **L — Linear:** The true relationship between x and y in the population is linear.
   Verification: The scatterplot shows a roughly linear pattern, and the residual plot exhibits random scatter around the horizontal zero line with no curved pattern.

2. **I — Independent:** Individual observations are independent.
   Verification: When sampling without replacement, verify $n \leq 0.10N$. In an experiment, verify subjects were randomly assigned to treatments.

3. **N — Normal Residuals:** For any fixed x, responses y vary normally about the line.
   Verification: Construct a histogram, boxplot, or normal probability plot of the residuals. Confirm there is no extreme skewness and no severe outliers.

4. **E — Equal Variance (Homoscedasticity):** The standard deviation of y (σ) is constant across all values of x.
   Verification: In the residual plot, the vertical spread of the residuals is roughly uniform across the range of x values (no fan-shaped or funnel pattern).

5. **R — Random:** Data were collected using a well-designed random sample or randomized experiment. *Verification:* Quote the problem stem confirming an SRS or randomized experimental protocol.

Lesson 9.2: Confidence Interval for Slope β

A $C\%$ confidence interval for the true population slope β is:

$$b_1 \pm t^* \cdot \text{SE}(b_1) \tag{9.3}$$

where t^* is the critical value from the t-distribution with $df = n - 2$.

Score-4 Template: Interpreting a Confidence Interval for Slope

"We are [C%] confident that for each additional 1 [unit of x] increase in [explanatory variable x], the true population mean [response variable y] increases/decreases by an amount between [Lower Bound] and [Upper Bound] [units of y]."

Lesson 9.3: Hypothesis Test for Slope β ($H_0 : \beta = 0$)

To test whether there is a statistically significant linear association:

$H_0 : \beta = 0$ (There is no linear association between x and y in the po

$H_a : \beta \neq 0$ or $\beta > 0$ or $\beta < 0$

The test statistic is:

$$t = \frac{b_1 - 0}{\text{SE}(b_1)}, \qquad df = n - 2 \tag{9.4}$$

If p-value $\leq \alpha$, reject H_0 and conclude that convincing evidence exists of a linear relationship between x and y.

Lesson 9.4: Deciphering Computer Regression Output

AP exams regularly provide regression computer output rather than raw data. You must extract key statistics instantly:

Predictor	**Coef**	**SE Coef**	**T**	**P**
Constant	b_0 (y-intercept)	SE(b_0)	$t = b_0/\text{SE}(b_0)$	p-value
Explanatory Variable (x)	b_1 **(slope)**	**SE**(b_1)	$t = b_1/\textbf{SE}(b_1)$	**Two-sided p-value**

Additional Output Statistics:

- $S = s$: The standard deviation of the residuals. Measures the typical prediction error.
- R-sq $= r^2$: The coefficient of determination.
- **Direction of r:** The computer output reports r^2 as a positive percentage. To find r, calculate $\pm\sqrt{r^2}$, giving r the same sign as the slope coefficient b_1!
- **One-Sided p-Value Warning:** Computer tables report **two-sided** p-values. If your alternative hypothesis is one-sided ($H_a : \beta > 0$ or $H_a : \beta < 0$), divide the table p-value by 2 (provided the sample slope is in the hypothesized direction)!

9.3 Worked Step-by-Step Examples

Example 9.1: Regression Output Analysis & Interpretation

Problem: An automotive research team investigates the relationship between vehicle curb weight (x, in thousands of pounds) and highway fuel efficiency (y, in miles per gallon). An SRS of $n = 18$ passenger cars yields:

Predictor	**Coef**	**SE Coef**	**T**	**P**
Constant	48.24	2.18	22.13	0.000
Weight	−5.12	0.64	−8.00	0.000

$S = 1.86$ mpg, $\quad R$-sq $= 80.0\%$.
(a) Write the equation of the least-squares regression line.
(b) Interpret the slope $b_1 = -5.12$ in context.
(c) Interpret $S = 1.86$ mpg in context.
(d) Construct a 95% confidence interval for the true slope β.

Solution:

- **Part (a):** $\widehat{\text{Fuel Efficiency}} = 48.24 - 5.12(\text{Weight})$.
- **Part (b):** For each additional 1,000-pound increase in vehicle weight, the predicted highway fuel efficiency decreases by approximately 5.12 miles per gallon.
- **Part (c):** When using vehicle weight to predict highway fuel efficiency with this model, predictions will typically be off by about 1.86 miles per gallon from the actual observed fuel efficiency.

- **Part (d):** Degrees of freedom: $df = n - 2 = 18 - 2 = 16$. From the t-table, for 95% confidence with $df = 16$, $t^* = 2.120$.

  CI $= b_1 \pm t^* \cdot \text{SE}(b_1) = -5.12 \pm 2.120(0.64) = -5.12 \pm 1.3568 = (\mathbf{-6.477},$

  "We are 95% confident that for each additional 1,000-pound increase in vehicle curb weight, the true mean highway fuel efficiency decreases by between 3.76 and 6.48 miles per gallon."

Example 9.2: Hypothesis Test for Positive Linear Association

Problem: Using the tree dataset from Example 9.1 where $n = 16$, $b_1 = 2.85$, and $\text{SE}(b_1) = 0.42$, carry out a formal hypothesis test to determine if there is convincing evidence that taller trees have larger diameters ($\beta > 0$) at $\alpha = 0.01$.

Solution:

- **Step 1 (Hypotheses):** $H_0 : \beta = 0$ (No positive linear relationship between diameter and height).
  $H_a : \beta > 0$ (There is a positive linear relationship between diameter and height).

- **Step 2 (Conditions):** Assume all LINER conditions are met as verified by scatterplots and residual plots.

- **Step 3 (Mechanics):**

$$t = \frac{b_1 - 0}{\text{SE}(b_1)} = \frac{2.85 - 0}{0.42} = 6.786 \approx 6.79$$

  Degrees of freedom: $df = 16 - 2 = 14$.

p-value: $P(t_{14} \geq 6.79) = \text{tcdf}(6.79, 10^{99}, 14) \approx 0.000004 < 0.0001$.

- **Step 4 (Conclusion):** Because the p-value (< 0.0001) is much less than $\alpha = 0.01$, we reject H_0. There is convincing statistical evidence of a positive linear association between pine tree diameter and tree height.

9.4 TI-84 CE Calculator Procedures for Slope Inference

TI-84 Playbook: Linear Regression Inference

- **Linear Regression *t*-Test (`LinRegTTest`):** 1. Enter explanatory data into `L1` and response data into `L2`.
  2. Press `[STAT]` → Arrow to `TESTS` → Select `F:LinRegTTest`.
  3. Enter `Xlist: L1`, `Ylist: L2`, `Freq: 1`.
  4. Select the alternative hypothesis for β and ρ ($\neq 0, < 0$, or > 0).
  5. Select `Calculate`. Displays t-statistic, p-value, $df = n-2, a, b, s, r^2$, and r.

- **Linear Regression *t*-Interval (`LinRegTInt`):** 1. Press `[STAT]` → `TESTS` → Select `G:LinRegTInt`.
  2. Enter `Xlist: L1`, `Ylist: L2`, and desired `C-Level` (e.g., 0.95).
  3. Select `Calculate`. Displays the confidence interval for β, $b_1, df, s, \text{SE}(b_1), a, r^2$, and r.

9.5 Comprehensive Practice Problem Set

Part I: 20 Multiple-Choice Questions with Explanations

1. What are the degrees of freedom for a linear regression t-test with a sample size of $n = 25$?

   (A) 25

   (B) 24

   (C) 23

   (D) 2

   (E) 1

   Answer: (C). Regression inference for slope uses $df = n - 2 = 25 - 2 = 23$.

2. The null hypothesis in a regression significance test evaluating linear association is:

   (A) $H_0 : b_1 = 0$

   (B) $H_0 : \beta = 0$

   (C) $H_0 : \mu = 0$

   (D) $H_0 : r = 1$

   (E) $H_0 : \alpha = \beta$

   Answer: (B). Hypotheses must be stated in terms of the population parameter β, never the sample statistic b_1.

3. Which of the following conditions ensures that the standard deviation of residuals is constant across all x?

(A) Linear

(B) Independent

(C) Normal residuals

(D) Equal variance

(E) Random

Answer: (D). The Equal Variance condition (homoscedasticity) specifies constant vertical spread of y about the line.

4. In a regression computer output, the standard error of the slope SE(b_1) appears under which column header?

(A) Coef

(B) SE Coef

(C) T

(D) P

(E) R-sq

Answer: (B). The standard error of the slope is located in the "SE Coef" column on the row for the explanatory variable.

5. A 95% confidence interval for the slope β is $(0.45, 1.85)$. Based on this interval:

(A) There is no linear relationship between x and y.

(B) We fail to reject $H_0 : \beta = 0$ at $\alpha = 0.05$.

(C) We reject $H_0 : \beta = 0$ at $\alpha = 0.05$ because zero is not contained in the interval.

(D) The correlation coefficient must be negative.

(E) $r^2 > 0.95$.

Answer: (C). Because the entire interval is strictly positive and excludes 0, zero is not a plausible value for β.

6. If the residual plot exhibits a distinct fan shape (narrow on the left, wide on the right), which condition is violated?

   (A) Linearity

   (B) Independence

   (C) Normality

   (D) Equal variance

   (E) Randomness

   Answer: (D). A fan or funnel shape violates the Equal Variance assumption.

7. What does the statistic s represent in a regression output?

   (A) The slope of the line

   (B) The correlation coefficient

   (C) The estimated standard deviation of the residuals

   (D) The sample size

   (E) The standard error of the intercept

   Answer: (C). $s = \sqrt{\frac{\sum e_i^2}{n-2}}$ estimates the typical distance that observed y-values deviate from the predicted line.

8. A regression yields $b_1 = 3.60$ with $\text{SE}(b_1) = 0.90$ for $n = 20$. The t-statistic for $H_0 : \beta = 0$ is:

   (A) 0.25

(B) 4.00

(C) 2.70

(D) 3.24

(E) 18.0

Answer: (B). $t = \frac{b_1 - 0}{\text{SE}(b_1)} = \frac{3.60}{0.90} = 4.00$.

9. If computer output reports R-sq $= 64\%$ and the slope coefficient is negative, what is the correlation coefficient r?

(A) 0.64

(B) 0.80

(C) -0.80

(D) -0.64

(E) -0.4096

Answer: (C). $r = -\sqrt{0.64} = -0.80$. The sign of r matches the sign of the slope b_1.

10. How is the Normality of residuals condition checked in practice?

(A) By checking whether $n \geq 30$

(B) By inspecting a histogram, boxplot, or normal probability plot of the residuals

(C) By testing r^2

(D) By plotting y versus x

(E) By checking $n \leq 0.10N$

Answer: (B). Normality applies to the distribution of residuals, checked via an NPP or histogram of residuals.

11. If the p-value for the slope t-test is 0.003 and $\alpha = 0.01$, the correct conclusion is:

    (A) Fail to reject H_0

    (B) Reject H_0; there is convincing evidence of a linear association

    (C) Accept H_0

    (D) Conclude the relationship is non-linear

    (E) Extrapolate predictions

    Answer: (B). Since p-value $= 0.003 < 0.01$, we reject H_0.

12. When constructing a 90% confidence interval for slope with $n = 12$, what critical value t^* is used?

    (A) t^* with $df = 12$

    (B) t^* with $df = 11$

    (C) t^* with $df = 10$

    (D) $z^* = 1.645$

    (E) $z^* = 1.960$

    Answer: (C). $df = n - 2 = 12 - 2 = 10$.

13. The point of averages $(\bar{x}, \bar{y})$ in simple linear regression:

    (A) Is always an outlier

    (B) Always lies exactly on the least-squares regression line

    (C) Has residual equal to 1

    (D) Has leverage equal to 1

    (E) Is never on the line

Answer: (B). The LSRL always passes through $(\bar{x}, \bar{y})$ by mathematical definition of the intercept $b_0 = \bar{y} - b_1\bar{x}$.

14. In regression, which value measures the percentage of variation in y explained by the linear model?

(A) r

(B) r^2

(C) s

(D) $\mathrm{SE}(b_1)$

(E) t

Answer: (B). The coefficient of determination r^2 measures explained variation.

15. A two-sided p-value in a computer printout is 0.040. For $H_a : \beta > 0$ with positive b_1, the one-sided p-value is:

(A) 0.040

(B) 0.080

(C) 0.020

(D) 0.960

(E) 0.010

Answer: (C). A one-sided p-value in the direction of the sample slope is half the two-sided value: $0.040/2 = 0.020$.

16. If $n = 10$, what are the degrees of freedom for the slope t-statistic?

(A) 8

(B) 9

(C) 10

(D) 18

(E) 20

Answer: (A). $df = n - 2 = 10 - 2 = 8$.

17. The sum of the residuals $\sum(y_i - \hat{y}_i)$ for any least-squares regression line is:

(A) Always positive

(B) Always negative

(C) Always exactly zero

(D) Equal to r^2

(E) Equal to s

Answer: (C). A mathematical property of the LSRL is that positive and negative residuals sum to zero.

18. What does a residual plot with a curved U-shape indicate?

(A) A linear model is inappropriate

(B) The correlation is 1

(C) The slope is zero

(D) Conditions are satisfied

(E) Extrapolation was performed

Answer: (A). Any systematic curve in the residual plot indicates that the relationship is non-linear.

19. If the confidence interval for slope β is $(-1.5, 0.5)$, what can be concluded at the corresponding α?

(A) β is significantly positive

(B) β is significantly negative

(C) Fail to reject $H_0 : \beta = 0$ because 0 is inside the interval

(D) $r = 0$

(E) Sample size was too large

Answer: (C). Since 0 is contained within the confidence interval, zero is a plausible value for β.

20. Which acronym summarizes the regression inference conditions?

(A) BINS

(B) BITS

(C) CUSS

(D) PANIC

(E) LINER

Answer: (E). LINER: Linear, Independent, Normal residuals, Equal variance, Random.

Part II: Score-4 Free-Response Problem with Complete Rubric

FRQ 9.1: Comprehensive Regression Inference on Computer Output

Problem: A sports physiologist studies the relationship between weekly training volume (x, in miles) and marathon completion time (y, in minutes) for an SRS of $n = 20$ competitive amateur runners. The regression analysis produces:

Predictor	Coef	SE Coef	T	P
Constant	310.50	14.25	21.79	0.000
Miles	−1.85	0.35	−5.29	0.000

$S = 8.42$ min, R-sq = 60.8%.

(a) State the equation of the least-squares regression line.

(b) Interpret the slope −1.85 in context.

(c) Construct and interpret a 95% confidence interval for the true slope β.

(d) Explain what $S = 8.42$ minutes means in the context of this study.

Grader-Approved Score-4 Solution:

Part (a):

$\widehat{\text{Marathon Time}} = 310.50 - 1.85(\text{Weekly Miles})$.

Part (b):

For each additional 1 mile of weekly training volume, the predicted marathon completion time decreases by approximately 1.85 minutes.

Part (c):

Step 1: Parameter & Name: We construct a 95% confidence interval for the true slope β of the population regression line relating weekly training miles to marathon finish time. Procedure: Linear regression t-interval for slope.

Step 2: Conditions: Stated as an SRS of 20 runners ($n = 20 < 0.10(\text{All runners})$). Assume scatterplot and residual plot verify linearity, normal residuals, and equal variance.

Step 3: Interval Calculations: $df = n - 2 = 20 - 2 = 18$. For 95% confidence with $df = 18$, $t^* = 2.101$.

From output: $b_1 = -1.85, \text{SE}(b_1) = 0.35$.

$\text{CI} = -1.85 \pm 2.101(0.35) = -1.85 \pm 0.7354 = (\mathbf{-2.5854}, \ \mathbf{-1.1146})$ min

Step 4: Interpretation: We are 95% confident that for each additional 1 mile per week of training, the true mean marathon completion time decreases by between 1.11 minutes and 2.59 minutes.

Part (d):

When using weekly training volume to predict marathon completion time with this linear model, predictions will typically deviate from actual observed race times by approximately 8.42 minutes.

9.6 Unit 9 Master Summary

Unit 9 Essential Review Checklist

- **LINER:** Linear, Independent, Normal residuals, Equal variance, Random.
- **Degrees of Freedom:** $df = n - 2$ for all slope t-intervals and t-tests.
- **Slope CI Formula:** $b_1 \pm t^* \cdot \text{SE}(b_1)$.
- **Slope Test Statistic:** $t = (b_1 - 0)/\text{SE}(b_1)$.
- **Computer Output:** Extract b_1 and $\text{SE}(b_1)$ from second row; remember r has the same sign as b_1.

Chapter 10

Practice Exam 1

Exam Overview & Instructions

This authentic practice exam simulates the official 2027 digital AP Statistics Exam.

- **Section I:** 42 Multiple-Choice Questions (90 Minutes — 50% of Total Score).
- **Section II:** 4 Free-Response Questions (90 Minutes — 50% of Total Score).
- Calculator active throughout both sections (TI-84 Plus CE or Bluebook Desmos Graphing Calculator). Nongraphing scientific calculators are strictly prohibited. Standard normal, t, and χ^2 tables provided in Appendix D.

10.1 Section I: Multiple-Choice Questions (Questions 1–42)

1. A set of 20 exam scores has Mean = 75 and SD = 8. A score of 91 corresponds to a z-score of:

(A) 1.50

(B) 2.00

(C) 2.25

(D) 1.75

(E) 1.25

2. Which of the following is a categorical variable?

(A) Annual income

(B) Blood type (A, B, AB, O)

(C) Systolic blood pressure

(D) Commute time in minutes

(E) High school GPA

3. A scatterplot exhibits a strong negative linear association with $r = -0.92$. The coefficient of determination r^2 is:

(A) -0.846

(B) 0.846

(C) -0.92

(D) 0.92

(E) 0.080

4. In a modified boxplot, the whiskers extend to:

(A) The outlier boundary fences $Q_1 - 1.5(\text{IQR})$ and $Q_3 + 1.5(\text{IQR})$

(B) The absolute minimum and maximum data values

(C) The most extreme observations that fall within the boundary fences

(D) Q_1 and Q_3

(E) $\bar{x} \pm 2s$

5. An observational study finds people who take multivitamins have fewer colds. Can we conclude multivitamins prevent colds?

(A) Yes, if $n > 1000$

(B) No, because confounding variables cannot be controlled

(C) Yes, if $p < 0.05$

(D) No, because vitamins are categorical

(E) Yes, if double-blind

6. If $P(A) = 0.5, P(B) = 0.4$, and A and B are independent, what is $P(A \cup B)$?

(A) 0.90

(B) 0.70

(C) 0.20

(D) 0.50

(E) 0.80

7. A random variable $X \sim B(50, 0.20)$. The mean and standard deviation are:

(A) $\mu = 10, \sigma = 2.83$

(B) $\mu = 10, \sigma = 8.00$

(C) $\mu = 25, \sigma = 5.00$

(D) $\mu = 10, \sigma = 2.00$

(E) $\mu = 50, \sigma = 10$

8. The Central Limit Theorem applies to which of the following?

    (A) The shape of the population

    (B) The shape of a single sample

    (C) The sampling distribution of sample means $\bar{x}$ as $n \geq 30$

    (D) The sampling distribution of $\hat{p}$

    (E) The variance of a binomial variable

9. In testing $H_0 : p = 0.5$ vs. $H_a : p > 0.5$, a sample gives $z = 2.33$. The p-value is:

    (A) 0.0198

    (B) 0.0099

    (C) 0.9901

    (D) 0.0500

    (E) 0.0200

10. A 95% confidence interval for μ with $n = 16$ uses which critical value t^* ($df = 15$)?

    (A) 1.960

    (B) 2.131

    (C) 1.753

    (D) 2.602

    (E) 2.947

11. Which test evaluates whether categorical proportions are identical across 3 hospital networks?

(A) 1-sample t-test

(B) Chi-Square Test for Goodness-of-Fit

(C) Chi-Square Test for Homogeneity

(D) Chi-Square Test for Independence

(E) ANOVA

12. What are the degrees of freedom for a regression t-test with $n = 28$?

(A) 28

(B) 27

(C) 26

(D) 14

(E) 2

13. If the residuals of a model exhibit a curved U-shape, this indicates:

(A) A linear model is inappropriate

(B) The correlation is positive

(C) The slope is 0

(D) The sample size is too large

(E) All residuals are zero

14. A fair 6-sided die is rolled until a 4 appears. What is the probability it takes more than 3 rolls?

(A) $(1/6)^3$

(B) $(5/6)^3 \approx 0.5787$

(C) $1 - (5/6)^3$

(D) $3(1/6)$

(E) $(5/6)(1/6)^2$

15. When constructing a confidence interval for p, which condition justifies assuming independence?

(A) $np \geq 10$

(B) $n \leq 0.10N$

(C) $n \geq 30$

(D) $r^2 > 0.8$

(E) Double-blinding

16. A 99% confidence interval for $\mu_1 - \mu_2$ is $(1.2, 5.8)$. Based on this interval:

(A) μ_1 is significantly greater than μ_2 at $\alpha = 0.01$ because the entire interval is positive.

(B) $\mu_1 = \mu_2$.

(C) $\mu_2 > \mu_1$.

(D) No conclusion is possible.

(E) p-value > 0.05.

17. The power of a hypothesis test can be increased by:

(A) Decreasing sample size n

(B) Decreasing α

(C) Increasing sample size n

(D) Making H_0 false

(E) Pooling variances

18. In a completely randomized experiment with 60 subjects and 3 treatments, how many subjects per treatment if balanced?

    (A) 10

    (B) 20

    (C) 30

    (D) 60

    (E) 180

19. A student scores at the 84^{th} percentile on a normal exam with $\mu = 500, \sigma = 100$. The student's score is approximately:

    (A) 584

    (B) 600

    (C) 650

    (D) 700

    (E) 750

20. Which of the following is true about standard deviation s_x?

    (A) It is resistant to outliers.

    (B) It can be negative.

    (C) It is measured in the same units as the original data.

    (D) It equals the variance.

    (E) It is unaffected by multiplication.

21. The LSRL passes through which point?

    (A) $(0, 0)$

(B) $(\bar{x}, \bar{y})$

(C) (s_x, s_y)

(D) (Q_1, Q_3)

(E) $(1, 1)$

22. What type of bias occurs when voluntary callers participate in a radio poll?

(A) Nonresponse bias

(B) Voluntary response bias

(C) Stratification bias

(D) Wording bias

(E) Randomization bias

23. Two independent events A and B have $P(A) = 0.3$ and $P(B) = 0.7$. What is $P(A \cap B)$?

(A) 1.00

(B) 0.40

(C) 0.21

(D) 0.50

(E) 0.00

24. The p-value in a significance test represents:

(A) The probability that H_0 is true.

(B) The probability of obtaining a test statistic as extreme as observed, assuming H_0 is true.

(C) The margin of error.

(D) The probability of a Type II error.

(E) The power.

25. In a 2×3 contingency table, degrees of freedom for Chi-Square test of independence is:

    (A) 6

    (B) 5

    (C) 3

    (D) 2

    (E) 1

26. If $s_x = 5, s_y = 10$, and $r = -0.6$, what is the slope of the regression line?

    (A) -1.2

    (B) -0.3

    (C) 1.2

    (D) -0.6

    (E) -3.0

27. What is the minimum value for an expected count in a Chi-Square test?

    (A) 1

    (B) 5

    (C) 10

    (D) 30

    (E) 50

28. If $\bar{x} = 50, \mu_0 = 45, s = 10, n = 25$, what is the test statistic t?

    (A) 1.0

    (B) 2.5

    (C) 5.0

    (D) 0.5

    (E) 2.0

29. A double-blind experiment prevents which sources of bias?

    (A) Both subject expectation bias and evaluator measurement bias

    (B) Nonresponse bias

    (C) Undercoverage

    (D) Extrapolation

    (E) Confounding from weather

30. The sampling distribution of $\hat{p}$ has mean:

    (A) $\bar{x}$

    (B) μ

    (C) p

    (D) np

    (E) $\sqrt{p(1-p)}$

31. If $z = -1.96$, the area to the left under the standard normal curve is:

    (A) 0.05

(B) 0.025

(C) 0.95

(D) 0.975

(E) 0.50

32. Which of the following conditions is required for regression inference?

(A) LINER

(B) BINS

(C) BITS

(D) CUSS

(E) PANIC

33. An experiment that compares each participant's heart rate while resting versus after exercising is:

(A) Observational study

(B) Matched pairs design

(C) Completely randomized design

(D) Stratified survey

(E) Cluster trial

34. If $P(A) = 0.8$, what is $P(A^c)$?

(A) 0.8

(B) 0.2

(C) 0.0

(D) 1.0

(E) 0.64

35. The sum of residuals $\sum(y - \hat{y})$ for a least-squares regression line is:

    (A) 1
    (B) -1
    (C) Always zero
    (D) r^2
    (E) s_y

36. A 90% confidence interval for μ is $(12.4, 18.6)$. What is the margin of error?

    (A) 6.2
    (B) 3.1
    (C) 15.5
    (D) 1.96
    (E) 0.90

37. What type of variable is the number of cars passing an intersection in one hour?

    (A) Categorical
    (B) Continuous quantitative
    (C) Discrete quantitative
    (D) Qualitative
    (E) Identifier

38. If a significance test rejects H_0 at $\alpha = 0.05$, the p-value must be:

(A) ≤ 0.05

(B) > 0.05

(C) 0.05 exactly

(D) Negative

(E) 1.0

39. When constructing a confidence interval, increasing the confidence level from 90% to 99%:

(A) Narrows the interval

(B) Widens the interval

(C) Has no effect on width

(D) Cuts sample size in half

(E) Decreases z^*

40. The standard error of the slope b_1 measures:

(A) The true population slope β

(B) The estimated standard deviation of the sampling distribution of sample slopes b_1

(C) The correlation coefficient r

(D) The y-intercept

(E) The r^2 value

Directions for Questions 41–42: Use the following information to answer Questions 41 and 42.

A public health agency studies the relationship between daily physical activity (x, in minutes) and resting pulse rate (y, in beats per minute) for a random sample of 25 adults. The least-squares regression line is $\hat{y} = 78.4 - 0.32x$, with $r = -0.68$ and standard error of the slope $\mathrm{SE}(b_1) = 0.075$.

41. What percentage of the variability in resting pulse rate is explained by the linear relationship with daily physical activity?

    (A) 32.0%

    (B) 46.2%

    (C) 68.0%

    (D) 82.5%

    (E) 64.0%

42. In a hypothesis test of $H_0 : \beta = 0$ versus $H_a : \beta < 0$, what is the calculated value of the test statistic t and its degrees of freedom?

    (A) $t = -4.27,\ df = 23$

    (B) $t = -4.27,\ df = 24$

    (C) $t = -0.32,\ df = 25$

    (D) $t = -2.13,\ df = 23$

    (E) $t = -1.96,\ df = 24$

10.2 Section II: Free-Response Questions (Questions 1–4)

Time: 90 Minutes — 4 Questions — 50% of Exam Score

Question 1: Multi-Focus on Practices 1 & 2 (Formulate Questions & Collect Data)

A school district administration investigates whether introducing dynamic standing desks into high school classrooms improves student academic attentiveness. A total of 40 standard classrooms across 4 high schools are available for an educational trial.

(a) **Skill 1.A:** Formulate a precise, valid statistical investigative question that can be addressed by this comparative study.

(b) **Skill 2.B:** Describe an ethical and valid random assignment protocol to allocate 20 classrooms to receive standing desks and 20 classrooms to maintain traditional seated desks using a random number generator.

(c) **Skill 2.D:** Identify a potential confounding variable (e.g., subject matter or time of day) and explain how a randomized block design (blocking by academic discipline, such as STEM vs. Humanities) would control for this variable.

(d) **Skill 2.E:** State the null and alternative hypotheses in words and mathematical symbols for a significance test evaluating whether standing desks produce higher mean attentiveness ratings.

Question 2: Multi-Focus on Practices 3 & 4 (Analyze Data & Interpret Results)

A pediatric nutritional researcher examines the sugar content (x, in grams per serving) and caloric density (y, in calories) for $n = 14$ popular commercial breakfast cereals. Summary statistics for sugar content are:

$$\text{Min} = 4, \quad Q_1 = 7, \quad \text{Median} = 11.5, \quad Q_3 = 14, \quad \text{Max} = 26$$

A computer regression package outputs the least-squares regression equation relating calories to sugar content:

$$\widehat{\text{Calories}} = 95.20 + 3.85(\text{Sugar})$$

with $s = 8.64$ calories and $r^2 = 72.8\%$.

(a) **Skill 3.B:** Determine the outlier boundary fences for sugar content using the $1.5 \times \text{IQR}$ rule. Is the cereal with 26 grams of sugar a confirmed outlier? Justify mathematically.

(b) **Skill 4.A:** Construct and describe the distribution of sugar content addressing Center, Unusual features, Shape, and Spread (C-U-S-S) with context.

(c) **Skill 3.B:** Predict the caloric content of a cereal containing 12.0 grams of sugar per serving. If an observed cereal with 12.0 grams of sugar actually has 148 calories, calculate and interpret the residual.

(d) **Skill 4.B:** Interpret the slope 3.85 and the coefficient of determination $r^2 = 72.8\%$ in complete context.

Question 3: Statistical Inference (Hypothesis Test for Proportions)

A university admissions office claims that at least 65% of admitted students will accept their offer and enroll. In an SRS of $n = 300$ admitted high school seniors, 180 enroll. Conduct a complete significance test at the $\alpha = 0.05$ level following the PHANTOM framework:

(a) State the null and alternative hypotheses, clearly defining the parameter of interest in context.

(b) Identify the correct inference procedure and verify all necessary conditions (Random, 10%, Large Counts) with explicit numerical evidence.

(c) Calculate the standardized test statistic z and determine the associated p-value.

(d) State your conclusion in the context of university enrollment at the $\alpha = 0.05$ significance level.

Question 4: Multi-Focus Synthesis on Practices 2, 3, & 4 (Comprehensive Investigation)

An educational research team models the relationship between daily study hours (x) and standardized exam performance (y, on a 100-point scale). The preliminary simple linear model is $\hat{y} = 42.0 + 7.5x$.

(a) **Practices 3 & 4:** A student who studied 4.0 hours scored 78 points. Calculate and interpret the residual in context. Explain what the slope 7.5 means.

(b) **Practices 2 & 3:** The researchers suspect omitted variable bias due to sleep duration (w, in hours). When sleep is included, the estimated model is $\hat{y} = 20.0 + 6.0x + 3.5w$. Explain how omitted variable bias affected the initial slope estimate in part (a).

(c) **Practice 3 (Probability Model):** Suppose exam score gains G follow a normal distribution with $\mu_G = 12.5$ points and $\sigma_G = 3.2$ points. What is the probability that a randomly chosen student gains more than 15.0 points?

(d) **Practices 2, 3, & 4 (Optimization under Constraint):** A student has a total combined daily budget of 14 hours for study and sleep ($x + w \leq 14$). Assuming minimum biological sleep required is 6 hours ($w \geq 6$), find the daily schedule that maximizes predicted exam score and calculate the maximum predicted score.

Chapter 11

Practice Exam 2

Exam Overview & Instructions

This second full-length practice exam provides comprehensive final review across the 5 curriculum units under strict 2027 digital testing conditions.

- **Section I:** 42 Multiple-Choice Questions (90 Minutes — 50% of Total Score).
- **Section II:** 4 Free-Response Questions (90 Minutes — 50% of Total Score).
- Graphing calculator active throughout (TI-84 Plus CE or Bluebook Desmos Graphing Calculator). Nongraphing scientific calculators are strictly prohibited.

11.1 Section I: Multiple-Choice Questions (Questions 1–42)

1. In a dataset of 100 values, the mean is 50 and standard deviation is 0. This implies:

(A) Half the values are 0 and half are 100.

(B) Every single value in the dataset equals 50.

(C) The variance is 50.

(D) The median is 0.

(E) The data is normal.

2. If a distribution is skewed to the right, which is true?

(A) Mean $>$ Median

(B) Mean $<$ Median

(C) Mean $=$ Median

(D) Range is zero

(E) $s_x = 0$

3. A scatterplot has $r = 0.0$. This means:

(A) There is no linear relationship between x and y.

(B) There is no relationship of any kind.

(C) The data forms a vertical line.

(D) All points lie on the y-axis.

(E) $b_1 = 1$.

4. In an experiment, the factor is:

(A) The explanatory variable manipulated by the researcher.

(B) The response variable.

(C) The subjects.

(D) The error rate.

(E) The randomizer.

5. If $P(A) = 0.6, P(B) = 0.3$, and $P(A \cap B) = 0.18$, events A and B are:

   (A) Disjoint

   (B) Independent

   (C) Dependent

   (D) Impossible

   (E) Complements

6. What is the expected value of a geometric random variable with $p = 0.25$?

   (A) 0.25

   (B) 1.0

   (C) 2.0

   (D) 4.0

   (E) 16.0

7. The sampling distribution of $\bar{x}$ will be approximately normal if:

   (A) $n \geq 30$ by the Central Limit Theorem

   (B) The population has $N \geq 30$

   (C) $p = 0.5$

   (D) $s_x = \sigma$

   (E) $\mu = 0$

8. For a 95% confidence interval for a proportion, the margin of error is:

(A) $1.96\sqrt{\hat{p}(1-\hat{p})/n}$

(B) $\hat{p}/n$

(C) $1.96/\sqrt{n}$

(D) $\sqrt{p(1-p)}$

(E) 2σ

9. When testing $H_0 : \mu = 50$ vs. $H_a : \mu \neq 50$, $t = -2.00$ with $df = 20$. The two-sided p-value is:

(A) $2 \times P(t_{20} \leq -2.00)$

(B) $P(t_{20} \leq -2.00)$

(C) $1 - P(t_{20} \leq -2.00)$

(D) 0.05

(E) 0.025

10. Degrees of freedom for a Chi-Square goodness-of-fit test with 5 categories is:

(A) 5

(B) 4

(C) 25

(D) 1

(E) 10

11. In a regression analysis, the slope $b_1 = -2.5$ and $\mathrm{SE}(b_1) = 0.5$. The t-statistic for $H_0 : \beta = 0$ is:

(A) -5.0

(B) -1.25

(C) 5.0

(D) 0.20

(E) -0.20

12. If an observation has a negative residual, the actual data point:

    (A) Lies below the regression line

    (B) Lies above the line

    (C) Lies on the line

    (D) Is an outlier

    (E) Has $z = 0$

13. What happens to the power of a test when the sample size n decreases?

    (A) Increases

    (B) Decreases

    (C) Remains constant

    (D) Becomes 1

    (E) Becomes 0

14. A z-score of 0 indicates that the observation:

    (A) Equals the mean

    (B) Equals 0

    (C) Is an outlier

    (D) Has a probability of 0

    (E) Is negative

15. A survey draws an SRS of 100 students from a school of 1000. The 10% condition is:

    (A) Satisfied ($100 \leq 0.10(1000) = 100$)

    (B) Violated

    (C) Not applicable

    (D) Only for means

    (E) Only for experiments

16. Which of the following is an unbiased estimator of the population proportion p?

    (A) Sample proportion $\hat{p}$

    (B) $\hat{p} + 0.05$

    (C) Sample mean $\bar{x}$

    (D) s_x

    (E) z^*

17. The sum of squared residuals $\sum(y - \hat{y})^2$ is minimized by:

    (A) The least-squares regression line

    (B) The horizontal line through $\bar{y}$

    (C) The median line

    (D) Any line with positive slope

    (E) A polynomial curve

18. If $P(A) = 0.4$, what is $P(A \cap A^c)$?

    (A) 0.00

    (B) 0.40

(C) 0.60

(D) 1.00

(E) 0.24

19. For a 1-sample t-interval with $n = 30$, degrees of freedom is:

(A) 29

(B) 30

(C) 31

(D) 28

(E) 15

20. In an experiment evaluating three pain relievers, patients are assigned by age group. Age is the:

(A) Blocking variable

(B) Response variable

(C) Treatment

(D) Placebo

(E) Dependent variable

21. If $z^* = 2.576$, the corresponding confidence level is:

(A) 90%

(B) 95%

(C) 98%

(D) 99%

(E) 99.9%

22. A distribution of house prices has $\mu = \$450,000$ and Median $=$ $\$320,000$. The shape is:

    (A) Skewed right

    (B) Skewed left

    (C) Symmetric

    (D) Uniform

    (E) Normal

23. What is the probability of flipping 4 heads in a row with a fair coin?

    (A) 0.50

    (B) 0.25

    (C) 0.125

    (D) 0.0625

    (E) 0.03125

24. A high leverage point in regression:

    (A) Has an extreme x-value

    (B) Always has a large residual

    (C) Has $y = 0$

    (D) Is always an error

    (E) Cannot affect the slope

25. If $n = 64$ and $\sigma = 16$, the standard deviation of the sample mean $\bar{x}$ is:

    (A) 2

(B) 4

(C) 8

(D) 16

(E) 0.25

26. A significance test with $\alpha = 0.05$ produces p-value $= 0.082$. The decision is:

    (A) Fail to reject H_0

    (B) Reject H_0

    (C) Accept H_a

    (D) Prove H_0 is true

    (E) Increase α

27. In a Chi-Square test, the expected counts represent:

    (A) Frequencies expected if the null hypothesis is true

    (B) Actual counts collected

    (C) Percentages of total

    (D) Standard errors

    (E) Degrees of freedom

28. A matched pairs t-test is mathematically equivalent to:

    (A) A 1-sample t-test on differences

    (B) A 2-sample independent t-test

    (C) A 1-prop z-test

    (D) An ANOVA

    (E) A Chi-Square test

29. What percentage of observations fall within $\pm 1\sigma$ of the mean in a normal distribution?

    (A) 50%

    (B) 68%

    (C) 95%

    (D) 99.7%

    (E) 100%

30. Which sampling method divides the population into clusters and samples all members of selected clusters?

    (A) Cluster sampling

    (B) Stratified sampling

    (C) Simple random sampling

    (D) Systematic sampling

    (E) Convenience sampling

31. If $P(A \mid B) = P(A)$, then A and B are:

    (A) Independent

    (B) Disjoint

    (C) Mutually exclusive

    (D) Dependent

    (E) Equal

32. Standard deviation of a binomial distribution is:

    (A) $\sqrt{np(1-p)}$

    (B) np

(C) $\sqrt{p(1-p)/n}$

(D) $s/\sqrt{n}$

(E) σ^2

33. What does a residual plot with random scatter around zero indicate?

(A) A linear model is appropriate

(B) The model is curved

(C) High bias

(D) Outliers exist

(E) Extrapolation error

34. A 95% confidence interval for p is $(0.40, 0.60)$. The point estimate $\hat{p}$ is:

(A) 0.50

(B) 0.40

(C) 0.60

(D) 0.10

(E) 0.20

35. When testing $H_0 : \mu = 0$ vs. $H_a : \mu > 0$, the test is:

(A) One-sided (right-tailed)

(B) One-sided (left-tailed)

(C) Two-sided

(D) Non-directional

(E) Chi-square

36. A placebo is used in an experiment to:

    (A) Control for the psychological effect of receiving a treatment

    (B) Increase the sample size

    (C) Ensure double-blinding

    (D) Eliminate random assignment

    (E) Guarantee significance

37. If $n = 100$ and $p = 0.50$, is the Large Counts condition met?

    (A) Yes ($np = 50 \geq 10$ and $n(1-p) = 50 \geq 10$)

    (B) No

    (C) Only for means

    (D) Only if $N > 10,000$

    (E) Depends on σ

38. In regression, *df* for the slope t-test is:

    (A) $n - 2$

    (B) $n - 1$

    (C) n

    (D) $2n$

    (E) $(r-1)(c-1)$

39. The interquartile range (IQR) is:

    (A) $Q_3 - Q_1$

    (B) Max − Min

    (C) $2s$

(D) $\mu \pm \sigma$

(E) $\bar{x} - M$

40. An experimental design where each subject receives two treatments in random order is:

(A) Matched pairs design

(B) Completely randomized design

(C) Observational study

(D) Cluster trial

(E) Stratified survey

Directions for Questions 41–42: Use the following information to answer Questions 41 and 42.

An automotive engineering team examines the relationship between curb weight (x, in thousands of pounds) and highway fuel economy (y, in miles per gallon) for an SRS of $n = 18$ vehicles. Computer regression output yields: $\hat{y} = 48.24 - 5.12x$, with $\text{SE}(b_1) = 0.640$, $s = 1.86$ mpg, and $R\text{-sq} = 80.0\%$.

41. What is the value of the correlation coefficient r between vehicle curb weight and fuel economy?

(A) $+0.800$

(B) -0.800

(C) $+0.894$

(D) -0.894

(E) -0.640

42. Which expression represents a 95% confidence interval for the true slope β of the population regression line?

(A) $-5.12 \pm 2.120(0.640)$

(B) $-5.12 \pm 2.101(0.640)$

(C) $-5.12 \pm 1.960(0.640)$

(D) $-5.12 \pm 2.120(1.86)$

(E) $48.24 \pm 2.120(0.640)$

11.2 Section II: Free-Response Questions (Questions 1–4)

Time: 90 Minutes — 4 Questions — 50% of Exam Score

Question 1: Multi-Focus on Practices 1 & 2 (Formulate Questions & Collect Data)

A regional orthopedic medical network evaluates two post-surgical rehabilitation regimens: Protocol Alpha (incorporating hydrotherapy) and Protocol Beta (standard land-based physical therapy). A total of 50 patients scheduled for identical knee replacement surgeries agree to participate.

(a) **Skill 1.A:** Formulate a precise statistical investigative question that can be evaluated using the patient recovery time data.

(b) **Skill 2.B:** Justify why a randomized comparative experiment is necessary to evaluate the recovery protocols rather than simply allowing patients to select their preferred rehabilitation regimen.

(c) **Skill 2.D:** The researchers recognize that patient age (under 60 vs. 60 and older) is strongly associated with recovery rate. Explain how a randomized block design blocking on age controls for this source of variation.

(d) **Skill 2.E:** State the null and alternative hypotheses in words and parameters (μ_α, μ_β) for a significance test to determine if Protocol Alpha results in a faster mean recovery time.

Question 2: Multi-Focus on Practices 3 & 4 (Analyze Data & Interpret Results)

A marine research expedition measures water depth (x, in hundreds of meters) and underwater ambient light intensity (y, in lumens/m^2) across $n = 22$ oceanic sampling sites. The raw scatterplot exhibits a strong non-linear decay. Applying a natural logarithm transformation to the response variable yields:

$$\widehat{\ln(y)} = 6.42 - 0.58x$$

with $s = 0.145$, $\text{SE}(b_1) = 0.042$, and $r^2 = 94.1\%$.

(a) **Skill 3.B:** Predict the light intensity y (in lumens/m^2) at a depth of 500 meters ($x = 5.0$).

(b) **Skill 3.B & 4.D:** At depth $x = 3.0$ (300 meters), observed light intensity is 120 lumens/m^2. Calculate and interpret the residual in the natural logarithm scale.

(c) **Skill 4.A:** Explain why examining the residual plot of the raw (x, y) data versus the transformed $(x, \ln(y))$

data is crucial before concluding that the linear model is appropriate.

(d) **Skill 4.B:** Interpret the coefficient of determination $r^2 = 94.1\%$ in context.

Question 3: Statistical Inference (Two-Proportion Confidence Interval)

A technology consumer research firm investigates whether user satisfaction differs between two prominent mobile operating systems: OS-Alpha and OS-Beta.

- OS-Alpha: In an independent random sample of 400 active users, 328 report being highly satisfied.
- OS-Beta: In an independent random sample of 500 active users, 375 report being highly satisfied.

(a) Define the parameters of interest and state the appropriate inference procedure.

(b) Verify all necessary conditions for constructing a two-proportion confidence interval with explicit numbers.

(c) Construct a 95% confidence interval for the true difference in satisfaction proportions $(p_\alpha - p_\beta)$.

(d) Based on your confidence interval, does the data provide convincing statistical evidence that OS-Alpha achieves a higher satisfaction rate than OS-Beta? Justify your claim.

Question 4: Multi-Focus Synthesis on Practices 2, 3, & 4 (Decision Analysis Under Uncertainty)

A clinical diagnostic startup develops a rapid screening test kit for an asymptomatic cardiovascular condition. The disease prevalence in the screening population is $P(D) = 0.02$. Laboratory trials establish:

- Sensitivity (True Positive Rate): $P(+ \mid D) = 0.95$
- Specificity (True Negative Rate): $P(- \mid D^c) = 0.98$

(a) **Skill 3.C:** Calculate the overall probability that a randomly chosen individual tests positive, $P(+)$, showing complete probability tree or total probability calculations.

(b) **Skill 3.C & 4.D:** Calculate the Positive Predictive Value (PPV), $P(D \mid +)$, and interpret this conditional probability in context. Explain why PPV is surprisingly low despite high sensitivity and specificity.

(c) **Practices 2, 3, & 4 (Decision Payoff Matrix):** The public health department evaluates a protocol where positive test results trigger immediate preventive therapy. Net payoffs per individual are:

- True Positive (Treated correctly): $+\$10,000$ (avoided hospitalization benefit)
- False Positive (Unnecessary treatment): $-\$2,500$ (medication side effects cost)
- False Negative (Missed case): $-\$50,000$ (severe disease progression cost)

- True Negative (Correct non-treatment): \$0

Calculate the expected net payoff per 1,000 screened individuals under this protocol versus a policy of screening no one (where all diseased cases go untreated). Provide a clear policy recommendation.

Chapter 12

Practice Exam 1: Detailed Solutions & Grading Rubrics

12.1 Section I: Multiple-Choice Detailed Explanations

1. **Answer: (B)**
   Explanation: The z-score formula is $z = \frac{x-\mu}{\sigma}$. Substituting the given values:

   $$z = \frac{91 - 75}{8} = \frac{16}{8} = 2.00$$

   A z-score of 2.00 signifies that an exam score of 91 lies exactly 2.00 standard deviations above the class mean. *Distractor analysis:* (A) 1.50 results from dividing 16 by 10 or calculating $(91-75)/10.67$. (C) 2.25 is $(93-75)/8$. (D) 1.75 is $(89-75)/8$. (E) 1.25 is $(85-75)/8$.

2. **Answer: (B)**
   Explanation: A categorical (qualitative) variable places an individual into one of several groups or categories. Blood

type (A, B, AB, O) is defined by qualitative categories with no meaningful numerical arithmetic. Annual income, blood pressure, commute time, and GPA are all quantitative variables measured on continuous or numerical scales.

3. **Answer: (B)**
   Explanation: The coefficient of determination is r^2. Squaring the correlation coefficient gives:

$$r^2 = (-0.92)^2 = 0.8464 \approx 0.846$$

   By mathematical definition, r^2 is always non-negative ($0 \leq r^2 \leq 1$), so option (A) -0.846 is impossible. Interpretation: approximately 84.6% of the variation in the response variable is explained by the linear relationship with the explanatory variable.

4. **Answer: (C)**
   Explanation: In a modified boxplot, individual outliers are plotted as separate dots, asterisks, or points beyond the fences. Whiskers do not extend indefinitely to the fences themselves; rather, they extend to the most extreme actual observations in the sample that fall inside the lower fence $Q_1 - 1.5(\text{IQR})$ and upper fence $Q_3 + 1.5(\text{IQR})$.

5. **Answer: (B)**
   Explanation: Observational studies observe individuals and record variables of interest without imposing treatments. Because no random assignment was utilized, lurking and confounding variables (e.g., diet, exercise, socioeconomic status, sleep habits) cannot be controlled. Thus, observational studies cannot establish cause-and-effect relationships.

6. **Answer: (B)**
   Explanation: By the general addition rule: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Because events A and B are independent, $P(A \cap B) = P(A) \times P(B) = (0.5)(0.4) = 0.20$. Therefore:

   $$P(A \cup B) = 0.50 + 0.40 - 0.20 = 0.70$$

   Distractor analysis: Option (A) 0.90 incorrectly assumes the events are mutually exclusive (disjoint) and fails to subtract $P(A \cap B)$.

7. **Answer: (A)**
   Explanation: For a binomial distribution $X \sim B(n, p)$:

   $$\mu = np = 50(0.20) = 10$$

   $$\sigma = \sqrt{np(1-p)} = \sqrt{50(0.20)(0.80)} = \sqrt{8.00} \approx 2.828 \approx 2.83$$

   Option (A) is correct.

8. **Answer: (C)**
   Explanation: The Central Limit Theorem (CLT) states that for any population distribution with finite mean μ and finite standard deviation σ, the sampling distribution of the sample mean $\bar{x}$ becomes approximately normal as the sample size n becomes sufficiently large ($n \geq 30$). The CLT does not alter the shape of the underlying population or individual samples.

9. **Answer: (B)**
   Explanation: For a right-tailed alternative hypothesis $H_a : p > 0.50$ with test statistic $z = 2.33$, the p-value is the area

under the standard normal curve to the right of 2.33:

$$p\text{-value} = P(Z \geq 2.33) = 1 - P(Z < 2.33) = 1 - 0.9901 = 0.0099$$

Using TI-84: `normalcdf(2.33, 9999, 0, 1)` $= 0.009903$.

10. **Answer: (B)**
    Explanation: With $n = 16$, the degrees of freedom are $df = n - 1 = 15$. For a 95% confidence interval, the area in each tail is $(1 - 0.95)/2 = 0.025$. Looking up $df = 15$ with central area 0.95 gives $t^* = 2.131$ (or on TI-84: `invT(0.975, 15)` $= 2.1314$). Critical value 1.960 corresponds to the standard normal z^* distribution.

11. **Answer: (C)**
    Explanation: A Chi-Square Test for Homogeneity evaluates whether the distribution of a single categorical variable is identical across two or more independent populations or treatment groups (here, 3 hospital networks). A test of independence uses a single sample categorized by two variables. Goodness-of-fit evaluates a single sample against a hypothesized distribution.

12. **Answer: (C)**
    Explanation: In simple linear regression inference for the slope β, degrees of freedom are given by $df = n - 2$ because two parameters (the intercept α and slope β) are estimated from the sample data. For $n = 28$:

$$df = 28 - 2 = 26$$

13. **Answer: (A)**

Explanation: An appropriate linear model displays a random cloud of points scattered evenly above and below the zero residual line with constant vertical spread. Any clear curved or parabolic pattern (such as a U-shape) in the residual plot indicates that the relationship between x and y is non-linear and that a straight line is not an appropriate model.

14. **Answer: (B)**
Explanation: Let X be the number of rolls until a 4 appears ($p = 1/6$). The event that $X > 3$ means that the first three consecutive rolls must all be failures (not a 4). The probability of failure on any single roll is $q = 1 - 1/6 = 5/6$. By the multiplication rule for independent trials:

$$P(X > 3) = (5/6)^3 = \frac{125}{216} \approx 0.5787$$

15. **Answer: (B)**
Explanation: When sampling without replacement from a finite population, the observations are technically dependent. However, if the sample size n does not exceed 10% of the population size N ($n \leq 0.10N$), the dependence between trials is negligible and the standard deviation formula $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$ remains sufficiently accurate.

16. **Answer: (A)**
Explanation: Since the entire 99% confidence interval $(1.2, 5.8)$ is strictly greater than zero, zero is not a plausible value for the true difference $\mu_1 - \mu_2$. This provides statistically significant evidence at $\alpha = 0.01$ that $\mu_1 - \mu_2 > 0$, or $\mu_1 > \mu_2$.

17. **Answer: (C)**

Explanation: Power is the probability of correctly rejecting a false null hypothesis ($1 - \beta$). Increasing the sample size n decreases the standard error of the estimator, narrowing the sampling distribution and reducing the overlap between the null and alternative distributions, which directly increases power.

18. **Answer: (B)**

Explanation: In a balanced completely randomized design with $N = 60$ subjects divided equally among $k = 3$ treatment groups, each group receives:

$$\frac{N}{k} = \frac{60}{3} = 20 \text{ subjects}$$

19. **Answer: (B)**

Explanation: In a normal distribution, the Empirical Rule states that approximately 68% of values lie within $\pm 1\sigma$ of the mean, leaving 32% in the two tails (16% below $\mu - 1\sigma$ and 16% above $\mu + 1\sigma$). Therefore, the area below $\mu + 1\sigma$ is $50\% + 34\% = 84\%$, corresponding to the 84$^{\text{th}}$ percentile:

$$x = \mu + 1\sigma = 500 + 100(1) = 600$$

20. **Answer: (C)**

Explanation: Standard deviation s_x is computed in the exact same measurement units as the original data (e.g., inches, grams, dollars), making it readily interpretable. Variance is measured in squared units. Standard deviation is nonresistant to outliers, can never be negative, and scales linearly when multiplied by a constant.

21. **Answer: (B)**
Explanation: A fundamental mathematical property of the least-squares regression line (LSRL) $\hat{y} = b_0 + b_1 x$ is that it always passes through the point of averages, $(\bar{x}, \bar{y})$. This is derived from the formula for the intercept: $b_0 = \bar{y} - b_1 \bar{x}$.

22. **Answer: (B)**
Explanation: Voluntary response bias occurs when sample members self-select themselves into the survey by choosing whether to respond (such as callers responding to a call-in radio poll or online survey). People with strong negative opinions are heavily overrepresented.

23. **Answer: (C)**
Explanation: For two independent events A and B, the multiplication rule states:

$$P(A \cap B) = P(A) \times P(B) = 0.3 \times 0.7 = 0.21$$

24. **Answer: (B)**
Explanation: The p-value is the conditional probability, assuming the null hypothesis H_0 is true, of observing a test statistic as extreme as or more extreme than the value actually observed in the sample, in the direction of the alternative hypothesis H_a.

25. **Answer: (D)**
Explanation: For a two-way contingency table with r rows and c columns, the degrees of freedom for a Chi-Square test of independence or homogeneity are:

$$df = (r-1)(c-1) = (2-1)(3-1) = 1 \times 2 = 2$$

26. **Answer: (A)**
    Explanation: The slope of the least-squares regression line is calculated as:
    $$b_1 = r\left(\frac{s_y}{s_x}\right) = -0.6\left(\frac{10}{5}\right) = -0.6(2) = -1.2$$

27. **Answer: (B)**
    Explanation: The Large Counts condition for Chi-Square tests requires that all individual expected cell counts must be at least 5 ($E_i \geq 5$). This ensures that the discrete multinomial distribution is sufficiently well approximated by the continuous Chi-Square distribution.

28. **Answer: (B)**
    Explanation: The 1-sample t-test statistic is:
    $$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{50 - 45}{10/\sqrt{25}} = \frac{5}{10/5} = \frac{5}{2} = 2.5$$

29. **Answer: (A)**
    Explanation: In a double-blind experiment, neither the experimental subjects nor the researchers/evaluators who interact with them know which treatment each subject is receiving. This eliminates both the psychological placebo effect in subjects and diagnostic/evaluator bias in researchers.

30. **Answer: (C)**
    Explanation: The sample proportion $\hat{p}$ is an unbiased estimator of the true population proportion p. The mean of its sampling distribution is exactly equal to the parameter: $\mu_{\hat{p}} = p$.

31. **Answer: (B)**
    Explanation: For the standard normal distribution $Z \sim N(0, 1)$, the area to the left of $z = -1.96$ is:

    $$P(Z \leq -1.96) = 0.0250$$

    Together with the right tail above $+1.96$ (0.0250), exactly 5% of the total area lies in the two tails, leaving 95% in the center.

32. **Answer: (A)**
    Explanation: The five conditions required for linear regression inference are summarized by the acronym **LINER**: Linear relationship, Independent observations, Normal distribution of y-values for each x, Equal variance of residuals (σ constant), and Random sampling or assignment.

33. **Answer: (B)**
    Explanation: Because each subject receives both conditions (resting and post-exercise measurements) and the analysis focuses on the within-subject differences, this is an example of a matched pairs experimental design.

34. **Answer: (B)**
    Explanation: By the complement rule of probability:

    $$P(A^c) = 1 - P(A) = 1 - 0.8 = 0.2$$

35. **Answer: (C)**
    Explanation: By mathematical construction of the ordinary least-squares line, the positive and negative vertical deviations from the line perfectly cancel out: $\sum(y_i - \hat{y}_i) = 0$.

36. **Answer: (B)**
Explanation: The width of the confidence interval is Upper$-$Lower $= 18.6 - 12.4 = 6.2$. The margin of error is half the interval width:
$$\text{ME} = \frac{6.2}{2} = 3.1$$

37. **Answer: (C)**
Explanation: The number of cars passing an intersection is obtained by counting distinct individual vehicles $(0, 1, 2, 3, \dots)$. Values are non-negative integers with no fractional values possible, defining a discrete quantitative variable.

38. **Answer: (A)**
Explanation: In hypothesis testing, the decision rule is to reject the null hypothesis H_0 if and only if the p-value $\leq \alpha$. Therefore, if H_0 was rejected at $\alpha = 0.05$, the p-value must be less than or equal to 0.05.

39. **Answer: (B)**
Explanation: Margin of error is $\text{ME} = z^* \left(\frac{\sigma}{\sqrt{n}}\right)$. Increasing the confidence level from 90% ($z^* = 1.645$) to 99% ($z^* = 2.576$) increases the critical value z^*, which directly increases the margin of error and widens the confidence interval.

40. **Answer: (B)**
Explanation: The standard error of the slope, $\text{SE}(b_1)$, estimates the standard deviation of the sampling distribution of sample slopes b_1 across repeated random samples of size n drawn from the same population.

41. **Answer: (B)**
Explanation: The coefficient of determination is r^2. Given

the correlation coefficient $r = -0.68$:

$$r^2 = (-0.68)^2 = 0.4624 \implies 46.2\%$$

Interpretation: Approximately 46.2% of the variability in resting pulse rate is accounted for by the linear relationship with daily physical activity. Option (C) 68.0% confuses r with r^2.

42. **Answer: (A)**
Explanation: In simple linear regression, the test statistic for the slope is:

$$t = \frac{b_1 - 0}{\text{SE}(b_1)} = \frac{-0.32}{0.075} \approx -4.267 \approx -4.27$$

Degrees of freedom for inference on slope are $df = n - 2 = 25 - 2 = 23$. Option (B) incorrectly uses $df = n - 1 = 24$.

12.2 Section II: Free-Response Complete Scoring Guidelines

Question 1: Multi-Focus on Practices 1 & 2 — Model Solution & Rubric

Part (a) Skill 1.A: Formulate Question

"Does replacing traditional seated desks with dynamic standing desks lead to a statistically significant increase in the mean academic attentiveness rating of high school students?"

Part (b) Skill 2.B: Random Assignment Protocol

1. Number the 40 available classrooms sequentially from 01

to 40.

2. Using a random number generator or statistical software, generate 20 unique pseudo-random integers between 01 and 40 without replacement.

3. The classrooms corresponding to these 20 numbers will receive the dynamic standing desks (Treatment Group). The remaining 20 classrooms will retain traditional seated desks (Control Group).

4. Following an 8-week intervention period, record attentiveness using an identical standardized rubric.

5. Compare the mean attentiveness scores between the two treatment conditions.

Part (c) Skill 2.D: Confounding & Randomized Block Design

Academic subject matter (e.g., Mathematics vs. Visual Arts) is a potential confounding variable because student attentiveness and natural physical movement vary systematically across subjects. If one desk type happens to be assigned to more physical education or art classes, the effect of desk type is confounded with academic subject matter.

Blocking Remedy: Group the 40 classrooms into homogeneous blocks by academic discipline (e.g., Block 1: STEM classes; Block 2: Humanities/Arts). Within each block, randomly assign half the classrooms to standing desks and half to traditional desks. This eliminates between-subject variability from the experimental error.

Part (d) Skill 2.E: Null & Alternative Hypotheses

Let μ_S be the true mean attentiveness score of classrooms with standing desks, and let μ_T be the true mean attentiveness score of classrooms with traditional desks:

$$H_0 : \mu_S = \mu_T \quad (\text{or } \mu_S - \mu_T = 0)$$

$$H_a : \mu_S > \mu_T \quad (\text{or } \mu_S - \mu_T > 0)$$

Grading Rubric Breakdown:

- **Essentially Correct (E):** Clearly formulates an investigative comparative question with variable and population, provides a fully reproducible random allocation procedure, thoroughly explains confounding with a valid blocking design, and states hypotheses correctly defining parameters in context.
- **Partially Correct (P):** Omits sampling without replacement in part (b), or identifies confounding without explaining how blocking isolates the variation, or writes hypotheses using sample statistics ($\bar{x}_S = \bar{x}_T$).
- **Incomplete (I):** Major conceptual confusion between blocking and stratified sampling.

Question 2: Multi-Focus on Practices 3 & 4 — Model Solution & Rubric

Part (a) Skill 3.B: Outlier Identification

Ordered sugar values ($n = 14$): $4, 6, 7, 8, 9, 10, 11, 12, 12, 13, 14, 15, 18, 26$.

Median $= (11 + 12)/2 = 11.5$ g. Lower half (first 7): $Q_1 = 8$ g. Upper half (last 7): $Q_3 = 14$ g.

IQR $= Q_3 - Q_1 = 14 - 8 = 6$ g.

Lower Fence $= 8 - 1.5(6) = 8 - 9 = -1$ g.

Upper Fence $= 14 + 1.5(6) = 14 + 9 = 23$ g.

Because 26 g $>$ 23 g, 26 grams is a confirmed high outlier. There are no low outliers (Min $= 4 \geq -1$).

Part (b) Skill 4.A: C-U-S-S Distribution Description

"The distribution of sugar content among these 14 breakfast cereals is skewed to the right with an isolated high outlier at 26 grams per serving. The center of the distribution is represented by a median of 11.5 grams (or mean of approximately 12.07 grams). The spread is characterized by an IQR of 6 grams and an overall range of 22 grams (from 4 g to 26 g)."

Part (c) Skill 3.B: Prediction & Residual Calculation

For $x = 12.0$ g: $\widehat{\text{Calories}} = 95.20 + 3.85(12.0) = 95.20 + 46.20 = 141.40$ calories.

Residual $= y - \hat{y} = 148.0 - 141.40 = +6.60$ calories.

Interpretation: The cereal contains 6.60 more calories per serving than predicted by the linear regression model based on its sugar content.

Part (d) Skill 4.B: Parameter & Model Interpretations

Slope (3.85): For each additional 1 gram increase in sugar content per serving, the model predicts that caloric content increases by approximately 3.85 calories.

Coefficient of Determination ($r^2 = 72.8\%$): Approximately 72.8% of the total variation in caloric content among these cereals is accounted for by the linear model with sugar content.

Grading Rubric Breakdown:

- **Essentially Correct (E):** Correct 5-number summary and mathematical justification of outlier, complete C-U-S-S description in context, correct prediction with positive residual interpretation, and flawless slope/r^2 interpretations with context and units.
- **Partially Correct (P):** Forgets units in residual or slope interpretation, or omits one C-U-S-S component.
- **Incomplete (I):** Arithmetic errors leading to incorrect fences or states slope deterministically ("calories will always increase by 3.85").

Question 3: Statistical Inference (Hypothesis Test) — Model Solution & Rubric

Step 1 (P): Parameter & Hypotheses

Let p be the true proportion of all admitted high school seniors who accept their admission offer and enroll at the university.

$$H_0 : p = 0.65 \quad \text{vs.} \quad H_a : p < 0.65$$

Step 2 (H & A): Name Procedure & Check Conditions

Procedure: One-proportion z-test for p.

- **Random:** An SRS of $n = 300$ admitted students was selected. (Satisfied)
- **10% Condition:** $300 \leq 0.10(\text{Total cohort of admitted students})$, assuming the university admitted at least 3,000 students.

(Satisfied)

- **Large Counts:** Under H_0, $np_0 = 300(0.65) = 195 \geq 10$ and $n(1-p_0) = 300(0.35) = 105 \geq 10$. Both expected counts exceed 10. (Satisfied)

Step 3 (N & T): Mechanics & Calculations
Sample proportion: $\hat{p} = \frac{180}{300} = 0.60$. Standard error under H_0:

$$\mathrm{SE}_0 = \sqrt{\frac{p_0(1-p_0)}{n}} = \sqrt{\frac{(0.65)(0.35)}{300}} = \sqrt{\frac{0.2275}{300}} \approx 0.02754$$

Standardized test statistic:

$$z = \frac{\hat{p} - p_0}{\mathrm{SE}_0} = \frac{0.60 - 0.65}{0.02754} = \frac{-0.05}{0.02754} \approx -1.82$$

p-value: $P(Z \leq -1.82) = 0.0344$.
Step 4 (O & M): Decision & Contextual Conclusion
Because the p-value (0.0344) is strictly less than the significance level $\alpha = 0.05$, we **reject the null hypothesis** H_0. There is convincing statistical evidence at the 5% level that the true enrollment proportion among admitted students is less than 65%.
Grading Rubric Breakdown:

- **Essentially Correct (E):** Correct parameter definition, all 3 conditions verified with numbers, correct test statistic $z = -1.82$ and p-value 0.0344, and correct contextual conclusion explicitly comparing p-value to α.

- **Partially Correct (P):** Uses $\hat{p}$ instead of p_0 in standard error formula or states hypotheses without defining p.
- **Incomplete (I):** Omits conditions or makes an incorrect decision ("accept H_0").

Question 4: Multi-Focus Synthesis on Practices 2, 3, & 4 — Model Solution & Rubric

Part (a): Residual & Slope Interpretation

Predicted score for $x = 4.0$: $\hat{y} = 42.0 + 7.5(4.0) = 42.0 + 30.0 = 72.0$ points.

Residual $= y - \hat{y} = 78.0 - 72.0 = +6.0$ points.

Interpretation: The student scored 6.0 points higher on the exam than predicted by study hours alone. The slope indicates that for each additional hour of daily study, the predicted exam score increases by 7.5 points.

Part (b): Omitted Variable Bias Analysis

In the multiple regression model, the slope for study hours drops from 7.5 to 6.0. The simple linear regression model suffered from **omitted variable bias**: students who study more may also manage their sleep differently, or sleep duration (w) independently explains score variability. Omitting sleep attributed some of the collective gains to study time alone, overestimating its marginal effect.

Part (c): Probability Model Calculation

$G \sim N(12.5, 3.2)$. Standardizing $g = 15.0$:

$$z = \frac{15.0 - 12.5}{3.2} = \frac{2.5}{3.2} \approx 0.78$$

$$P(G > 15.0) = P(Z > 0.78) = 1 - 0.7823 = 0.2177 \quad \text{(or 21.8\%)}$$

Interpretation: Approximately 21.8% of students are expected to achieve score gains exceeding 15.0 points.

Part (d): Constrained Optimization & Policy Recommendation

Given constraint $x + w \leq 14 \implies w = 14 - x$, with $w \geq 6 \implies x \leq 8$. Substitute into multiple regression equation:

$$\hat{y} = 20.0 + 6.0x + 3.5(14 - x) = 20.0 + 6.0x + 49.0 - 3.5x = 69.0 + 2.5x$$

Because the effective derivative with respect to study time is $+2.5 > 0$, predicted score is maximized by choosing the largest feasible value of x. Subject to minimum sleep $w \geq 6$, the maximum allowable study time is $x = 14 - 6 = 8.0$ hours. Maximum predicted score:

$$\hat{y}_{\max} = 69.0 + 2.5(8) = 69.0 + 20.0 = 89.0 \text{ points}$$

Recommendation: Students should allocate 8 hours to studying and 6 hours to sleep to maximize predicted test scores under the 14-hour daily budget.

Grading Rubric Breakdown:

- **Essentially Correct (E):** Correct residual calculation with context, sound conceptual explanation of omitted variable bias, accurate normal distribution probability with work shown, and mathematically rigorous linear optimization under constraints.

- **Partially Correct (P):** Correctly calculates parts (a) and (c) but fails to incorporate the boundary constraint $w \geq 6$ in part (d).
- **Incomplete (I):** Major misunderstandings in regression mechanics or normal probability.

Chapter 13

Practice Exam 2: Detailed Solutions & Grading Rubrics

13.1 Section I: Multiple-Choice Detailed Explanations

1. **Answer: (B)**
   Explanation: The sample standard deviation is $s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$. For s to equal exactly zero, the sum of squared deviations from the mean must be zero, meaning $(x_i - \bar{x})^2 = 0$ for every observation i. Hence, every observation in the sample must be identical to the mean ($x_i = 50$ for all i).

2. **Answer: (A)**
   Explanation: In a distribution that is skewed to the right (positively skewed), extreme large values in the long right tail pull the non-resistant mean upward, while the median remains resistant and anchored near the center of the data. Thus, Mean > Median.

3. **Answer: (C)**

Explanation: A correlation coefficient of $r = 0.0$ signifies the complete absence of a linear relationship between the two quantitative variables. It does not rule out strong non-linear or curved relationships (e.g., a perfect parabola).

4. **Answer: (B)**
   Explanation: In stratified random sampling, the population is partitioned into homogeneous subgroups (strata) of individuals who are similar with respect to a characteristic known to influence the response variable. Independent simple random samples are then drawn from within each stratum.

5. **Answer: (A)**
   Explanation: Two events A and B are independent if knowing that event B has occurred provides no information about the probability of event A, which is expressed mathematically by the condition $P(A \mid B) = P(A)$ (or equivalently $P(A \cap B) = P(A)P(B)$).

6. **Answer: (A)**
   Explanation: The four defining requirements for a binomial setting are summarized by the acronym **BINS**: Binary outcomes (Success or Failure), Independent trials, Number of trials n is fixed in advance, and Same probability of success p for each trial.

7. **Answer: (B)**
   Explanation: By the Central Limit Theorem, if the sample size is sufficiently large ($n \geq 30$), the sampling distribution of the sample mean $\bar{x}$ is approximately normal regardless of the shape of the underlying population distribution.

8. **Answer: (A)**
   Explanation: A Type I error occurs when a researcher rejects a null hypothesis that is actually true (a false positive finding). The maximum probability of committing a Type I error is fixed by the significance level α.

9. **Answer: (B)**
   Explanation: The correct interpretation of a 95% confidence level is: If we were to take many random samples of the same size from the same population and construct a 95% confidence interval from each sample, approximately 95% of the resulting intervals would capture the true population parameter.

10. **Answer: (A)**
    Explanation: In a Chi-Square Goodness-of-Fit test with k distinct categorical levels, the degrees of freedom are $df = k-1$, representing the number of categories minus one linear constraint (that the sum of expected counts must equal the sample size).

11. **Answer: (A)**
    Explanation: For a one-sample t-interval or t-test based on a sample of size $n = 30$, the degrees of freedom are $df = n - 1 = 30 - 1 = 29$.

12. **Answer: (A)**
    Explanation: When subjects are grouped by age prior to treatment assignment to account for known physiological differences in pain tolerance, age is functioning as a blocking variable. Blocking reduces unexplained experimental variability.

13. **Answer: (D)**
    Explanation: For a 99% confidence interval, the central area under the standard normal curve is 0.99, leaving an area of 0.005 in each tail. The critical value is $z^* = \text{invNorm}(0.995) \approx 2.576$.

14. **Answer: (A)**
    Explanation: A distribution with mean $\$450,000$ and median $\$320,000$ has Mean $>$ Median. The substantially higher mean demonstrates that high-priced luxury homes are stretching the upper tail, indicating a right-skewed distribution.

15. **Answer: (D)**
    Explanation: For a fair coin, the probability of flipping heads on any single toss is 0.5. Because tosses are independent:
    $$P(4 \text{ heads}) = (0.5)^4 = 0.0625$$

16. **Answer: (A)**
    Explanation: A high-leverage point in regression is an observation whose explanatory variable value (x) is substantially farther from the mean $\bar{x}$ than the other data points. It exerts strong potential leverage over the slope of the regression line.

17. **Answer: (A)**
    Explanation: The standard deviation of the sampling distribution of the sample mean $\bar{x}$ is:
    $$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{16}{\sqrt{64}} = \frac{16}{8} = 2.0$$

18. **Answer: (A)**

Explanation: Because the p-value (0.082) exceeds the significance level $\alpha = 0.05$, the sample data do not provide sufficiently extreme evidence against the null hypothesis. The correct statistical decision is to fail to reject H_0. (We never "accept" H_0 or claim it has been proven true).

19. **Answer: (A)**
    Explanation: In Chi-Square tests, the expected cell counts represent the hypothetical counts that would be anticipated if the null hypothesis were true in the population.

20. **Answer: (A)**
    Explanation: A matched pairs t-test analyzes the single list of within-pair differences ($d_i = x_{1i} - x_{2i}$). Mathematically and computationally, it is identical to a one-sample t-test on the sample of differences against $H_0 : \mu_d = 0$.

21. **Answer: (B)**
    Explanation: Under the Empirical Rule (68-95-99.7 Rule) for normal distributions, approximately 68% of all observations fall within ± 1 standard deviation of the mean ($\mu \pm 1\sigma$).

22. **Answer: (A)**
    Explanation: In cluster sampling, the population is divided into heterogeneous clusters (often based on geographic location). A random sample of entire clusters is chosen, and all individuals within the selected clusters are surveyed.

23. **Answer: (A)**
    Explanation: By definition, events A and B are independent if the conditional probability $P(A \mid B)$ is identical to the unconditional probability $P(A)$.

24. **Answer: (A)**
    Explanation: For a binomial random variable $X \sim B(n, p)$, the standard deviation is given by $\sigma = \sqrt{np(1-p)}$.

25. **Answer: (A)**
    Explanation: When a residual plot exhibits random scatter with no discernible pattern, curve, or systematic change in spread around the horizontal line Residual $= 0$, it confirms that a linear model is appropriate for the data.

26. **Answer: (A)**
    Explanation: The sample point estimate $\hat{p}$ lies exactly at the center of the symmetric confidence interval:

$$\hat{p} = \frac{0.40 + 0.60}{2} = 0.50$$

27. **Answer: (A)**
    Explanation: An alternative hypothesis containing a strict greater-than inequality ($H_a : \mu > 0$) focuses rejection exclusively on the upper tail of the sampling distribution, making it a one-sided (right-tailed) test.

28. **Answer: (A)**
    Explanation: A placebo is an inactive or dummy treatment administered to the control group to equalize psychological expectations across treatment arms, isolating the true physiological efficacy of the active treatment from the psychological placebo effect.

29. **Answer: (A)**
    Explanation: The Large Counts condition requires $np \geq 10$

and $n(1-p) \geq 10$. Here, $n = 100$ and $p = 0.50$:

$$np = 100(0.50) = 50 \geq 10 \quad \text{and} \quad n(1-p) = 100(0.50) = 50 \geq 10$$

Both conditions are satisfied, so the sampling distribution of $\hat{p}$ is approximately normal.

30. **Answer: (A)**
Explanation: In simple linear regression with n data pairs, two degrees of freedom are consumed estimating the intercept and slope parameters, leaving $df = n - 2$ for the t-test of the slope β.

31. **Answer: (A)**
Explanation: The interquartile range is defined as the distance between the third quartile and the first quartile: $\text{IQR} = Q_3 - Q_1$, measuring the spread of the middle 50% of the data.

32. **Answer: (A)**
Explanation: In a matched pairs experimental design where each individual serves as their own control, every subject receives both experimental treatments in a randomized order to counterbalance sequence or carryover effects.

33. **Answer: (A)**
Explanation: The test statistic for regression slope is $t = \frac{b_1 - 0}{\text{SE}(b_1)} = \frac{-2.5}{0.5} = -5.0$.

34. **Answer: (A)**
Explanation: The residual is defined as $\text{Residual} = y - \hat{y}$. If the residual is negative ($y - \hat{y} < 0$), then $y < \hat{y}$, meaning the observed actual point lies strictly below the predicted regression line.

35. **Answer: (B)**
Explanation: Decreasing sample size increases standard errors and widens sampling distributions, increasing the overlap between null and alternative distributions. This inflates the Type II error rate β and directly decreases the power of the test $(1-\beta)$.

36. **Answer: (A)**
Explanation: By definition, $z = \frac{x-\bar{x}}{s}$. If $z = 0$, then $x - \bar{x} = 0 \implies x = \bar{x}$. The observation equals the mean.

37. **Answer: (A)**
Explanation: The 10% condition requires $n \leq 0.10N$. Here, $n = 100$ and $N = 1000$. $0.10(1000) = 100$. Because $100 \leq 100$, the condition is satisfied (barely meeting the upper boundary), justifying the independence assumption.

38. **Answer: (A)**
Explanation: The sample proportion $\hat{p}$ is an unbiased estimator of the population proportion p because the expected value of its sampling distribution equals p ($\mu_{\hat{p}} = p$).

39. **Answer: (A)**
Explanation: The least-squares regression line is mathematically derived specifically to minimize the sum of the squared vertical residuals, $\sum(y_i - \hat{y}_i)^2$.

40. **Answer: (A)**
Explanation: An event A and its complement A^c are mutually exclusive (disjoint) by definition, sharing no outcomes in common. Therefore, the intersection $A \cap A^c = \emptyset$, and $P(A \cap A^c) = 0.00$.

41. **Answer: (D)**

Explanation: The coefficient of determination is R-sq $=$ $80.0\% = 0.800$. The magnitude of the correlation coefficient is $|r| = \sqrt{0.800} \approx 0.8944$. Because the regression slope is negative ($b_1 = -5.12$), the correlation coefficient must have the same sign as the slope:

$$r = -\sqrt{0.800} \approx -0.894$$

Option (C) $+0.894$ fails to match the negative slope direction.

42. **Answer: (A)**

Explanation: For simple linear regression inference on slope β, degrees of freedom are $df = n - 2 = 18 - 2 = 16$. With $df = 16$ at a 95% confidence level, the critical value from the t-table is $t^* = 2.120$. The confidence interval formula is:

$$b_1 \pm t^* \cdot \text{SE}(b_1) \implies -5.12 \pm 2.120(0.640)$$

Option (B) incorrectly uses $df = n - 1 = 17$ ($t^* = 2.101$). Option (C) incorrectly uses normal critical value $z^* = 1.960$. Option (D) incorrectly uses s instead of $\text{SE}(b_1)$.

13.2 Section II: Free-Response Complete Scoring Guidelines

Question 1: Multi-Focus on Practices 1 & 2 — Model Solution & Rubric

Part (a) Skill 1.A: Formulate Question

"Does postoperative rehabilitation with Protocol Alpha (hydrotherapy) result in a statistically significant reduction in mean recovery time (in days) compared to Protocol Beta (land-based therapy) among adult knee replacement patients?"

Part (b) Skill 2.B: Justification for Randomized Experiment

Allowing patients to self-select their rehabilitation regimen introduces severe selection bias and confounding. For example, younger, more physically active, or highly motivated patients might disproportionately choose the hydrotherapy regimen. Any observed differences in recovery time could then be caused by pre-existing patient health, age, or fitness rather than the rehabilitation protocol itself. Random assignment balances these known and lurking confounding variables across both treatment groups, establishing a valid basis for causal inference.

Part (c) Skill 2.D: Blocking by Patient Age

Patient age is strongly associated with bone healing, tissue regeneration, and joint mobility. By grouping patients into homogeneous age blocks (e.g., Block 1: Under 60; Block 2: 60 and older) and randomly assigning Protocol Alpha and Protocol Beta within each block, the researchers isolate and remove the variability in recovery times attributable to age from the experimental error. This increases the statistical power of the experiment to detect a genuine treatment effect.

Part (d) Skill 2.E: Hypotheses

Let μ_α be the true mean recovery time (in days) for knee replacement patients using Protocol Alpha, and let μ_β be the true mean recovery time using Protocol Beta:

$$H_0 : \mu_\alpha = \mu_\beta \quad (\text{or } \mu_\alpha - \mu_\beta = 0)$$

$$H_a : \mu_\alpha < \mu_\beta \quad (\text{or } \mu_\alpha - \mu_\beta < 0)$$

Grading Rubric Breakdown:

- **Essentially Correct (E):** Precisely formulates an investigative comparative question with variable and population, clearly explains confounding and why random assignment enables causal inference, explains how blocking isolates age variability, and correctly states hypotheses defining parameters in context.
- **Partially Correct (P):** Explains random assignment without mentioning confounding, or states hypotheses without defining μ_α and μ_β.
- **Incomplete (I):** Confuses blocking with random sampling.

Question 2: Multi-Focus on Practices 3 & 4 — Model Solution & Rubric

Part (a) Skill 3.B: Prediction of Light Intensity

Given the transformed model $\widehat{\ln(y)} = 6.42 - 0.58x$. At depth $x = 5.0$ (500 meters):

$$\widehat{\ln(y)} = 6.42 - 0.58(5.0) = 6.42 - 2.90 = 3.52$$

Exponentiating both sides to solve for predicted light intensity $\hat{y}$:

$$\hat{y} = e^{3.52} \approx 33.78 \text{ lumens/m}^2$$

Part (b) Skills 3.B & 4.D: Residual in Transformed Scale

For $x = 3.0$: $\widehat{\ln(y)} = 6.42 - 0.58(3.0) = 6.42 - 1.74 = 4.68$.
Observed $y = 120$ lumens/m^2 $\implies \ln(120) \approx 4.7875$.

$$\text{Residual} = \text{Observed} - \text{Predicted} = 4.7875 - 4.68 = +0.1075$$

Interpretation: The observed natural logarithm of light intensity at 300 meters was 0.1075 units greater than predicted by the linear regression model.

Part (c) Skill 4.A: Importance of Residual Plot Analysis

Examining the residual plot of the raw (x, y) data versus the transformed $(x, \ln(y))$ data is crucial because a high r^2 value alone does not guarantee model adequacy. A raw scatterplot may exhibit a strong non-linear curve that still yields a high correlation. If the residual plot of the raw data displays a distinct curved pattern, a linear model is invalid. If the residual plot of the transformed data displays a random scatter of points with no discernible pattern and constant variance around the zero residual line, the log-transformed linear model is validated as appropriate.

Part (d) Skill 4.B: Interpretation of $r^2 = 94.1\%$

Approximately 94.1% of the total variation in the natural logarithm of underwater ambient light intensity is accounted for by the linear model with ocean depth (in hundreds of meters).

Grading Rubric Breakdown:

- **Essentially Correct (E):** Correctly computes $\hat{y} = e^{3.52} \approx 33.78$ with units, computes transformed residual with context, articulates why residual plots detect curvature where r^2 cannot, and interprets r^2 correctly referencing the transformed response.
- **Partially Correct (P):** Forgets to exponentiate in part (a) or calculates residual on untransformed scale $(120 - e^{4.68})$.
- **Incomplete (I):** Major misconceptions regarding logarithmic transformations.

Question 3: Statistical Inference (Two-Proportion CI) — Model Solution & Rubric

Step 1 (P): Parameter & Procedure

Let p_α be the true proportion of all active OS-Alpha users who are highly satisfied, and let p_β be the true proportion of all active OS-Beta users who are highly satisfied. Procedure: Two-sample z-interval for $p_\alpha - p_\beta$.

Step 2 (A): Condition Verification

- **Random:** Independent random samples of 400 OS-Alpha and 500 OS-Beta users were selected. (Satisfied)
- **10% Condition:** $400 \leq 0.10(\text{All OS-Alpha users})$ and $500 \leq 0.10(\text{All OS-Beta users})$, as both operating systems have millions of active users. (Satisfied)
- **Large Counts:** Sample successes and failures: $n_\alpha \hat{p}_\alpha = 328 \geq 10$, $n_\alpha(1-\hat{p}_\alpha) = 72 \geq 10$; $n_\beta \hat{p}_\beta = 375 \geq 10$, $n_\beta(1-\hat{p}_\beta) = 125 \geq 10$. All counts are ≥ 10. (Satisfied)

Step 3 (N): Interval Calculation

Sample proportions: $\hat{p}_\alpha = \frac{328}{400} = 0.82$, $\hat{p}_\beta = \frac{375}{500} = 0.75$. Point estimate: $\hat{p}_\alpha - \hat{p}_\beta = 0.82 - 0.75 = 0.070$. Standard error (unpooled for confidence intervals):

$$\text{SE} = \sqrt{\frac{(0.82)(0.18)}{400} + \frac{(0.75)(0.25)}{500}} = \sqrt{\frac{0.1476}{400} + \frac{0.1875}{500}} = \sqrt{0.000369}$$

For 95% confidence, critical value $z^* = 1.960$:

$$\text{Margin of Error} = 1.960 \times 0.02728 \approx 0.0535$$

$$\text{Confidence Interval} = 0.070 \pm 0.0535 \implies (0.0165,\ 0.1235)$$

Step 4 (I): Contextual Interpretation & Claim Evaluation

Interpretation: We are 95% confident that the true difference in the proportion of satisfied users between OS-Alpha and OS-Beta ($p_\alpha - p_\beta$) is between 0.0165 and 0.1235 (or between 1.65% and 12.35%).

Claim Justification: Because the entire 95% confidence interval consists entirely of positive values and does not include 0, there is convincing statistical evidence at the 5% significance level that OS-Alpha achieves a higher satisfaction rate than OS-Beta.

Grading Rubric Breakdown:

- **Essentially Correct (E):** Correct parameter definitions, all 3 conditions verified with numbers, correct unpooled SE and interval, and correct contextual interpretation with justification based on zero exclusion.
- **Partially Correct (P):** Uses pooled proportions in SE formula (pooled is strictly for hypothesis tests, unpooled for confidence intervals) or states generic conclusion without context.
- **Incomplete (I):** Formula or condition errors.

Question 4: Multi-Focus Synthesis on Practices 2, 3, & 4 — Model Solution & Rubric

Part (a) Skill 3.C: Total Probability of Testing Positive $P(+)$

Using the Law of Total Probability:

$$P(+) = P(D) \times P(+ \mid D) + P(D^c) \times P(+ \mid D^c)$$

Given $P(D) = 0.02 \implies P(D^c) = 0.98$. Sensitivity $P(+ \mid$

$D) = 0.95$. False positive rate $P(+ \mid D^c) = 1 - \text{Specificity} = 1 - 0.98 = 0.02$.

$P(+) = (0.02)(0.95) + (0.98)(0.02) = 0.0190 + 0.0196 = 0.0386$ (or 3.86

Interpretation: Approximately 3.86% of individuals in the population will receive a positive test result.

Part (b) Skills 3.C & 4.D: Positive Predictive Value (PPV)

By Bayes' Theorem:

$$P(D \mid +) = \frac{P(D \cap +)}{P(+)} = \frac{0.0190}{0.0386} \approx 0.4922 \quad \text{(or 49.2\%)}$$

Interpretation: When an individual tests positive, the probability that they actually have the condition is approximately 49.2% (less than half).

Explanation: Because the condition is rare (prevalence of only 2%), the large healthy cohort (98%) generates false positives (0.0196) that exceed the true positives (0.0190) from the small infected cohort. High accuracy on healthy individuals still produces substantial false positives when the baseline base rate is very low.

Part (c) Practices 2, 3, & 4: Expected Payoff Comparison for 1,000 Individuals

In a cohort of $N = 1,000$ individuals:

- Truly diseased: $1,000 \times 0.02 = 20$ people.
- Truly healthy: $1,000 \times 0.98 = 980$ people.

Expected clinical outcomes under testing: 1. True Positives (treated): $20 \times 0.95 = 19$ individuals. Payoff:

$19 \times (+\$10,000) = +\$190,000$.

2. False Negatives (missed): $20 \times 0.05 = 1$ individual. Payoff: $1 \times (-\$50,000) = -\$50,000$.

3. False Positives (overtreated): $980 \times 0.02 = 19.6$ individuals. Payoff: $19.6 \times (-\$2,500) = -\$49,000$.

4. True Negatives (correct non-treatment): $980 \times 0.98 = 960.4$ individuals. Payoff: $960.4 \times \$0 = \$0$.

Net Expected Payoff with Testing:

$$E(\text{Payoff}_{\text{Test}}) = +\$190,000 - \$50,000 - \$49,000 + \$0 = +\$91,000$$

Payoff Treating No One (Baseline): All 20 diseased individuals remain untreated and progress to severe illness:

$$E(\text{Payoff}_{\text{No Screening}}) = 20 \times (-\$50,000) = -\$1,000,000$$

Policy Recommendation: The health department should adopt the rapid screening test. The screening protocol delivers a net positive payoff of $+\$91,000$ per 1,000 people, generating a relative net health and financial gain of $\$1,091,000$ over the untreated baseline.

Grading Rubric Breakdown:

- **Essentially Correct (E):** Correct probability tree/total probability for $P(+) = 0.0386$, correct PPV $\approx 49.2\%$ with insightful explanation of base-rate fallacy, and comprehensive payoff matrix calculation comparing testing ($+\$91,000$) to the untreated baseline ($-\$1,000,000$).

- **Partially Correct (P):** Computes $P(+)$ and PPV correctly but miscalculates the payoff matrix or omits the baseline comparison.
- **Incomplete (I):** Confusion regarding sensitivity and conditional probabilities.

Appendix A

Appendix A: AP Statistics Formula Reference Sheet

A.1 Descriptive Statistics & One-Variable Data

- **Sample Mean:**

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

- **Sample Standard Deviation:**

$$s_x = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

- **Standardized Score (z-Score):**

$$z = \frac{x-\mu}{\sigma} \quad \text{or} \quad z = \frac{x-\bar{x}}{s_x}$$

- **Interquartile Range & Outlier Fences:**

$$\text{IQR} = Q_3 - Q_1$$

Lower Fence $= Q_1 - 1.5(\text{IQR})$, Upper Fence $= Q_3 + 1.5(\text{IQR})$

A.2 Two-Variable Data & Linear Regression

- **Least-Squares Regression Line (LSRL):**

$$\hat{y} = b_0 + b_1 x$$

- **Slope of Regression Line:**

$$b_1 = r\left(\frac{s_y}{s_x}\right)$$

- **y-Intercept of Regression Line:**

$$b_0 = \bar{y} - b_1\bar{x}$$

- **Correlation Coefficient:**

$$r = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{x_i - \bar{x}}{s_x}\right)\left(\frac{y_i - \bar{y}}{s_y}\right)$$

- **Residual:**

$$e_i = y_i - \hat{y}_i = \text{Observed } y - \text{Predicted } y$$

- **Standard Deviation of Residuals:**

$$s = \sqrt{\frac{\sum(y_i - \hat{y}_i)^2}{n-2}}$$

A.3 Probability & Random Variables

- **Addition Rule for Disjoint Events:**

$$P(A \cup B) = P(A) + P(B)$$

- **General Addition Rule:**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- **Conditional Probability:**

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \quad (P(B) > 0)$$

- **Multiplication Rule for Independent Events:**

$$P(A \cap B) = P(A) \times P(B)$$

- **Discrete Random Variable Parameters:**

$$\mu_X = E(X) = \sum x_i \cdot P(X = x_i)$$

$$\sigma_X^2 = \mathrm{Var}(X) = \sum (x_i - \mu_X)^2 \cdot P(X = x_i)$$

- **Linear Transformation $Y = a + bX$:**

$$\mu_Y = a + b\mu_X, \qquad \sigma_Y = |b|\sigma_X, \qquad \sigma_Y^2 = b^2\sigma_X^2$$

- **Linear Combination of Independent Random Vari-**

ables X and Y:

$$\mu_{X\pm Y} = \mu_X \pm \mu_Y, \qquad \sigma^2_{X\pm Y} = \sigma^2_X + \sigma^2_Y, \qquad \sigma_{X\pm Y} = \sqrt{\sigma^2_X + \sigma^2_Y}$$

- **Binomial Distribution $X \sim B(n,p)$:**

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$\mu_X = np, \qquad \sigma_X = \sqrt{np(1-p)}$$

- **Geometric Distribution $X \sim$ Geom(p):**

$$P(X = k) = (1-p)^{k-1} p$$

$$\mu_X = \frac{1}{p}, \qquad \sigma_X = \frac{\sqrt{1-p}}{p}$$

A.4 Sampling Distributions & Inference Foundations

- **General Confidence Interval Structure:**

$$\text{Statistic} \pm (\text{Critical Value}) \times (\text{Standard Error of Statistic})$$

- **General Significance Test Statistic Structure:**

$$\text{Test Statistic} = \frac{\text{Statistic} - \text{Null Value}}{\text{Standard Error of Statistic}}$$

- **Proportion Parameters & Standard Errors:**
  - *One Proportion:* $\mu_{\hat{p}} = p$, $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$, $\text{SE}(\hat{p}) =$

$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

- *Two Proportions (Interval):* $\mathrm{SE}(\hat{p}_1-\hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$
- *Two Proportions (Test, Pooled):* $\hat{p}_c = \frac{x_1+x_2}{n_1+n_2}$, $\mathrm{SE}_c = \sqrt{\hat{p}_c(1-\hat{p}_c)\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}$

- **Mean Parameters & Standard Errors:**
  - *One Mean:* $\mu_{\bar{x}} = \mu$, $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$, $\mathrm{SE}(\bar{x}) = \frac{s_x}{\sqrt{n}}$, $df = n-1$
  - *Paired Differences:* $\mu_{\bar{x}_d} = \mu_d$, $\mathrm{SE}(\bar{x}_d) = \frac{s_d}{\sqrt{n}}$, $df = n-1$
  - *Two Independent Means:* $\mathrm{SE}(\bar{x}_1-\bar{x}_2) = \sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}$, $df = \min(n_1-1, n_2-1)$

- **Chi-Square Test Statistic:**

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

$$\text{Expected Count (Two-Way Tables)} = \frac{(\text{Row Total}) \times (\text{Column Total}}{\text{Table Total}}$$

- **Inference for Regression Slope β:**

$$\text{Confidence Interval:} \quad b_1 \pm t^* \cdot \mathrm{SE}(b_1), \quad df = n-2$$

$$\text{Test Statistic:} \quad t = \frac{b_1 - 0}{\mathrm{SE}(b_1)}, \quad df = n-2$$

Appendix B

Appendix B: Master AP Statistics Glossary

Alpha (α) Level

The threshold probability chosen by the researcher below which the p-value warrants rejecting the null hypothesis; represents the maximum probability of committing a Type I error.

Alternative Hypothesis (H_a)

The claim about the population parameter that the researcher seeks to establish evidence in favor of; can be one-sided ($>$, $<$) or two-sided ($\neq$).

Back-to-Back Stemplot

A graphical display comparing two quantitative distributions using a single common stem in the center with leaves extending left and right.

Bayes' Theorem

A mathematical formula for determining conditional probability: $P(A \mid B) = \frac{P(B|A)P(A)}{P(B)}$.

Bell Curve

The characteristic symmetric, unimodal, bell-shaped density curve representing the Gaussian or normal probability distribution.

Bias

Systematic favoritism that causes sample results or parameter estimates to deviate consistently in one direction from the true population value.

Bimodal

A distribution displaying two prominent distinct peaks or local maxima.

Binomial Distribution

The probability distribution of the number of successes k in n independent trials, each with constant success probability p.

BINS Checklist

The four requirements for a binomial setting: Binary outcomes, Independent trials, Number of trials fixed, Same probability of success.

BITS Checklist

The four requirements for a geometric setting: Binary outcomes, Independent trials, Trials continue until first success, Same probability of success.

Blinding

An experimental control procedure where subjects (single-blind) or both subjects and evaluators (double-blind) do not know which treatment is assigned.

Blocking

The grouping of experimental units into homogeneous subsets based on a known extraneous source of variability prior to random assignment of treatments.

Boxplot

A visual display representing the five-number summary (Min, Q_1, Median, Q_3, Max) of a quantitative dataset.

Categorical Variable

A variable that places an individual into one of several non-numerical categories or qualitative groups.

Census

An exhaustive study that attempts to collect data from every single member of an entire population.

Central Limit Theorem (CLT)

The theorem stating that for any population with finite mean and variance, the sampling distribution of the sample mean $\bar{x}$ becomes approximately normal as sample size $n \geq 30$.

Chi-Square Distribution

A family of right-skewed, strictly positive continuous distributions parameterized by degrees of freedom, used for testing categorical counts and variances.

Chi-Square Goodness-of-Fit Test

A significance test evaluating whether a single categorical sample conforms to a hypothesized population distribution across k categories ($df = k - 1$).

Chi-Square Test for Homogeneity

A test evaluating whether the distribution of a single categorical variable is identical across two or more independent populations or treatment groups.

Chi-Square Test for Independence

A test evaluating whether two categorical variables measured on a single sample from one population are statistically independent.

Cluster Sampling

A probability sampling design where the population is partitioned into heterogeneous geographic clusters, a random sample of clusters is chosen, and all units in selected clusters are observed.

Coefficient of Determination (r^2)

The fraction of total variation in the response variable y that is explained by the linear relationship with explanatory variable x.

Complement Rule

The probability that an event A does not occur is $P(A^c) = 1 - P(A)$.

Completely Randomized Design

An experimental design where all experimental units are allocated among all treatment groups purely through a chance mechanism.

Conditional Probability

The probability of an event A occurring given that another event B is already known to have occurred: $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$.

Confidence Interval

An interval of plausible values for an unknown population parameter, computed from sample data in the form Estimate $\pm$ Margin of Error.

Confidence Level (C)

The success rate of the estimation method: the percentage of all possible samples of size n that produce intervals containing the true parameter.

Confounding Variable

An extraneous variable that is associated with the explanatory variable and also influences the response variable, obscuring whether the treatment caused the observed outcome.

Continuous Quantitative Variable

A numerical variable that can take on any real value within a continuous interval (e.g., height, weight, time, temperature).

Control Group

A baseline comparison group in an experiment that receives an inactive placebo, standard treatment, or no treatment at all.

Convenience Sampling

A non-probability sampling method that selects individuals who are easiest to reach, introducing substantial sampling bias.

Correlation Coefficient (r)

A numerical measure between -1 and $+1$ of the strength and direction of a linear relationship between two quantitative variables.

Critical Value

The number of standard errors separating the parameter from the boundary of the rejection region in a hypothesis test or confidence interval (z^*, t^*, χ^*).

CUSS Framework

The mandatory AP acronym for describing quantitative distributions: Center, Unusual features, Shape, Spread.

Degrees of Freedom (df)

The number of independent pieces of information remaining in a statistic after estimating one or more sample parameters.

Density Curve

A mathematical curve with total area equal to 1.0 that models the probability distribution of a continuous random variable.

Discrete Quantitative Variable

A numerical variable that can take on only isolated, countable values (typically integers, e.g., number of siblings, defects).

Disjoint Events

Events that have no outcomes in common and cannot occur simultaneously: $P(A \cap B) = 0$. Also called mutually exclusive.

Dotplot

A simple graphical display where each data value is represented as a dot plotted above its numerical coordinate on an axis.

Double-Blind

An experimental protocol where neither the participating subjects nor the clinical evaluators know which treatment each subject is receiving.

Empirical Rule (68-95-99.7)

For normal distributions, approximately 68% of values lie within $\mu \pm \sigma$, 95% within $\mu \pm 2\sigma$, and 99.7% within $\mu \pm 3\sigma$.

Expected Value ($E(X)$)

The long-run theoretical weighted average value of a random variable across infinite repeated trials: $\mu_X = \sum x_i P(x_i)$.

Experiment

A scientific investigation where researchers deliberately impose specific treatments on experimental units to observe the resulting responses.

Explanatory Variable

The independent variable (x) that researchers suspect may explain, predict, or cause changes in the response variable.

Extrapolation

The hazardous practice of using a regression model to make predictions for x-values outside the range of the observed training data.

Five-Number Summary

The foundational summary of a quantitative dataset comprising: Minimum, First Quartile (Q_1), Median, Third Quartile (Q_3), and Maximum.

Geometric Distribution

The probability distribution of the number of Bernoulli trials required to achieve the first success ($X \sim \text{Geom}(p)$).

Histogram

A bar-like display for continuous quantitative data where the horizontal axis is divided into contiguous equal-width class intervals and bar heights represent frequencies.

Independence (Events)

Events A and B are independent if $P(A \mid B) = P(A)$ or $P(A \cap B) = P(A)P(B)$.

Influential Point

A data point in regression that substantially alters the slope, intercept, or correlation coefficient when removed.

Interquartile Range (IQR)

The distance between the upper and lower quartiles ($Q_3 - Q_1$), capturing the spread of the central 50% of the data.

Law of Large Numbers

As the number of repetitions of a chance experiment increases, the empirical relative frequency of an event approaches its theoretical probability.

Least-Squares Regression Line (LSRL)

The unique line that minimizes the sum of squared vertical residuals: $\min \sum (y_i - \hat{y}_i)^2$.

Leverage

A measure of how far an observation's explanatory variable value (x) lies from the mean $\bar{x}$; high-leverage points have great potential to steer the regression line.

Linear Transformation

Converting a variable by multiplying by a constant and/or adding a constant: $Y = a + bX$.

LINER Conditions

The five conditions required for linear regression inference: Linear, Independent, Normal residuals, Equal variance, Random sampling.

Margin of Error

The half-width of a confidence interval representing the maximum expected sampling error: ME = (Critical Value) $\times$ (SE).

Matched Pairs Design

A randomized block experiment where subjects are paired in sets of two based on common characteristics, or each subject receives both treatments in random order.

Mean ($\bar{x}$)

The arithmetic average of a dataset, calculated as the sum of all values divided by the total count n; non-resistant to outliers.

Median (M)

The middle value of an ordered dataset that divides the upper 50% of observations from the lower 50%; resistant to outliers.

Nonresponse Bias

Bias introduced when individuals selected for a sample fail or refuse to provide responses, leading to systematic nonresponse errors.

Normal Distribution

A continuous symmetric, unimodal probability distribution completely specified by its mean μ and standard deviation σ ($N(\mu, \sigma)$).

Null Hypothesis (H_0)

The skeptical baseline statement asserting that no difference, no effect, or no relationship exists in the population.

Observational Study

A study in which researchers observe individuals and record variables without attempting to influence or manipulate responses.

One-Sided Test

A hypothesis test where the alternative hypothesis specifies a directional effect ($H_a : \theta > \theta_0$ or $H_a : \theta < \theta_0$).

Outlier

An extreme observation that lies significantly outside the general pattern of the data; quantitatively identified by the $1.5 \times \text{IQR}$ rule.

***p*-Value**

The probability, assuming the null hypothesis is true, of observing a test statistic as extreme as or more extreme than the observed sample value.

Paired *t*-Test

A one-sample t-test conducted on the list of differences between matched pairs of quantitative measurements.

PANIC Framework

The standard AP rubric for constructing confidence intervals: Parameter, Assumptions/Conditions, Name test, Interval, Conclusion.

Parameter

A fixed numerical characteristic of an entire population (e.g., μ, σ, p, β), typically unknown.

PHANTOM Framework

The standard AP rubric for hypothesis tests: Parameter, Hypotheses, Assumptions, Name test, Test statistic, Obtain p-value, Make decision.

Placebo

An inert, harmless dummy treatment designed to mimic the active experimental intervention without pharmacological effect.

Placebo Effect

The well-documented psychological phenomenon where patients experience genuine perceived improvement simply from believing they are receiving treatment.

Point Estimate

A single numerical statistic computed from sample data that serves as the best guess for an unknown population parameter.

Power $(1 - \beta)$

The probability that a significance test will correctly reject a false null hypothesis at a chosen significance level α.

Quartiles

The values that partition an ordered dataset into four quarters: Q_1 (25^{th} percentile) and Q_3 (75^{th} percentile).

Random Assignment

The allocation of experimental units to treatments purely by chance, creating comparable treatment groups and permitting causal inference.

Random Sampling

Selecting individuals from a population by chance, giving each unit a known non-zero probability of inclusion, permitting generalization.

Random Variable

A numerical variable whose values are determined by the outcomes of a random phenomenon.

Range

The difference between the maximum and minimum values in a dataset (Max − Min); highly non-resistant to outliers.

Replication

The practice of assigning each treatment to many experimental units to ensure that effects can be distinguished from chance variation.

Residual

The vertical difference between an observed data value and the value predicted by the regression equation: $e = y - \hat{y}$.

Residual Plot

A scatterplot of residuals plotted on the vertical axis against the explanatory variable (or predicted values) on the horizontal axis.

Resistant Measure

A numerical summary statistic (such as median or IQR) that is not substantially affected by extreme outliers or skewness.

Response Bias

Systematic error caused by misleading question wording, interviewer appearance, or dishonest answers to sensitive questions.

Response Variable

The dependent outcome variable (y) whose values researchers seek to measure, predict, or explain.

Sample

A representative subset of individuals selected from a population to gather information.

Sampling Distribution

The probability distribution of values taken by a sample statistic across all possible random samples of the same size n from the population.

Sampling Variability

The natural variation observed in statistics from one random sample to another.

Scatterplot

A two-dimensional Cartesian graph plotting pairs of quantitative observations (x, y) to display association patterns.

Significance Level (α)

The predetermined probability cutoff used as the standard for rejecting H_0.

Simple Random Sample (SRS)

A sampling design where every subset of n individuals from the population has an equal chance of being selected.

Skewed Left

A distribution whose left tail extends farther than its right tail; typically Mean < Median.

Skewed Right

A distribution whose right tail extends farther than its left tail; typically Mean > Median.

Slope (b_1)

The rate of change in the predicted response variable for each one-unit increase in the explanatory variable: $b_1 = r\left(\frac{s_y}{s_x}\right)$.

Standard Deviation (s_x)

The square root of sample variance, measuring the typical distance that observations deviate from the mean.

Standard Error (SE)

An estimate of the standard deviation of a statistic's sampling distribution based on sample data.

Standard Normal Distribution

The unique normal distribution with mean $\mu = 0$ and standard deviation $\sigma = 1$, denoted $Z \sim N(0, 1)$.

Statistic

A numerical measure calculated from sample data (e.g., $\bar{x}, s_x, \hat{p}, b_1$), used to estimate population parameters.

Stemplot

A graphical display where each observation is separated into a stem (leading digits) and a leaf (final digit).

Stratified Random Sampling

A sampling method where the population is divided into homogeneous strata and an SRS is drawn from within each stratum.

Systematic Sampling

A probability sampling method where individuals are selected at regular fixed intervals after a random starting point (every k^{th} person).

t-Distribution

A family of symmetric, bell-shaped continuous probability distributions parameterized by degrees of freedom, having heavier tails than the normal curve.

Two-Sided Test

A hypothesis test where the alternative hypothesis posits that the parameter differs from the null value in either direction ($H_a : \theta \neq \theta_0$).

Type I Error

The error committed by rejecting a true null hypothesis (false positive); probability is α.

Type II Error

The error committed by failing to reject a false null hypothesis (false negative); probability is β.

Undercoverage Bias

Bias introduced when certain members of the population are systematically omitted from the sampling frame.

Unimodal

A distribution displaying exactly one single prominent peak.

Variance (s^2)

The mean squared deviation of observations from their sample mean; equal to standard deviation squared.

Voluntary Response Bias

Bias that occurs when sample individuals self-select themselves by volunteering to participate.

z-Score

The standardized value indicating how many standard deviations an observation lies above or below the mean: $z = \frac{x-\mu}{\sigma}$.

Appendix C

Appendix C: TI-84 CE Graphing Calculator Master Playbook

C.1 Essential Calculator Setups & Data Management

- **Diagnostic Mode Setup:** Press `[2nd] [0]` (CATALOG) → Scroll to `DiagnosticOn` → Press `[ENTER]` twice. Ensures r and r^2 are displayed when computing linear regressions.
- **Data Entry into Lists:** Press `[STAT]` → Select `1:Edit...` → Type values into `L1`, `L2`, etc. To clear a list, highlight the list name at the top, press `[CLEAR]`, then press `[ENTER]`.
- **One-Variable Statistics:** Press `[STAT]` → Arrow right to `CALC` → Select `1:1-Var Stats`. Set `List: L1`, leave `FreqList` blank (unless using frequency table) → `Calculate`. Displays $\bar{x}, \sum x, s_x, \sigma_x, n$, Min, Q_1, Med, Q_3, Max.

C.2 Two-Variable Regression Procedures

- **Scatterplot Configuration:** Press `[2nd] [Y=]` (STAT PLOT) → Select `1:Plot1` → Turn `On`. Choose the first icon (Scatterplot). Set `Xlist: L1`, `Ylist: L2`. Press `[ZOOM] [9]` (ZoomStat) to automatically auto-scale the viewing window.

- **Least-Squares Regression Line:** Press `[STAT]` → Arrow to `CALC` → Select `8:LinReg(a+bx)`. Set `Xlist: L1`, `Ylist: L2`. To automatically store the regression equation in Y_1, set `Store RegEQ: Y1` (press `[VARS]` → `Y-VARS` → `1:Function` → `1:Y1`).

- **Residual Plot Generation:** After computing regression, the calculator automatically populates list `RESID`. Press `[2nd] [Y=]` → Select `Plot1` → Change `Ylist` to `RESID` (press `[2nd] [STAT]` (LIST) → Select `RESID`). Press `[ZOOM] [9]`.

C.3 Probability & Distribution Functions

- **Normal Curve Area (`normalcdf`):** Press `[2nd] [VARS]` (DISTR) → Select `2:normalcdf`. Syntax: `normalcdf(lower, upper,` μ`,` σ`)`. For standard normal, default $\mu = 0, \sigma = 1$. Use -10^{99} (`-1E99`) for $-\infty$ and 10^{99} (`1E99`) for $+\infty$.

- **Inverse Normal Values (`invNorm`):** Press `[2nd] [VARS]` → Select `3:invNorm`. Syntax: `invNorm(area,` μ`,` σ`)`. Finds the value x having the specified cumulative area to its left.

- **Student's t Distribution (`tcdf & invT`):** `tcdf(lower, upper, df)` computes area under t-curve. `invT(area, df)` finds the critical value t^* for a given left-tail area and degrees of freedom.

- **Binomial Probability Functions:**
  - `binompdf(n, p, k)`: Computes exact probability of k successes, $P(X = k)$.
  - `binomcdf(n, p, k)`: Computes cumulative probability of up to k successes, $P(X \leq k)$.
- **Geometric Probability Functions:**
  - `geometpdf(p, k)`: Probability that the first success occurs exactly on trial k, $P(X = k)$.
  - `geometcdf(p, k)`: Cumulative probability of first success on or before trial k, $P(X \leq k)$.

C.4 Statistical Inference Tests & Confidence Intervals

- **Proportion Inference:**
  - *One-Proportion z-Test:* `[STAT]` $\rightarrow$ `TESTS` $\rightarrow$ `5:1-PropZTest`. Enter p_0, x, n, choose alternative, `Calculate`.
  - *One-Proportion z-Interval:* `[STAT]` $\rightarrow$ `TESTS` $\rightarrow$ `A:1-PropZInt`. Enter x, n, C-Level.
  - *Two-Proportion z-Test:* `[STAT]` $\rightarrow$ `TESTS` $\rightarrow$ `6:2-PropZTest`.
  - *Two-Proportion z-Interval:* `[STAT]` $\rightarrow$ `TESTS` $\rightarrow$ `B:2-PropZInt`.
- **Mean Inference:**
  - *One-Sample t-Test:* `[STAT]` $\rightarrow$ `TESTS` $\rightarrow$ `2:T-Test`.

  - *One-Sample t-Interval:* `[STAT]` → `TESTS` → `8:TInterval`.
  - *Two-Sample t-Test:* `[STAT]` → `TESTS` → `4:2-SampTTest`. Always select `Pooled: NO` for AP Statistics.
  - *Two-Sample t-Interval:* `[STAT]` → `TESTS` → `0:2-SampTInt`. Always select `Pooled: NO`.

- **Chi-Square Tests:**
  - *Goodness-of-Fit Test:* Enter observed in `L1`, expected in `L2`. Press `[STAT]` → `TESTS` → `D:`χ^2`GOF-Test`. Set $df = k - 1$.
  - *Two-Way Table Test:* Enter observed matrix in `[2nd]` `[x`$^{-1}$`]` (MATRIX) → `[A]`. Press `[STAT]` → `TESTS` → `C:`χ^2`-Test`. Expected counts are automatically calculated and stored in `[B]`.

- **Regression Inference:** Press `[STAT]` → `TESTS` → `F:LinRegTTest`. Enter `Xlist: L1`, `Ylist: L2`, choose alternative ($\beta \neq 0$), `Calculate`. Outputs t-statistic, p-value, $df = n-2, a, b, s, r^2, r$.

Appendix D

Appendix D: Official AP Statistics Statistical Tables

Table Usage & Exam Instructions

The following tables are modeled directly on the official tables provided in the AP Statistics Exam booklet. Use these tables when working through practice problem sets and simulated exams.

D.1 Table A: Standard Normal Cumulative Probabilities ($z \leq 0$)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
−1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
−1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
−1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
−1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
−1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
−1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
−1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
−0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
−0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
−0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
−0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
−0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
−0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
−0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
−0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
−0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
−0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

D.2 Table A (Cont.): Standard Normal Cumulative Probabilities ($z \geq 0$)

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

D.3 Table B: t-Distribution Critical Values (t^*)

	Tail Probability p (Upper Tail)							
	.25	.20	.15	.10	.05	.025	.01	.005
	Confidence Level C							
df	50%	60%	70%	80%	90%	95%	98%	99%
1	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032
6	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707
7	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499
8	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355
9	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250
10	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169
11	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106
12	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055
13	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012
14	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977
15	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947
16	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921
17	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898
18	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878
19	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861
20	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845
21	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831
22	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819
23	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807
24	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797
25	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787
26	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779

D.4 Table C: χ^2 Distribution Critical Values (χ^*)

	Tail Probability p (Upper Tail Area)							
df	.25	.20	.15	.10	.05	.025	.01	.005
1	1.32	1.64	2.07	2.71	3.84	5.02	6.63	7.88
2	2.77	3.22	3.79	4.61	5.99	7.38	9.21	10.60
3	4.11	4.64	5.32	6.25	7.81	9.35	11.34	12.84
4	5.39	5.99	6.74	7.78	9.49	11.14	13.28	14.86
5	6.63	7.29	8.12	9.24	11.07	12.83	15.09	16.75
6	7.84	8.56	9.45	10.64	12.59	14.45	16.81	18.55
7	9.04	9.80	10.75	12.02	14.07	16.01	18.48	20.28
8	10.22	11.03	12.03	13.36	15.51	17.53	20.09	21.95
9	11.39	12.24	13.29	14.68	16.92	19.02	21.67	23.59
10	12.55	13.44	14.53	15.99	18.31	20.48	23.21	25.19
11	13.70	14.63	15.77	17.28	19.68	21.92	24.72	26.76
12	14.85	15.81	16.99	18.55	21.03	23.34	26.22	28.30
13	15.98	16.98	18.20	19.81	22.36	24.74	27.69	29.82
14	17.12	18.15	19.41	21.06	23.68	26.12	29.14	31.32
15	18.25	19.31	20.60	22.31	25.00	27.49	30.58	32.80
16	19.37	20.47	21.79	23.54	26.30	28.85	32.00	34.27
17	20.49	21.61	22.98	24.77	27.59	30.19	33.41	35.72
18	21.60	22.76	24.16	25.99	28.87	31.53	34.81	37.16
19	22.72	23.90	25.33	27.20	30.14	32.85	36.19	38.58
20	23.83	25.04	26.50	28.41	31.41	34.17	37.57	40.00
21	24.93	26.17	27.66	29.62	32.67	35.48	38.93	41.40
22	26.04	27.30	28.82	30.81	33.92	36.78	40.29	42.80
23	27.14	28.43	29.98	32.01	35.17	38.08	41.64	44.18
24	28.24	29.55	31.13	33.20	36.42	39.36	42.98	45.56
25	29.34	30.68	32.28	34.38	37.65	40.65	44.31	46.93
26	30.43	31.79	33.43	35.56	38.89	41.92	45.64	48.29
27	31.53	32.91	34.57	36.74	40.11	43.19	46.96	49.65

Final AP Statistics Exam Readiness Checklist

The 2027 Score-5 Checklist

Review this final checklist 24 hours before your examination:

- ☐ **TI-84 Fresh Batteries / Fully Charged:** Put new batteries or charge overnight; ensure `DiagnosticOn` is enabled.
- ☐ **C-U-S-S Framework Memorized:** Center, Unusual Features, Shape, Spread always in context.
- ☐ **P-A-N-I-C for Confidence Intervals:** Parameter, Assumptions, Name interval, Interval calculation, Conclusion in context.
- ☐ **P-H-A-N-T-O-M for Significance Tests:** Parameter, Hypotheses, Assumptions, Name test, Test statistic, Obtain p-value, Make decision in context.
- ☐ **Variances Always Add:** $\sigma_{X\pm Y} = \sqrt{\sigma_X^2 + \sigma_Y^2}$. Never subtract standard deviations!
- ☐ **Large Counts for χ^2:** All **expected counts** ≥ 5 (never check observed counts!).
- ☐ **Degrees of Freedom Precision:** $df = n - 1$ for 1-sample t; $df = n - 2$ for slope t; $df = (r-1)(c-1)$ for two-way tables.
- ☐ **Never Say "Prove":** Fail to reject H_0; never claim H_0 is proven or accepted.